# Electric Current and Circuits

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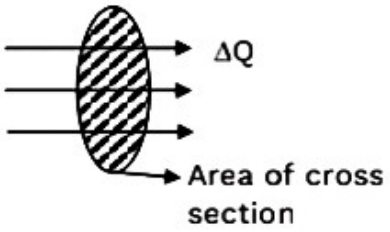
# Electric Current

**INTRODUCTION**
Unlike static electricity, in current electricity there is a continuous flow of electrons along a path. This passage of electric current through a material is referred to as conduction or simply electricity.

Time rate of flow of charge through a cross-sectional area is called current. If $\Delta q$ charge flows in time interval $\Delta t$ then average current is given by

$$I_{avg} = \frac{\Delta q}{\Delta t}$$

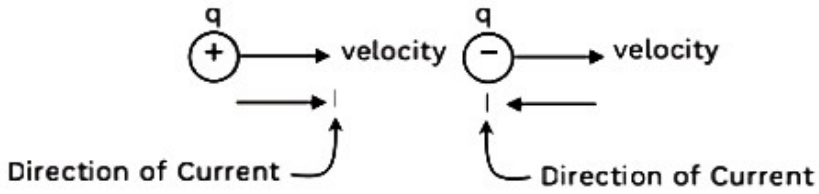
$\Delta q$ = Net charge flow through a cross-section in $\Delta t$ time.




**Instantaneous Current :**

$$I = \lim_{\Delta t \to 0} \frac{\Delta q}{\Delta t} = \frac{dq}{dt}$$

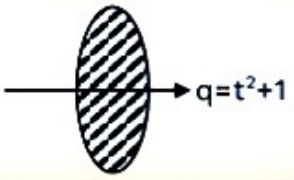
**Direction of Current :**




SI unit is ampere.

$$I = \frac{q}{t} = \frac{1\text{Columb}}{1\text{sec}} = \frac{1C}{\text{sec}} = 1 \text{ amp.}$$

**Q.** **For the given figure ($q = t^2 + 1$) find;**
**(1) Average Current in first two second.**
**(2) Instantaneous Current at, t = 0, t = 1, t = 2 sec.**




**Sol.** (1) $t = 0$ $\quad q = 0^2 + 1$ $\quad q_i = 1\ C$

$t = 2$ sec $\quad q = (2)^2 + 1$ $\quad q_f = 5\ C$

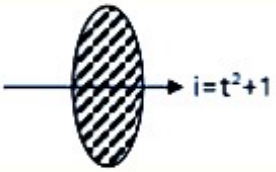
$$I_{avg} = \frac{\Delta q}{\Delta t} = \frac{5-1}{2} = \frac{4}{2} = 2\,\text{amp}$$

(2) $q = t^2+1$

$$i = \frac{dq}{dt} = 2t + 0$$

$i = 2t$ ;
$i\ (t = 0) = 0$ amp
$i\ (t = 1) = 2$ amp
$i\ (t = 2) = 4$ amp

**Q.** **(a) Find I; t = 0, t = 1 sec**
**(b) Average current for first 2 second.**
**Current through area of cross-section.**




**Sol.** (a) $I = t^2 + 1 \Rightarrow I(0) = 0^2 + 1 = 1 \text{ amp} \Rightarrow I(1) = 1^2 + 1 = 2 \text{ amp.}$

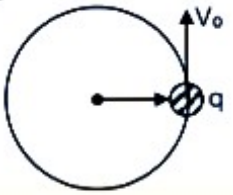
(b) $\text{Avg current } I_{avg} = \frac{\Delta q}{\Delta t} = \frac{\text{charge flow in first 2 sec}}{\text{Time}}$

$$\Delta q = \int_0^2 i.dt$$

$$\Delta q = \int_0^2 (t^2 + 1).dt = \left|\frac{t^3}{3}\right|_0^2 + (t)_0^2 = \frac{8}{3} + 2 = \frac{8+6}{3} = \frac{14}{3} \text{ coulomb}$$

$$I_{avg} = \frac{\Delta q}{\Delta t} = \frac{14}{3 \times 2} = \frac{7}{3} \text{amp} \quad \Rightarrow \quad I_{avg} = \frac{7}{3} \text{amp}$$

**Q.** **Find average current in the loop.**

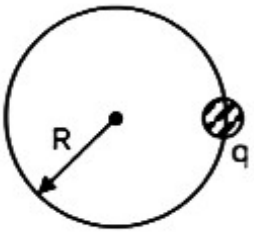



**Sol.**

$$T = \frac{2\pi R}{v_0}$$

T = time period

$$I_{avg} = \frac{\Delta q}{\Delta T} = \frac{q}{T} = \frac{q}{\frac{2\pi R}{v_0}}; \quad I_{avg} = \frac{q \times v_0}{2\pi R} \text{ amp}$$




**Q.** **Find equivalent average current of ring having charge q and angular speed ω.**

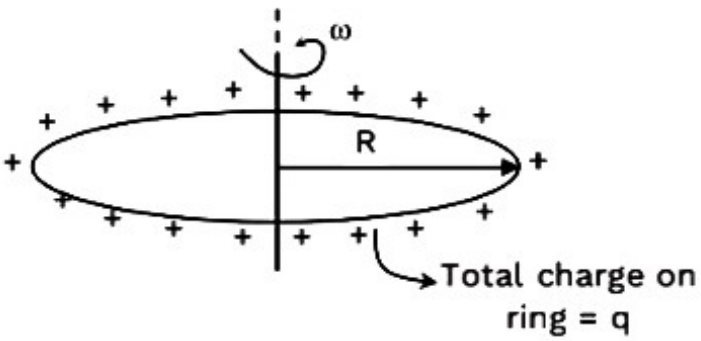
**Sol.**

$$I_{avg} = \frac{\Delta q}{\Delta t} = \frac{\Delta q}{T} = \frac{q}{T} = q \times f$$

$$I_{avg} = q \times f$$

$$f = \frac{1}{T} = \frac{1}{\frac{2\pi}{\omega}} = \frac{\omega}{2\pi}$$

$$I_{avg} = q \times \frac{\omega}{2\pi}$$



**Q.** **One billion electrons pass from A to B in a conductor in time 1ms. Find the direction and magnitude of current**

**Sol.** $i = \frac{Ne}{t} = \frac{10^9 \times 1.6 \times 10^{-19}}{10^{-3}} = 1.6 \times 10^{-7}$ amp $\Rightarrow$ i = 0.16 μA.

The current flows from B to A.

- It is a scalar quantity.

**Behaviour of Conductor in Absence of Applied Electric Field**

If we do not apply battery across conductor, electrons move in random direction.

- The average displacement and average velocity is zero.

The electron which is present in the conductor gains energy due to temperature and moves in random directions.

The speed gained by virtue of temperature is called as thermal speed and the motion is known as thermal motion of an electron.

$$\frac{1}{2}mv_{rms}^2 = \frac{3}{2}kT$$

So, thermal speed $V_{rms} = \sqrt{\frac{3KT}{m}}$ where m is mass of electron.

At room temperature T = 300 K, $V_{rms} \simeq 10^5$ m / sec.

**Two important definitions :**

(1) **Mean free path (λ) :** The average distance travelled by an electron between two successive collision is know as mean free path. ($\lambda \sim 10A^\circ$)

$$\lambda = \frac{\text{Total distance travelled by free electron}}{\text{number of collision}}$$

(2) **Relaxation Time :** The time taken by an electron between two successive collision is called as relaxation time (t). ($t \sim 10^{-14}$ sec)

$$\text{Relaxation time : } \tau = \frac{\text{total time taken}}{\text{number of collision}}$$

**Behaviour of conductor in presence of applied electric field :**

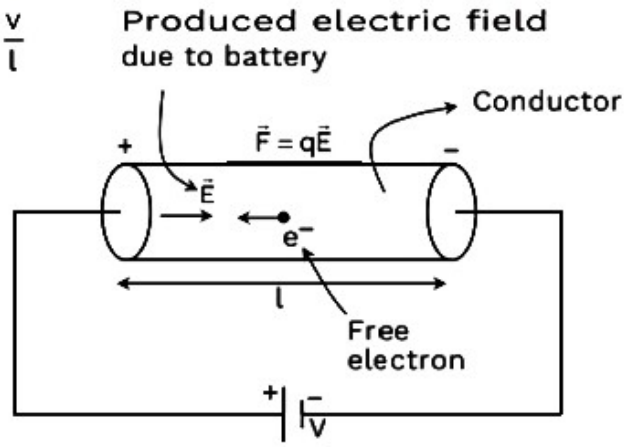
(1) Electric field is developed in conductor $E = \frac{v}{l}$

(2) Force on electron :

$\vec{F} = q\vec{E} \Rightarrow \vec{F} = -e\vec{E}$

(3) Acceleration of electron :

$\vec{a} = \frac{\vec{F}}{m} \Rightarrow \vec{a} = -\frac{e\vec{E}}{m}$

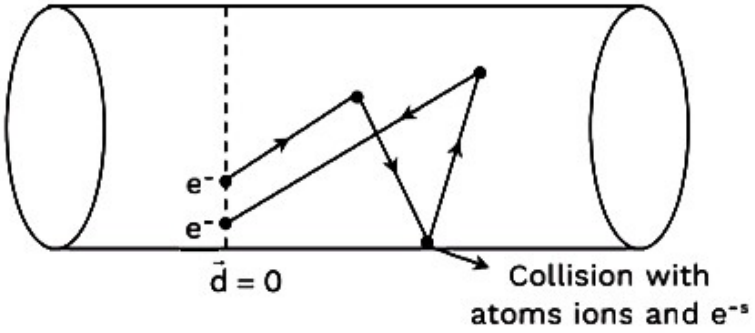


# Drift Velocity

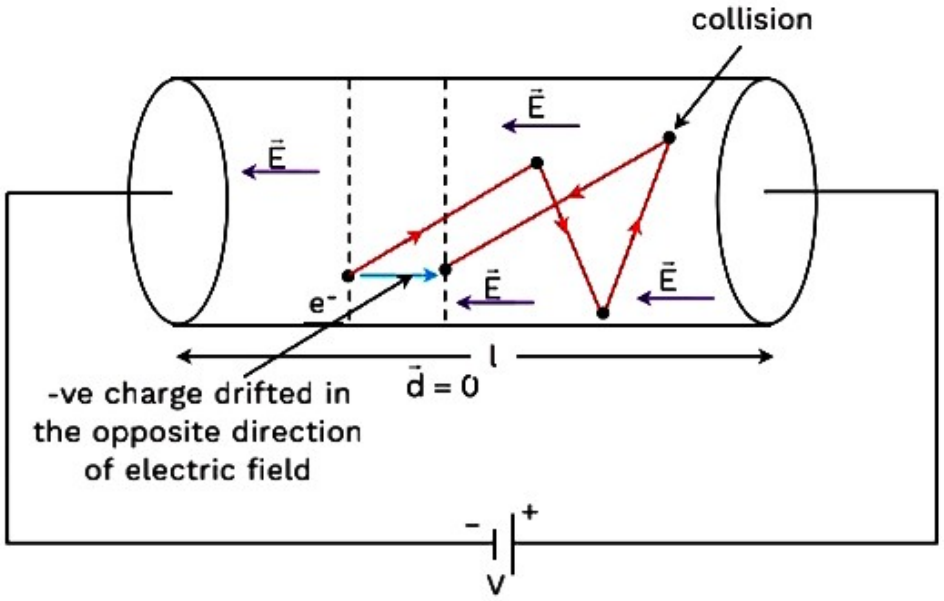
Drift velocity is defined as the velocity with which the free electron gets drifted towards the positive terminal under the effect of applied electric field.

**Path of electron in absence of $\vec{E}$ field ($\vec{E} = 0$) :**

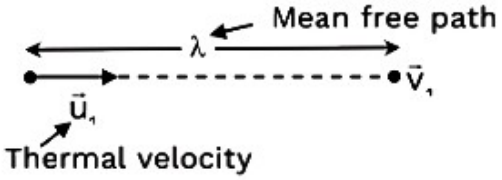
In absence of applied potential difference, electrons have random motion. The average displacement is zero.




**Path of electron in presence of $\vec{E}$ field :**




In addition to its thermal velocity, due to acceleration given by applied electric field, the electron acquires a velocity component in a direction opposite to the direction of the electric field. The gain in velocity due to the applied field is very small and is lost in the next collision.



$$\vec{v}_1 = \vec{u}_1 + \vec{a}\tau_1 \quad ...(1)$$

Where $\vec{a}$ = acceleration of electron

$\tau$ = Relaxation time

Similarly for other electrons :

$$\vec{v}_2 = \vec{u}_2 + \vec{a}\tau_2 \quad ...(2)$$

$$\vec{v}_3 = \vec{u}_3 + \vec{a}\tau_3 \quad ...(3)$$

$$\vec{v}_n = \vec{u}_n + \vec{a}\tau_n \quad ...(n)$$

The average velocity of all the free electrons in the conductor is equal to the drift velocity $\vec{v}_d$ of the free electrons.

$$\vec{v}_d = \frac{\vec{v}_1 + \vec{v}_2 + \vec{v}_3 ..... \vec{v}_n}{n} = \frac{\vec{u}_1 + \vec{a}\tau_1 + \vec{u}_2 + \vec{a}\tau_2 + \vec{a}\tau_2 + .....\vec{u}_n + \vec{a}\tau_n}{n}$$

$$= \left(\frac{\vec{u}_1 + \vec{u}_2 + \vec{u}_3 + ...\vec{u}_n}{n}\right) + a\left(\frac{\tau_1 + \tau_2 + \tau_3 + ...\tau_n}{n}\right)$$

Since average thermal speed = 0

So, $\frac{\vec{u}_1 + \vec{u}_2 + \vec{u}_3 + ...\vec{u}_n}{n} = 0$

and $\frac{\tau_1 + \tau_2 + \tau_3 + ...\tau_n}{n} = \tau$ = average relaxation time.

So, $\vec{v}_d = \vec{a}\tau$

$$\vec{v}_d = \frac{-e\vec{E}}{m_e}(\tau)$$

Where;

$\vec{v}_d$ = Drift velocity of electron

$\vec{E}$ = Electric field applied

$m_e$ = Mass of electron

$\tau$ = Average relaxation time

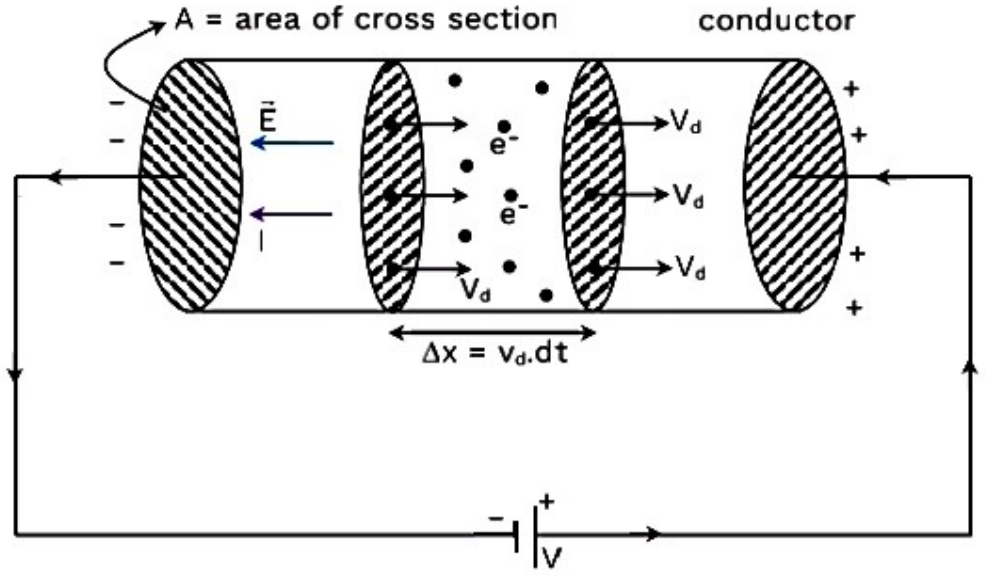
**Note :** (1) The order of drift velocity is $10^{-4}$ m/sec

(2) Drift speed of electron.

$$|v_d| = \frac{eE}{m_e}(\tau)$$

**Relation Between Current & Drift Velocity**

Let the number of free electrons per unit volume in a conductor is = n.




Number of charge particles confined in $\Delta x$ distance = number of electron per unit volume × volume.

$= n \times (A \times \Delta x)$ ...(1)

$(\Delta x) = v_d \times dt$ ...(2)

"Total charge flowing in dt time $= (e^-) \times$ no of charge particle

$= e^- \times (n \times A \times v_d \times dt)$

Total charge $(dq) = neAv_d \times dt$

So, Electric current $i = \frac{dq}{dt} = \frac{d}{dt}(ne^-Av_d dt)$ $\boxed{i = neAv_d}$

Where, i = electric current

n = number of charge particle per unit volume

e = electric charge

A = Area of cross section

$v_c$ = drift velocity

**Q. A current of 1.34 A exits in a copper wire of cross section 1.0 mm². Assume each copper atom contribute one free electron calculate the drift speed of the free electron in the wire. The density of copper is 8990 kg/m³ and atomic mass = 63.50.**

**Sol.** Mass of 1 $m^3$ volume of the copper is = 8990 kg = $8990 \times 10^3$g

Number of mole in 1 $m^3$ = $\frac{8990 \times 10^3}{63.5} = 1.4 \times 10^5$

As each mole contains $6 \times 10^{23}$ atoms therefore number of atoms in 1 $m^3$.

$n = 1.4 \times 10^5 \times (6 \times 10^{23}) = (8.4 \times 10^{28})$

$\because \quad I = neAv_d$

$\therefore \quad v_d = \frac{I}{neA} = \frac{1.34}{8.4 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}} = 10^{-4}$ m/sec

$v_d = 0.1$ mm / sec

**Current Density**

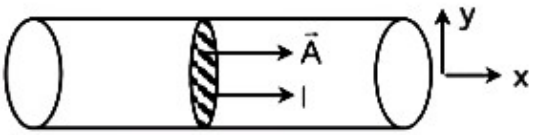
Current density is a vector quantity and it is defined as current per unit normal area at that point and direction is along the direction of the current at that point.

$$\vec{J} = \frac{di}{ds}\hat{n} \Rightarrow I = \int \vec{J}.d\vec{s}$$

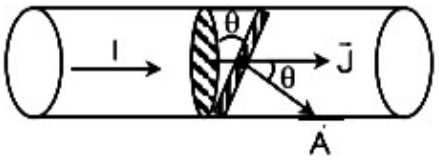
$\hat{n}$ = unit vector in the direction of current.

$$\vec{J} = \frac{i}{|A_{\perp r}|}(\hat{n})$$

$|A_{\perp r}|$ = magnitude of area which is perpendicular to current direction.




$|\vec{A}| = A, \quad \vec{J} = \frac{I}{|A|}(\hat{i})$




$I = \vec{J}.\vec{ds}$

$= |\vec{J}||A|\cos\theta$

$I = JA\cos\theta$

$|J| = \frac{I}{A\cos\theta} = \frac{I}{A_{\perp r}}$

Important points

(i) $\vec{J} = \left(\frac{I}{A_{\perp r}}\right)(\hat{n})$

$\hat{n}$ = unit vector along direction of current.

(ii) $I = \int \vec{J}.\vec{ds}$

**Note :** Current density $\vec{J}$ is a vector quantity. It's direction is same as that of $\vec{E}$. It's SI unit is ampere /$m^2$ and dimension $[L^{-2}A]$.

**Q.** **The current density at a point is $\vec{J} = \left[\left(2\times10^4\right)\hat{j}\right] Jm^{-2}$. As what rate the charge flows through a cross sectional area $\vec{s} = (2\hat{i} + 3\hat{j})cm^2$.**

**Sol.** The rate of flow of charge = current = $I = \vec{J}.\vec{ds}$

$\Rightarrow \quad I = \vec{J}.\vec{s} = (2 \times 10^4)\left[\hat{j}.(2\hat{i} + 3\hat{j})\right] \times 10^{-4} A \Rightarrow I = 6$ A.

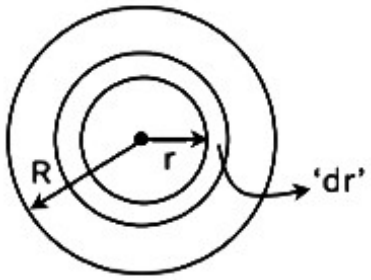
**Q.** **A potential difference applied to the end of a wire made up of an allow drives a current through it. The variation of current density is given as J = (3 + 2r), where r is the distance of the point from the axis. It R be the radius of the wire, then the total current through any cross section of the wire.**

**Sol.** $dI = \vec{J}.\vec{ds} = (3 + 2r)(2\pi r dr)\cos 0^\circ$

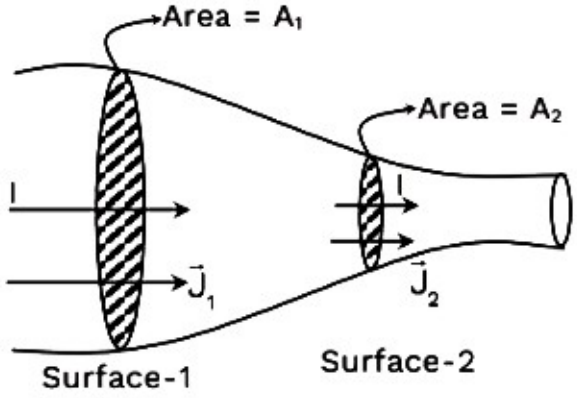
$= (2\pi)\left(3r + 2r^2\right)dr$

$$I = \int_0^R (2\pi)\left(3r + 2r^2\right)dr = 2\pi\left[\frac{3r^2}{2} + \frac{2}{3}r^3\right]_0^R$$

$$I = 2\pi\left[\frac{3R^2}{2} + \frac{2R^3}{3}\right] \text{ units}$$



**Relation between current density and area**




Since current is same at surface 1 and surface 2.

$$I_1 = I_2$$

$$J_1A_1 = J_2A_2$$

Also, we can find relation between $\vec{E}$ and $\vec{J}$ fixed at these two surfaces area.

By ohm's law (vector form),

$\vec{J} = \sigma\vec{E}$ [will be discussed later in this chapter]

$$\left|\vec{J}_2\right| = \left|\vec{J}_1\right|\frac{A_1}{A_2}$$

Since $A_2 < A_1$

$$\left|\vec{J}_2\right| > \left|\vec{J}_1\right|$$

$$\left|\sigma\vec{E}_2\right| > \left|\sigma\vec{E}_1\right|$$

$$\left|\vec{E}_2\right| > \left|\vec{E}_1\right|$$

Where,

$E_2$ = Electric field at surface $S_2$

$E_1$ = Electric field at surface $S_1$

**Mobility**

**We can defind mobility as** the drift velocity per unit electric field.

$$\mu = \frac{v_d}{E} = \frac{|v_d|}{E}.$$

Unit : $\frac{m^2}{\text{volt. sec}}$.

**Q.** **Consider a conductor of length 40 cm where a potential difference of 10V is maintained between the ends of conductor. Find out the mobility of the electrons provided the drift velocity of the electrons is $5 \times 10^{-4}$ $ms^{-1}$**

**Sol.** Given L = 40 cm, V = 10V, $V_d = 5 \times 10^{-4}$ $ms^{-1}$

To find the electrons mobility, we need the value of the electric field which can be obtained using the following formula.

$$E = \frac{V}{L} \Rightarrow E = \frac{10}{0.4} = 25V/m$$

Also the formula used for electron mobility.

$$\mu = \frac{V_d}{E} = \frac{5 \times 10^{-4}}{25} \frac{1}{5} \times 10^{-5}$$

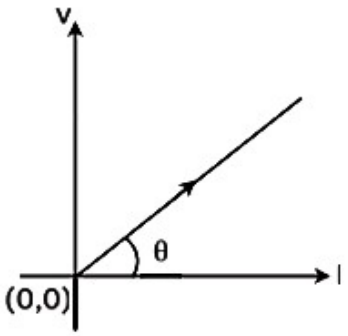
$$= 0.2 \times 10^{-6} = 2 \times 10^{-7}\ m^2V^{-1}s^{-1}$$

# Ohm's Law

**Statement of ohm's Law :**

If the physical conditions of the conductor (length, temperature, mechanical strain etc.) remain same, then the current flow flowing through the conductor is directly proportional to the potential difference across its two ends e.g., $I \propto V \Rightarrow V = IR$

Graph :

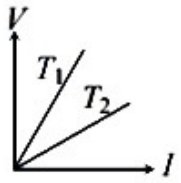



Ohm's law is not a universal law. The substances which obey Ohm's law are known as ohmic substances.

$\Rightarrow$ Slope of the graph : $\tan\theta = \frac{V}{I} = R =$ electrical resistance

Unit : voltage/ampere ; also called ohm ($\Omega$).

**Q.** **The current–voltage (I-V) graphs for a given metallic wire at two different temperatures $T_1$ and $T_2$ are shown in the figure. It follows from the graphs that**




**(A) $T_1 > T_2$ (B) $T_1 < T_2$ (C) $T_1 = T_2$**

**(D) $T_1$ is greater or less than $T_2$ depending on whether the resistance R of the wire is greater or less than the ratio V/I.**

**Sol.** $R = \frac{V}{I}$ = Slope of V-I curve

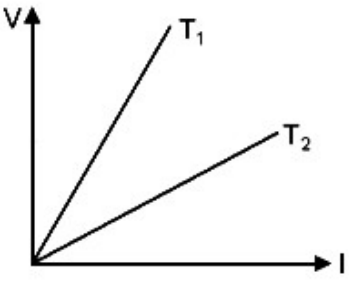
Slope of V-I curve at $T_1$ > Stope of V-I curve at $T_2$

$R_1 > R_2$

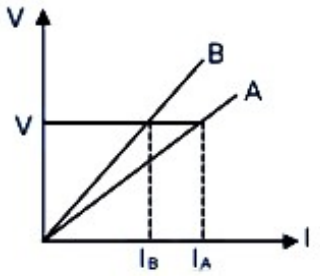
From Relation ; $R = R_0 (1 + \alpha\Delta T)$

$R = f(T)$, T increases $\Rightarrow$ R increases, $T_1 > T_2$

So, (A) is correct option.



**Q.** **The Voltage-current graphs for two resistors of the same material and the same radius with lengths, $L_1$ and $L_2$ are shown in figure. If $L_1 > L_2$, state with reason which of these graphs represents voltage-current change for $L_1$.**




**Sol.** As $R_A = V/I_A$ and $R_B = V/I_B$,

$$\frac{I_A}{I_B} = \frac{R_B}{R_A}$$

From figure, as $I_A > I_B$, so $R_B > R_A$. Since, $R \propto L$, graph B represents the voltage-current change for $L_1$.

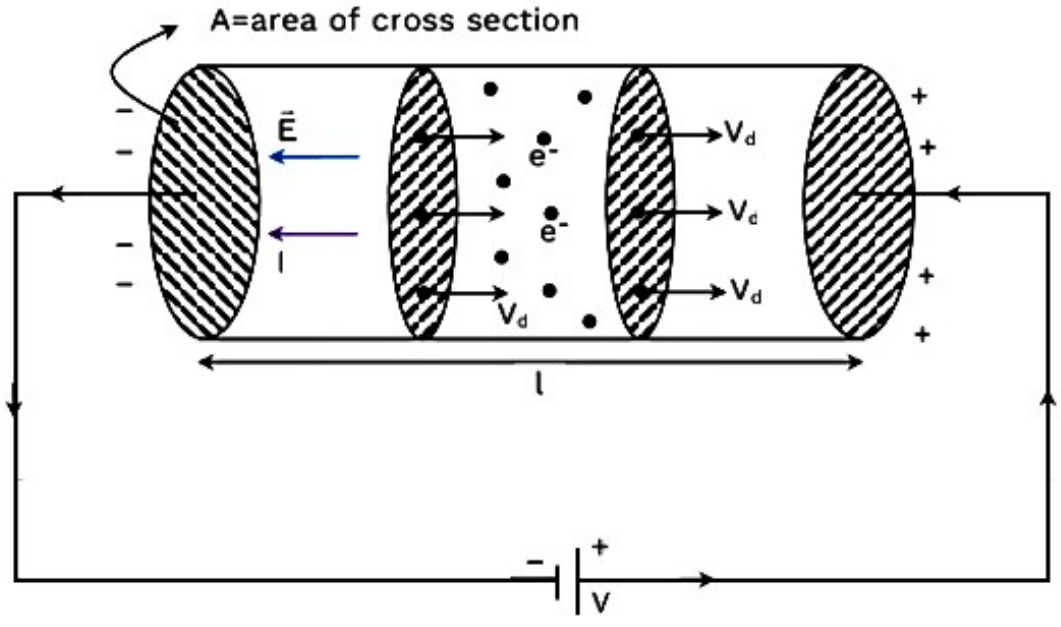
Ohm's law can be expressed in two forms.
(1) microscopic ohm's law (vector form)
(2) macroscopic ohm's law (scalar form)

**Microscopic Ohm's law (Vector form)**

Statement: In many materials, the current density is linearly dependent on the external electric $\vec{E}$, with their relation expressed as

$$\vec{J} = \sigma\vec{E}$$

Where $\sigma$ = conductivity of material.



We know the relation : $I = neAv_c$ ...(1)

Drift speed : $|\vec{v}_d| = \left|\frac{e\vec{E}\tau}{m_e}\right|$ ...(2)

From equation (1) and (2)

$$I = neA\left|\frac{e\vec{E}\tau}{m_e}\right| \Rightarrow I = \left(\frac{ne^2\tau}{m_e}\right)(|\vec{E}|)A$$

$$\frac{I}{A} = \left(\frac{ne^2\tau}{m_e}\right)|\vec{E}| \Rightarrow J = \left(\frac{ne^2\tau}{m_e}\right)|\vec{E}| \Rightarrow \vec{J} = \left(\frac{ne^2\tau}{m_e}\right)\vec{E}$$

For a given conductor; if physical condition of conductor is constant that is

(1) Length (l)

(2) Temperature (as temperature is constant, $\tau$ is also constant)

(3) Area of cross section (A)

Then, $\frac{ne^2\tau}{m_e} = \text{constant}$. So, $\vec{J} \propto \vec{E}$

(current density) $\propto$ (electric field)

$\boxed{\vec{J} = \sigma\vec{E}}$, this is known as ohm's law in vector form or known as microscopic ohm's law.

Where $\sigma$ = electrical conductivity of conductor.

$$\sigma = \frac{ne^2l}{m_e}$$

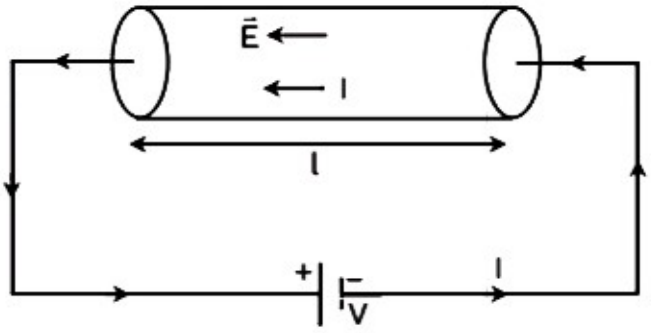
**Ohm's law in (scalar form)**

$$\vec{J} = \sigma\vec{E}$$

$$\frac{I}{A} = \sigma E$$

$$\frac{I}{A} = \frac{1}{\rho}E$$

$$I = \frac{1}{\rho} \times \left(\frac{V}{l}\right) \times A, \Rightarrow I = \left(\frac{A}{\rho \times l}\right)(V)$$




$$\sigma = \frac{1}{\rho}$$

$\rho$ = electrical resistivity of material

$$\rho = \frac{m_e}{ne^2\tau}$$

$$|\vec{E}| = \frac{V}{l}$$

$$I = \left(\frac{A}{\rho l}\right)(V)$$

$$V = \left(\frac{\rho l}{A}\right)(I)$$

For a given conductor : $M_e$ = mass of electron

If temperature is constant, $\tau$ and 'n' are also constant

Then, A = constant, l = constant, $\rho$ = constant

then $\rho\frac{l}{A}$ = constant called electrical resistance of the material.

Where, $\boxed{V = IR}$, "ohm's Law in scalar form".

Where R is called resistance of the material.

$$R = \frac{\rho l}{A}$$

where l = length of conductor along the direction of current.

A = area of cross section perpendicular to current.

R = Electrical resistance.

**Q.** **A certain electric conductor has a square cross section 2.0 mm on a side and is 12 m long. The resistance between its ends is 0.072 Ω.**

**(a)** **Find the resistivity of the material?**

**(b)** **If the electric field magnitude in the conductor is 0.12 V/m, what is the total current?**

**(c)** **If the material has $8 \times 10^{28}$ free electron per cubic meter, find the average drift velocity under the conditions of part (b).**

**Sol.** (a) $A = (2 \times 10^{-3})^2 = 4 \times 10^{-6}\ m^2$

$l = 12\ m,\ R = 0.072\ \Omega$

$$\rho = \frac{RA}{l} = \frac{0.072 \times 4 \times 10^{-6}}{12} = 2.4 \times 10^{-8}\ \Omega m\ .$$

(b) Electric field in the conductor = $\frac{V}{l} = 0.12\ V/m$

$V = E \times l \Rightarrow V = 0.12 \times 12 = 1.44$ volts

$i = \frac{V}{R} = \frac{1.44}{0.072} = 20A$

(c) Let drift velocity = $V_d$

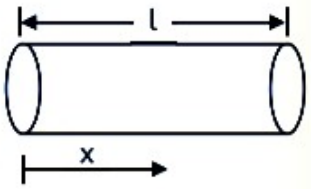
$n = 8 \times 10^{28}$ electron/$m^3$

current = $I = neAV_d$

$$V_d = \frac{i}{neA} = \frac{20}{1.6 \times 10^{-19} \times 8 \times 10^{28} \times 4 \times 10^{-6}}$$

$V_d = 3.9 \times 10^{-4}$ m / sec

**Q.** **A cylindrical wire has length l and area of cross section A. The conductivity of wire material changes with distance from left end as $\sigma = \sigma_0 \frac{l}{x}$, where $\sigma_0$ is a constant. Find the electric field inside the conductor as a function of x, when a battery of emf V is connected across its ends.**




**Sol.** Let us calculate the resistance of the cylindrical conductor.
Here, we can consider the cylinder to be made of a number of elemental discs.
Resistance of one such elemental disc (shown in figure) will be

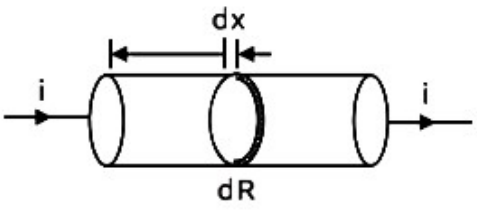
$$dR = \frac{dx}{\sigma A} = \frac{xdx}{A\sigma_0 l}$$

Hence, total resistance of the cylindrical wire

$$R = \int dR = \frac{1}{Al\sigma_0}\int_0^l xdx = \frac{1}{Al\sigma_0}\left[\frac{x^2}{2}\right]_0^l = \frac{l}{2A\sigma_0}$$

Current through the wire, $I = \frac{V}{R} = \frac{2VA\sigma_0}{l}$

Current density, $j = \frac{I}{A} = \frac{2V\sigma_0}{l}$



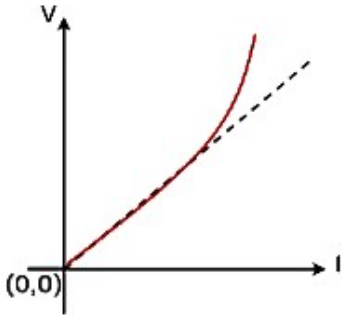

Using ohm's law in microscopic form,

$$E = \frac{j}{\sigma} = \frac{2V\sigma_0}{l}\frac{x}{\sigma_0 l} = \frac{2V}{l^2}x$$

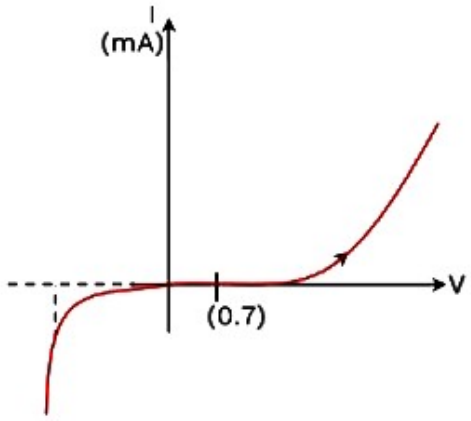
**Limitation of Ohm's Law :**

Although ohm's law has been found valid over a large class of materials, there do exist material and devices used in electric circuits where the proportionality of V and I does not hold.

**(a)** The solid line is the voltage V versus current I for good conductors.

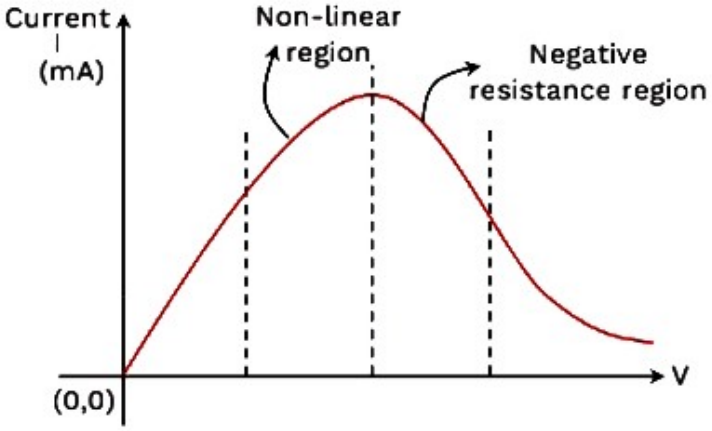



**(b)** V-I curve for non-ohmic resistance



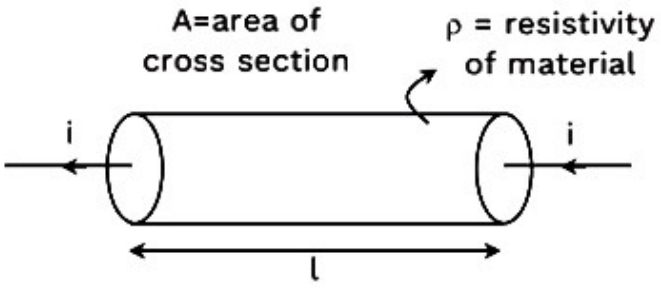

**Variation of Current $V_s$ voltage for GaAs.**

The relation between V and I is not unique i.e., there is more than one value of V for the same current I, For example V-I curve for GaAs. Such material is called non-ohmic material or non-ohmic devices. V = IR does not hold for such class of material and device.



**Resistance**

The resistance of a conductor is defined as the opposition which the conductor offers to the flow of charge(electrons in the case of conductor). When we apply a potential difference across a conductor, free electrons get accelerated and collide with position ions, atoms and electrons and their motion is thus opposed. The opposition offered by the ions, atoms and electrons is called resistance of the conductor.

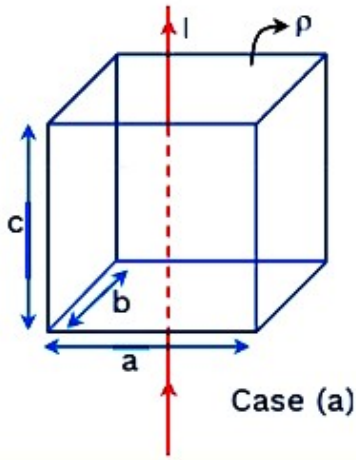



Resistance is the property of a conductor by virtue of which if opposes the flow of current in it.

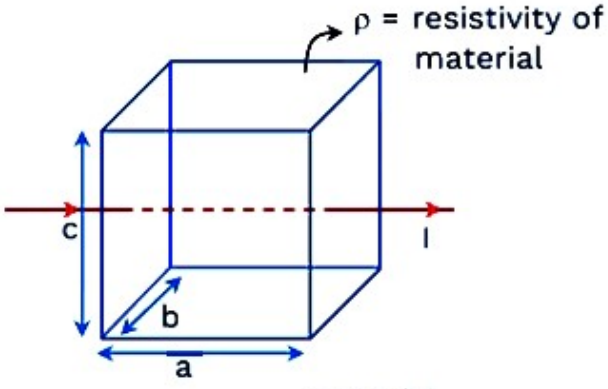
$$R = \rho \frac{\ell}{A}$$

Where, $\ell$ = length of conductor along the direction of current.
A = Area of cross section of the material, perpendicular to the direction of current.
ρ = resistivity of the material.

**Q.** **Find the resistance in the following cases :**




Case (a)

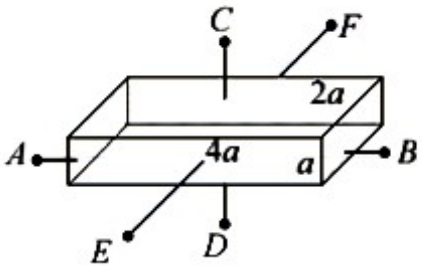



Case (b)

**Sol.** (a) $R = \rho \times \frac{l}{A} \Rightarrow R = \rho \times \frac{c}{a \times b}$

(b) $R = \rho \times \frac{l}{A} \Rightarrow R = \rho \times \frac{a}{(b \times c)}$

**Q.** **A conductor with rectangular cross-section has dimensions (a × 2a × 4a) as shown in figure. Resistance across AB is x, across CD is y and across EF is z. Then**




**(A) x = y = z** **(B) x > y > z** **(C) y > z > x** **(D) x > z > y**

**Sol.** $R = \frac{\rho l}{A}$, $x = \rho\frac{4(a)}{(a)\ (2a)} = \frac{2\rho}{a}$, $y = \frac{\rho(a)}{(4a)\ (2a)} = \frac{\rho}{8a}$

$z = \frac{\rho(2a)}{(4a)\ (a)} = \frac{\rho}{2a}$ then $x > z > y$ $\therefore$ (D) is correct option.

**Q.** **A 3000 km long cable consists of seven copper wires, each of diameter 0.73 mm, bundled together and surrounded by an insulting sheath. Calculate the resistance of the cable. Use $3 \times 10^{-6}\ \Omega$ cm for the resistivity of the copper.**

**Sol.** The resistance R of a conductor is related to the resistivity $\rho$ by $R = \frac{\rho l}{A}$, where l and A are the length of the conductor and the cross-sectional area, respectively. Since the cable consists of N = 7 copper wires, the total cross-sectional area is

$$A = N\pi r^2 = N\frac{\pi d^2}{4} = 7\frac{\pi(0.073\,\text{cm})^2}{4}$$

The resistance then becomes,

$$R = \frac{\rho l}{A} = \frac{(3\times10^{-6}\,\Omega\,\text{cm})(3\times10^{8}\,\text{cm})}{\frac{7\pi(0.073\,\text{cm})^2}{4}} = 3.1\times10^{4}\,\Omega \quad \Rightarrow\ R = 31\ \text{K}\Omega$$

**Recasting a wire of given mass**

Consider a conductor of mass m, density d, resistivity $\rho$ to be recast to a wire of length l, cross sectional area A (or radius r), then we have

$$R = \frac{\rho l}{A} \qquad ...(1)$$

Since, m = volume × density = Ald ...(2)

$$\Rightarrow A = \frac{m}{ld}$$

Then from (1), we get,

$$R = \frac{\rho l}{(m/ld)} = \left(\frac{\rho d}{m}\right) l^2 \qquad ...(3)$$

As $\rho$, d, m are constants, so we have

$$R \propto l^2$$

$$\Rightarrow \boxed{\frac{R_2}{R_1} = \left(\frac{l_2}{l_1}\right)^2} \qquad ...(4)$$

So, we conclude that if a given mass of wire is stretched (or drawn) to n times its original length, then the new resistance is $n^2$ times the original resistance of wire. Hence

$$\boxed{R_{new} = n^2 R_{old}}$$

Further if we put value of l from (2) i.e., $l = \frac{m}{Ad}$ in (1), then we get, $R = \frac{\rho m}{AdA} = \left(\frac{m\rho}{d}\right)\left(\frac{1}{A^2}\right)$

$$\Rightarrow \boxed{R \propto \frac{1}{A^2}} \qquad ...(5)$$

Since $A = \pi r^2$ $\Rightarrow$ $R \propto \frac{1}{r^4}$

**Remark(s) :**

Thus, if given mass of material is stretched to make its length l, then $R \propto l^2$ and if the given mass of material is stretched to make its radius r, then

$$R \propto \frac{1}{r^4} \quad \text{or} \quad \frac{R_2}{R_1} = \left(\frac{r_1}{r_2}\right)^4$$

So, we conclude that if a given mass of wire is stretched (or draw) such that new radius of wire is n times its old radius, then

$$R_{new} = \frac{1}{n^4} R_{old}$$

**Fractional change in resistance due to different parameter change like length, area and radius wire**

Since we know that The resistance of a wire depends on its length L, radius r and cross-sectional area A (where $A = \pi r^2$). Suppose the the wire is stretched by a small amount, its resistance R changes.

**Case 1 :**

Suppose the length L is increased by a small amount $\Delta L$ (<< L). Then, we can write

$$R = \frac{\rho L}{A} = \rho\left(\frac{L^2}{V}\right)$$ (As AL = volume (V), which is constant for given wire)

Now we calculate fractional change in resistance with a small change in length is

$$\boxed{\frac{\Delta R}{R} = 2\frac{\Delta L}{L}}$$

**Case 2 :**

If the wire is stretched so that the cross-sectional area A reduces by small amount $\Delta A$ (<< A). Then we can write

$$R = \frac{\rho L}{A} = \rho\left(\frac{V}{A^2}\right) \quad \Rightarrow \quad \boxed{\frac{\Delta R}{R} = -2\frac{\Delta A}{A}}$$

**Case 3 :**

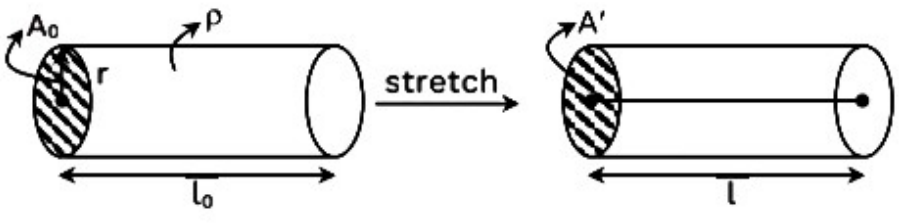
Suppose the wire is stretched so that the radius reduces by $\Delta r$ (<< r). Then, we can write

$$R = \frac{\rho L}{A} = \rho\left(\frac{LA}{A^2}\right) = \rho\frac{V}{(\pi r^2)^2} = \left(\frac{\rho V}{\pi^2}\right)\frac{1}{r^4} \Rightarrow \quad \boxed{\frac{\Delta R}{R} = -4\frac{\Delta r}{r}}$$

**Stretch Based Question**

**Q.** **Suppose wire is stretched to double its length, find out the new resistance of the wire if original resistance of the wire was 'R'.**

**Sol.**




$$R = \rho \times \frac{l_0}{A} \quad ...(i)$$

New resistance = R' = ?

**Concept :**

(1) Volume must be constant

$V_1 = V_2$ (here $V_1$ and $V_2$ are initial and final volume of wire)

$A_0 \times l_0 = A' \times nl_0$

$$A' = \frac{A_0 l_0}{nl_0} \Rightarrow A' = \frac{A_0}{n}$$

(2) $\rho$ = constant : Since material is same

$$R' = \rho \times \frac{l'}{A'} \Rightarrow R' = \frac{\rho \times (nl_0)}{\left(\frac{A_0}{n}\right)} = \left(\frac{\rho l_0}{A_0}\right)(n^2) \Rightarrow R' = n^2 R$$

Given n = 2; $R' = (2)^2 R \Rightarrow R' = 4R$

**Short Cut :**

If a wire is stretched such that its new length becomes $nl_0$ then new resistance of wire becomes $R' = n^2R$

**Q.** **A carbon resistance is stretched such that it becomes 0.1% larger. Then calculate the percentage change in the resistance.**

**Sol.** $$R = \rho \times \frac{l}{A} \Rightarrow R = \rho \times \frac{l}{A} \times \frac{l}{l}$$

$$R = \rho \times \frac{l^2}{(A \times l)} \Rightarrow R = \rho \times \frac{l^2}{V} = \rho \frac{l^2}{V}$$

Take log both side

$$\ln(R) = \ln\left(\rho \times \frac{l^2}{V}\right)$$

$$\ln(R) = \ln(\rho) + \ln(l^2) - \ln(V)$$

Now differentiate both side w.r.t. x

$$\frac{1}{R}\frac{dR}{dx} = \frac{1}{\rho}\frac{d\rho}{dx} + 2\frac{1}{l}\frac{dl}{dx} - \frac{1}{V}\frac{dV}{dx}$$

$$\frac{dR}{R} = \frac{d\rho}{\rho} + 2\frac{dl}{l} - \frac{dV}{V}$$

Since, $\rho$ = constant; V = constant

$$\frac{dR}{R} = 2\left(\frac{dl}{l}\right)$$

$$100 \times \frac{\Delta R}{R} = 2\left(\frac{dl}{l}\right) \times 100 \quad \Rightarrow \quad \left(\frac{\Delta R}{R} \times 100\right) = 2\left(\frac{\Delta l}{l} \times 100\right)$$

Percentage change

$$\%R = 2\left(\frac{\Delta l}{l} \times 100\right) = 2 \times (0.1\%)$$

$$\frac{\Delta R}{R} \times 100 = 0.2\%$$

**Q.** **A wire of mass m, length l, density d, and area of cross section A is stretched due to this length increased by 10% of its its original value. Express the changed resistance in percentage.**

**Sol.** Given mass m, length $l_1 = l$, density d, and area of cross section $A_1 = A$. Let $\rho$ be the resistivity and $R_1$ be the resistance of the wire. Mass of wire m = volume × density = Al × d = Ald.

Therefore, area of cross section is $A_1 = m/ld$. and the resistance of the wire is.

$$R_1 = \rho \frac{l}{A_1} = \rho \frac{l^2 d}{m}\left(\frac{\rho e}{m}\right) l^2 = kl^2$$

Let $l_2$ be the new length, then

$$l_2 = l + \frac{10}{100} l = l + 0.1l = 1.1l$$

Let $R_2$ be the resistance of the wire after stretching, then

$$R_2 = kl_2^2$$

Now Dividing Eq. (ii) by Eq. (i), we have $\frac{R_2}{R_1} = \frac{l_2^2}{l_1^2} = \frac{(1.1)^2 l_1^2}{l_1^2} = 1.21$

or $R_2 = 1.21\ R_1 = R_1 + 0.21\ R_1$

or $R_2 - R_1 = 0.21\ R_1$

Hence, the percentage change in the resistance is

$$\frac{R_2 - R_1}{R_1} \times 100 = 21\%$$

**Resistivity ('ρ') and Its Dependency on Temperature**

For a given conductor of uniform cross-sectional A and length l, it has been observed that the electrical resistance R is

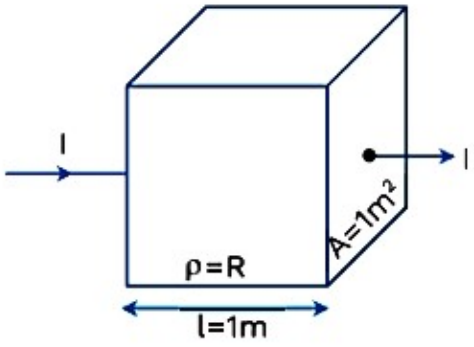
$$\Rightarrow R \propto \frac{l}{A} \Rightarrow R = \rho \frac{l}{A}$$

Where ρ is a constant of proportionality called specific resistance or resistivity of the material of the conductor at a given temperature.

$$\Rightarrow \rho = \frac{RA}{l}$$

If l = 1m, A = 1m², then ρ = R (Numerically) so, the specific resistance of the material of the conductor is defined as the resistance offered by the conductor of length 1m and cross sectional area 1m², when current flows normal to the area.

**Unit :** ohm-metre (Ω m).




**Resistivity depends on**

(1) Nature of Material
(2) Temperature of material
(3) It does not depend on the size and shape of the material because it is the property of the conductor material.

**Dependency of Resistivity on Temperature**

**Case 1:** Conductor : $\rho = \frac{m}{ne^2\tau}$

Temperature ↑ ⇒ vibration of $e^-$ ↑ ⇒ Collision ↑ ⇒ τ ↓ ⇒ ρ ↑ ⇒ R ↑
So, on increasing temperature ρ increases and also, R increases.

**Case 2:** Semi conductor :
Temperature ↑ ⇒ τ ↓ ⇒ ρ ↑
Temperature ↑ ⇒ n ↑ ⇒ ρ ↓
Temperature ↑ ⇒ nτ ↑ ⇒ ρ ↓
So, on increasing temperature ⇒ ρ decreases and also, R decreases.

**Temperature Dependency of Resistivity (Mathematical relation)**

Let α = fractional change in resistivity per unit change in temperature.

$$\Rightarrow \alpha = \frac{d\rho}{\rho dT}$$

$\Rightarrow \quad \frac{d\rho}{\rho} = \alpha dT$ ($\alpha$ can be assumed constant for small temperature variation)

Integrate both sides,

$$\Rightarrow \int_{\rho_0}^{\rho} \frac{d\rho}{\rho} = \alpha \int_{T_0}^{T} dT \Rightarrow \ln\left(\frac{\rho}{\rho_0}\right) = \alpha (T - T_0)$$

$$\Rightarrow \quad e^{\alpha(T-T_0)} = \frac{\rho}{\rho_0} \Rightarrow \rho = \rho_0 . e^{\alpha(T-T_0)} \quad \Rightarrow \quad \rho = \rho_0 . e^{\alpha(T-T_0)}$$

**Approximation**

$$\rho = \rho_0 \left(1 + \alpha (T - T_0) + \frac{\{\alpha (T - T_0)\}^2}{2} + \ldots\right)$$

Here we use $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$

Now : If $\alpha (T - T_0) <<< 1$ then we can Neglect higher order terms:

$$\boxed{\rho = \rho_0 (1 + \alpha (T - T_0))}$$

Where $\rho_0$ = Resistivity at temperature $T_0$

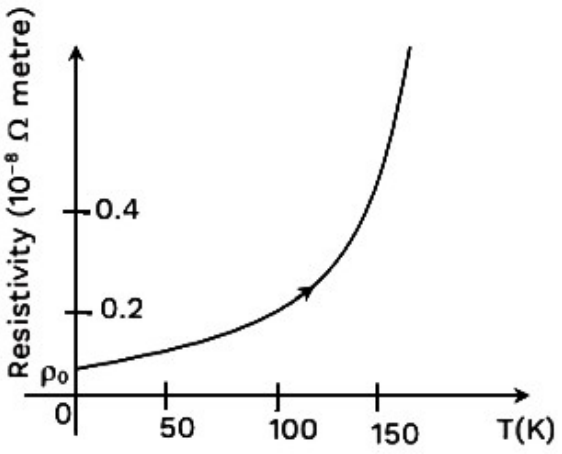
$\alpha$ = Coefficient of thermal resistivity

$T - T_0$ = Change in temperature

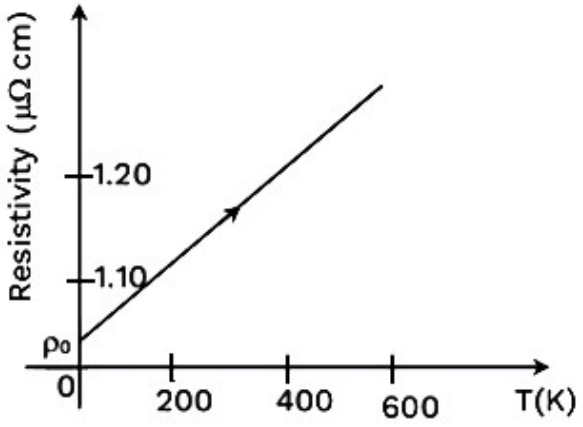
$\rho$ = Resistivity at temperature T

**Graph of Resistivity and Mathematical Relation**

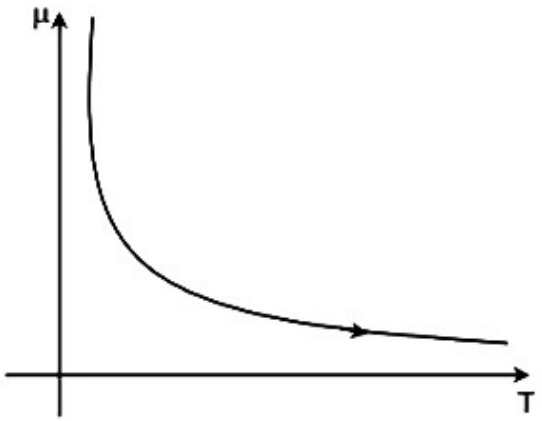
Resistivity $\rho_T$ of copper as a function of Temperature T.



Resistivity vs. temperature curve for Nichrome




Resistivity vs. temperature curve for Semiconductor




**Q.** **Given that A wire l = 8m long of uniform cross-sectional area A = 8 mm², has a conductance of G = 2.45 $\Omega^{-1}$. Find the resistivity of given material of the wire.**

**(A)** $2.1 \times 10^{-7}\ \Omega m$ **(B)** $3.1 \times 10^{-7}\ \Omega m$

**Sol.** **(C)** $4.1 \times 10^{-7}\ \Omega m$ **(D)** $5.1 \times 10^{-7}\ \Omega m$

$$\rho = \frac{RA}{l} = \frac{A}{Gl} = \frac{8 \times 10^{-6}}{2.45 \times 8} = 4.1 \times 10^{-7}\ \Omega \text{ meter}$$

$\therefore$ (C) is correct option.

**Variation of Resistance with Temperature**

We know that the resistance of a given conductor depends on its length and area of cross-section besides the resistivity.

**Approximation:** As temperature changes, the length and area also change but these changes are quite small and factor $\left(\frac{l}{A}\right)$ may be treated as constant.

$$R = \frac{\rho l}{A} \Rightarrow R = \rho \times \frac{l}{A}$$

$$\frac{l}{A} = \text{constant}$$

Take ln both side,

$$\ln(R) = \ln\left(\frac{l}{A}\right) + \ln(\rho)$$

$$\ln(R) = \ln(l) - \ln(A) + \ln(\rho)$$

Differentiate both side,

$$\frac{1}{R}dR = 0 - 0 + \frac{1}{\rho}d\rho \quad \Rightarrow \quad \frac{dR}{R} = \frac{d\rho}{\rho} \qquad ...(1)$$

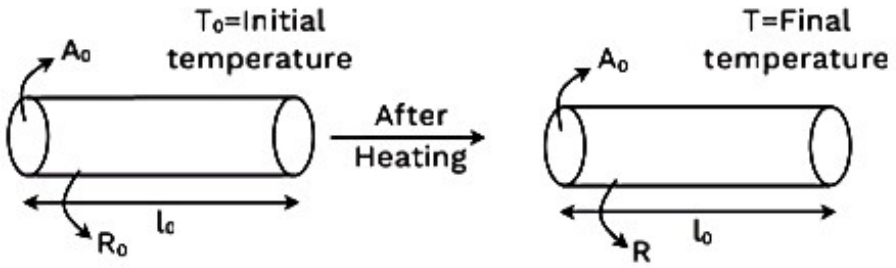
From relation : $\frac{d\rho}{\rho dT} = \alpha \Rightarrow \frac{d\rho}{\rho} = \alpha dT$ ...(2)

From equation (1) and (2)

Important relation : $\boxed{\frac{dR}{R} = \alpha dT}$

Where $\alpha$ = Temperature coefficient Resistance.

Integrate both side, $\int_{R_0}^{R} \frac{dR}{R} = \int_{T_0}^{T} \alpha dT$



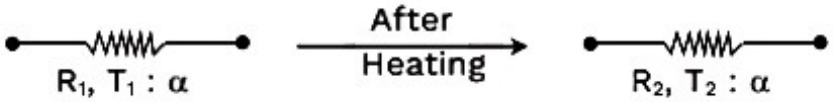

$$\left|\ln(R)\right|_{R_0}^{R} = \alpha \left|aT\right|_{T_0}^{T} \quad \Rightarrow \quad \ln\left(\frac{R}{R_0}\right) = \alpha(T - T_0)$$

$$R = R_0 . e^{\alpha(T - T_0)}, \text{ if } \left|\alpha(T - T_0)\right| << 1$$

$$R = R_0\left(1 + \alpha(T - T_0)\right)$$

**Temperature coefficient of Resistance**

The resistance change factor per degree Celsius of temperature change is called the temperature coefficient of resistance. It is denoted by 'α'.




$$\alpha = \left|\frac{R_2 - R_1}{R_1T_2 - R_2T_1}\right|.$$

**Q.** **A copper coil has a resistance of 20.0 Ω at 0°C and a resistance of 26.4 Ω at 80°C. Find out the temperature coefficient of resistance of copper.**

**Sol.** $R_{80^\circ C} = R_{0^\circ C}[1 + \alpha\Delta T]$

$\Rightarrow \quad 26.4 = 20.0\,[1 + \alpha \times (80 - 0)]$

$\Rightarrow \quad \frac{26.4}{20} = 1 + 80\alpha$

On solving, we get $\alpha = 4 \times 10^{-3}/^\circ C$

**Q.** **The resistance of the metallic wire is 120 Ω at 20°C. Find the temperature at which the resistance of same metallic wire rises to 240 Ω where the temperature coefficient of wire is $2 \times 10^{-40}\ C^{-1}$.**

**Sol.** Given $R_1 = R_{20} = 120\ \Omega$, $R_2 = R_T = 240\Omega$, $\alpha = 2 \times 10^{-40}\ C^{-1}$, $t_1 = 20^\circ C$, $t_2 = T = ?$

Formula used to find the temperature coefficient of resistance is

$$\alpha = \frac{R_2 - R_1}{R_1(t_2 - t_1)} \quad \Rightarrow \quad \alpha = \frac{R_T - R_{20}}{R_{20}(T - 20)}$$

$$2 \times 10^{-4} = \frac{240 - 120}{120(T - 20)} \quad \Rightarrow \quad T - 20 = \frac{120}{120 \times 2 \times 10^{-4}}$$

$$\Rightarrow \qquad T = \frac{1}{2 \times 10^{-4}} + 20 = 0.5 \times 10^4 + 20 = 5020^\circ C$$

**Q.** **This silver wire has resistance of 1.0 Ω at 20°C. The wire is placed in liquid bath & its resistance rises to 1.2 Ω. What is the temperature of the bath? (Here $\alpha = 10^{-2}$/°C).**

**Sol.** Here change in resistance is very small so that we can apply
$R = R_0 (1 + \alpha\Delta\theta)$

$\Rightarrow 1.2 = 1 \times (1 + 10^{-2}\Delta\theta)$

$\Rightarrow \Delta\theta = 20°C$

$\Rightarrow \theta - 20 = 20$

$\Rightarrow \theta = 40°C$

**Q.** **A conducting wire has resistance of 10 ohm at 0°C and a is $\frac{1}{273}$/°C, then determine its resistance at 273°C.**

**Sol.** In such type of problem, term $\alpha\Delta T$ will have a larger value so could not be used directly in $R = R_0(1 + \alpha\Delta T)$. We need to go for basic as

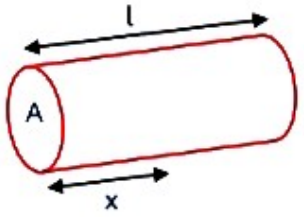
As we know that $\alpha = \frac{dR}{RdT}$

$\Rightarrow \int \frac{dR}{R} = \int \alpha dT \quad \Rightarrow \ln \frac{R_2}{R_1} = \alpha (T_2 - T_1)$

$\Rightarrow R_2 = R_1 e^{\alpha(T_2 - T_1)} \quad \Rightarrow R = 10e^1$

$\Rightarrow R_2 = 10\ e\ \Omega$

**Q.** **The cross-sectional area and length of cylindrical conductor are A and l respectively. The specific conductivity varies as $\sigma(x) = \sigma_0 l / \sqrt{x}$ where x is the distance along axis of the cylinder from one of its ends.**

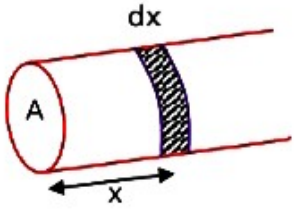



**(i)** **Compute the resistance of the system along the cylinder axis.**

**(ii)** **What is the electric field at each point in the cylinder in the above case?**

**Sol.** (i) $dR = \frac{1}{\sigma(x)}\frac{dx}{A} = \frac{\sqrt{x}dx}{Al\sigma_0}$

$\therefore \quad R = \int_0^l dR = \frac{1}{Al\sigma_0}\int_0^l \sqrt{x}dx = \frac{2\sqrt{l}}{3A\sigma_0}$




(ii) $\frac{V_0}{R} = \frac{3A\sigma_0 V_0}{2\sqrt{l}}$

The current density, $J = \frac{I}{A} = \frac{3\sigma_0 V_0}{2\sqrt{l}}$

Electric field, $E(x) = \frac{J}{\sigma(x)} = \frac{3V_0\sqrt{x}}{2l^{3/2}}$

**Q.** **Two wires A and B having resistivity $\rho_A = 3\times10^{-5}\,\Omega m$ and $\rho_B = 6\times10^{-5}\Omega m$ of same cross section area are connected end to end to form a single wire. If the resistance of the joined wire does not change in temperature, then find the ratio their lengths, given that temperature coefficient of resistivity of wire A and B is $\alpha_A = 4\times10^{-5}/°C$ and $\alpha_B = -6\times10^{-6}/°C$. You can Assume that mechanical dimensions do not change with temperature.**

**Sol.** $R_A' = R_A(1+\alpha_A\Delta T)$

$R_B' = R_B(1+\alpha_B\Delta T)$

$R' = R_A(1+\alpha_A\Delta T) + R_B(1+\alpha_B\Delta T)$

From given condition,

$R_A(1+\alpha_A\Delta T) + R_B(1+\alpha_B\Delta T) = R_A + R_B$

$\Rightarrow \quad R_A + R_A\alpha_A\Delta T + R_B + R_B\alpha_B\Delta T = R_A + R_B$

$\Rightarrow \quad R_A\alpha_A + R_B\alpha_B = 0$

$\frac{\rho_A l_A}{A_A}\alpha_A + \frac{\rho_B l_B}{A_B}\alpha_B = 0 \Rightarrow \frac{l_A}{l_B} = \frac{3}{10}$

**Q.** **The temperature coefficient of resistivity a is given by $\alpha = \left(\frac{1}{\rho}\right)\frac{d\rho}{dT}$, where r is the resistivity at temperature T.**

**Assume that $\alpha$ is not constant and is given by $\alpha = -\frac{a}{T}$, where T is the absolute temperature where a is a constant, show that the resistivity r is given by $\rho = \frac{b}{T^n}$, where b is another constant.**

**Sol.** (a) $\alpha = \frac{1}{\rho}\frac{d\rho}{dT}$

$\Rightarrow \frac{d\rho}{\rho} = \alpha dT = -a\frac{dT}{T}$

Let $\rho = \rho_0$ at $T = T_0$, then $\int_{\rho_0}^{\rho}\frac{d\rho}{\rho} = -a\int_{T_0}^{T}\frac{dT}{T}$

$\Rightarrow \log_e\left(\frac{\rho}{\rho_0}\right) = -a\log_e\left(\frac{T}{T_0}\right) = \log_e\left(\frac{T_0}{T}\right)^a$

$\Rightarrow \rho = \left(\rho_0 T_0^a\right)\frac{1}{T^a} = \frac{b}{T^a}$

Here, $b = \rho_0 T_0^a$

# Colour Code for Carbon Resistance

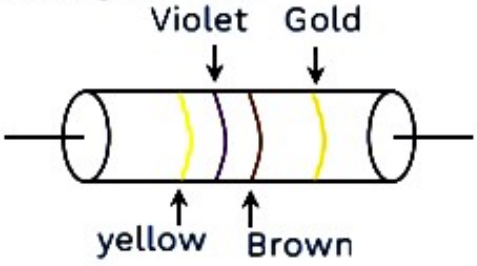
| Colour | Strip A | Strip B | Strip C | Strip D (Tolerance) |
|---|---|---|---|---|
| Block | 0 | 0 | $10^0$ | |
| Brown | 1 | 1 | $10^1$ | |
| Red | 2 | 2 | $10^2$ | |
| Orange | 3 | 3 | $10^3$ | |
| Yellow | 4 | 4 | $10^4$ | |
| Green | 5 | 5 | $10^5$ | |
| Blue | 6 | 6 | $10^6$ | |
| Violet | 7 | 7 | $10^7$ | |
| Grey | 8 | 8 | $10^8$ | |
| White | 9 | 9 | $10^9$ | |
| Gold | - | - | $10^{-1}$ | ± 5% |
| Silver | - | - | $10^{-2}$ | ± 10% |
| No Colour | - | - | - | ± 20% |

Must be remembered as : B B R O Y Great Britain very Good Wife

**Q.** **Draw a colour code for 42 k Ω ± 10% carbon resistance.**

**Sol.** The problem based on colour coding of resistance, According to colour code, colour for digit 4 is yellow for digit 2 it is red, for 3 colour is orange and 10% tolerance is represented by silver colour, so colour code should be yellow, red, orange, and silver.

**Q.** **What is resistance of following resistor.**




**Sol.** Number for yellow is 4, number for violet is 7.
We know Brown colour gives multiplier $10^1$ and gold gives a tolerance of ±5 %.
So resistance of resistor is $47 \times 10^1\ \Omega \pm 5\% = 470 \pm 5\%\ \Omega$.

**Electrical Conductivity**

The reciprocal of resistivity (ρ) is called the conductivity (σ)$\Rightarrow \quad \sigma = \frac{1}{\rho}$

The unit of conductivity is mho metre$^{-1}$ ($\Omega^{-1}$ m$^{-1}$).
Ohm's Law in alternative form may be expressed as J = σE, where J = Current Density and E = Electric Field strength.

**Electrical Conductance**

Conductance (G) is the reciprocal of resistance $\Rightarrow \quad G = \frac{1}{R}$

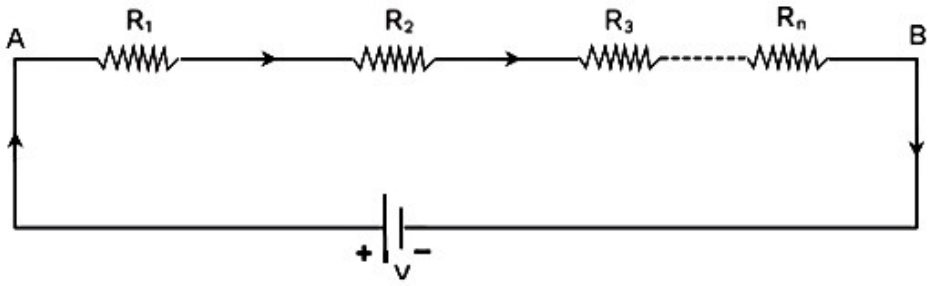
Unit of conductance is siemen or mho (= $\Omega^{-1}$).

# Combination of Resistances

### (a) Series Combination

$R_{series} = R_1 + R_2 + R_3 \ldots\ldots\ldots R_n$

Series connection means current is same in all the resistors.




Let the equivalent resistance between A and B is equal to $R_{eq}$. Then by definition,

$$R_{equivalent} = \frac{V}{I} \qquad \ldots(1)$$

Using Kirchhoff's 2nd Rule for the loop shown in figure,

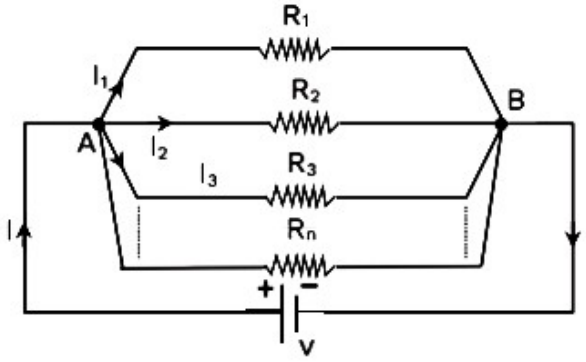
$$V = IR_1 + IR_2 + IR_3 + \ldots\ldots + IR_n \qquad \ldots(2)$$

From equation (1) and (2),

$$\boxed{R_{eq} = R_1 + R_2 + R_3 \ldots\ldots R_n}$$

### (b) Parallel Combination

In parallel combination, potential difference will remain same across each resistor.




Here again, $R_{eq} = \frac{V}{I}$ ...(1)

$$I = i_1 + i_2 + i_3 + \ldots\ldots + i_n = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} + \ldots + \frac{V}{R_n} \qquad \ldots(2)$$

From equation (1) and (2)

$$\boxed{\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \ldots\ldots \frac{1}{R_n}}$$

**Q.** **It is given that the resistance of the series combination of two resistances is S. When they are joined in parallel, the total resistance is become P. If S = nP, then find out the minimum possible value of n is**

**(A) 4 (B) 3 (C) 2 (D) 1**

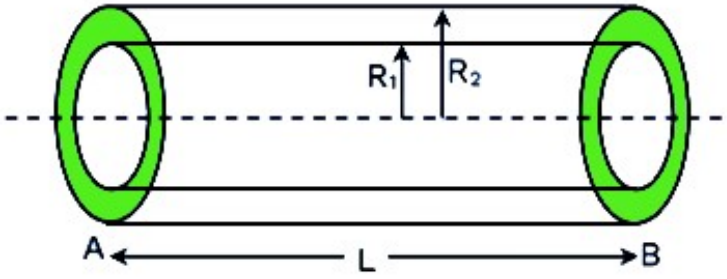
**Sol.** For two resistances $R_1$ and $R_2$

$S = R_1 + R_2$ (in series), $P = \frac{1}{R_1} + \frac{1}{R_2}$ (in parallel)

According to $S = nP$, $R_1 + R_2 = n\left(\frac{R_1R_2}{R_1 + R_2}\right)$

If n is minimum, $R_1 = R_2 = R$ then $n = 4$

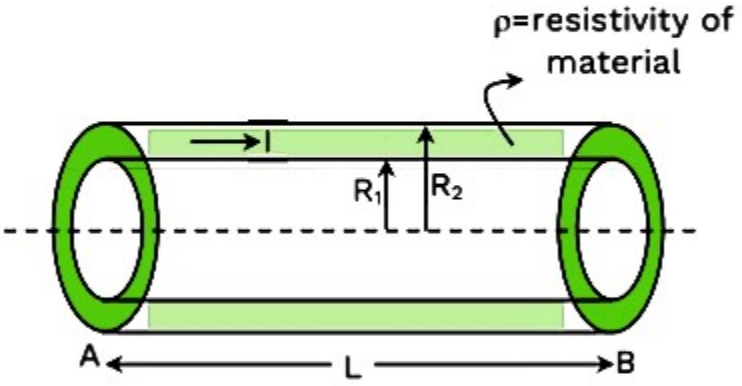
**Q.** **Given that a hollow cylinder of length L and inner radius $R_1$ and outer radius $R_2$, as shown in figure. The material has resistivity RO.**




**(a)** **Suppose a potential difference source (battery) is applied between the ends of the cylinder body and produces a current which is flowing parallel to the axis. Find the resistance?**

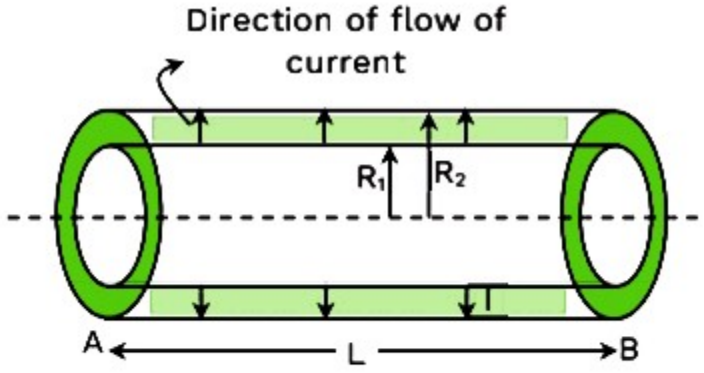
**(b)** **If the potential difference source (battery) is applied between the inner and outer surfaces so that current flows radially outward, find the resistance measured in this case?**

**Sol.** (a) we observe that when a potential difference is applied between the two ends of the given cylinderical body, current flows parallel to the axis of cylinder.




In this case, the cross-sectional area is $A = \pi\left(R_2^2 - R_1^2\right)$, and the resistance is given by $R = \frac{\rho L}{A} = \frac{\rho L}{\pi\left(R_2^2 - R_1^2\right)}$

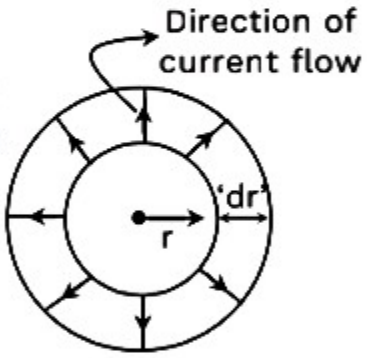
(b) Consider a differential element which is made up of a thin cylinder of inner radius r and outer radius r + dr and length L. Its contribution to the resistance of the system is given by



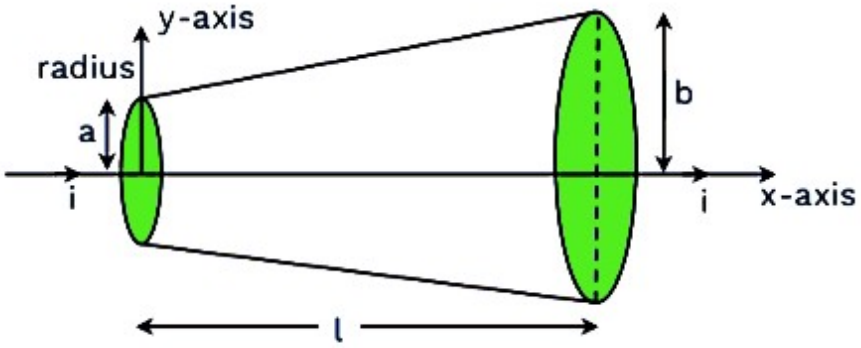

$$dR = \frac{\rho dl}{A} = \frac{\rho dr}{2\pi rL}$$

Where $A = 2\pi rL$ is the area normal to the current flow direction. The total resistance of the system becomes

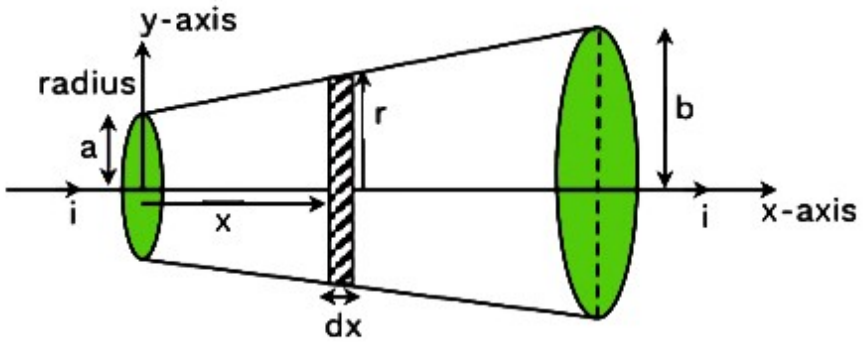
$$R = \int_{R_1}^{R_2} \frac{\rho dr}{2\pi rL} = \frac{\rho}{2\pi L}\log_e\left(\frac{R_2}{R_1}\right)$$



**Q.** **Consider a material of resistivity $r_0$ in a shape of a truncated cone of altitude l, and radii b and a, for the right and the left ends respectively, as shown in the figure.**
**Assuming that the current is distributed uniformly through the cross-section of the cone, what is the resistance between the two ends.**




**Sol.**



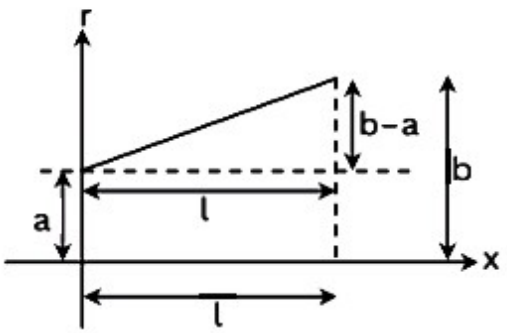

Consider a thin disk of radius 'r' at a distance x from the left end of the figure shown.

Equation of line :

$y = mx + c$

$$r = \frac{(b-a)}{l}(x) + a$$

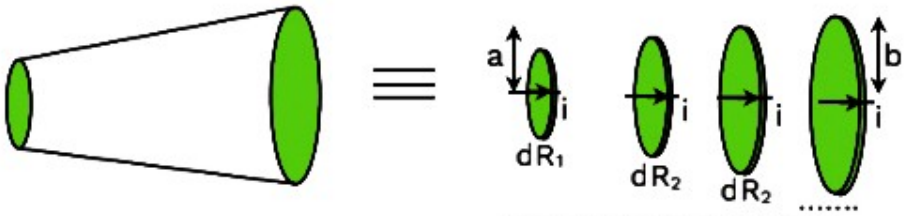
$$r = \frac{(b-a)}{l}(x) + a \qquad \ldots(1)$$




Now differentiate this equation :

$$dr = \frac{(b-a)}{l}(dx) + 0,\ dr = \frac{(b-a)}{l}(dx),\ dx = \frac{l\,dr}{(b-a)}$$

$$R_{Total} = dR_1 + dR_2 + dR_3 + \ldots dR_n$$




Combination of disks

$\sum_{i=1}^{n} dr$ {current is same in all disks so they must be connected in series}

$$R_{Total} = \int dR, \quad R_{Total} = \int dR = \int \frac{\rho_0 dx}{(\pi r^2)}, \quad R_{Total} = \int dR = \rho_0 \int \frac{dx}{\pi r^2}$$

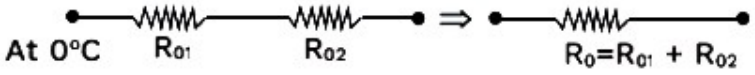
$$R_{Total} = \rho_0 \int \frac{l dr}{(b-a)\pi r^2} = \frac{\rho_0 \times l}{(b-a)\pi} \int_a^b \frac{dr}{r^2} = -\frac{\rho_0 l}{(b-a)\pi}\left(\frac{1}{a} - \frac{1}{b}\right)$$

$$= \frac{\rho_0 l}{(b-a)\pi}\left(\frac{1}{b} - \frac{1}{a}\right) = \frac{\rho_0 l}{(b-a)\pi}\left(\frac{b-a}{ab}\right)$$

$$R = \frac{\rho_0 l}{\pi(ab)} \text{ (Total resistance)}$$

**Q.** **Two resistors with temperature coefficient of resistance $\alpha_1$ and $\alpha_2$ have resistances $R_{01}$ and $R_{02}$ at 0°C. What is the temperature coefficient of the compound resistor consisting of the two resistors connected, (a) in series (b) in parallel.**

**Sol.** In series :




At t°C, $R_{01}(1+\alpha_1 t) + R_{02}(1+\alpha_2 t) = R_0(1+\alpha t)$

$$\Rightarrow \quad R_{01}(1+\alpha_1 t) + R_{02}(1+\alpha_2 t) = (R_{01} + R_{02})(1+\alpha t)$$

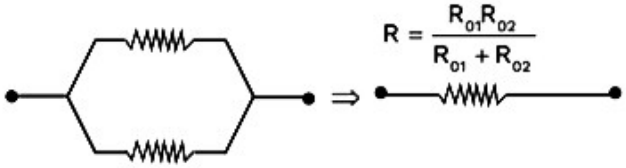
$$\Rightarrow \quad R_{01} + R_{01}\alpha_1 t + R_{02} + R_{02}\alpha_2 t = R_{01} + R_{02} + (R_{01} + R_{02})\alpha t$$

$$\Rightarrow \quad \alpha = \frac{R_{01}\alpha_1 + R_{02}\alpha_2}{R_{01} + R_{02}}$$

In Parallel :

At t °C, $\frac{1}{R_0(1+\alpha t)} = \frac{1}{R_{01}(1+\alpha_1 t)} + \frac{1}{R_{02}(1+\alpha_2 t)}$

$$\Rightarrow \quad \frac{R_{01}+R_{02}}{R_{01}R_{02}(1+\alpha t)} = \frac{1}{R_{01}(1+\alpha_1 t)} + \frac{1}{R_{02}(1+\alpha_2 t)}$$




Since $\alpha t << 1$, so by using the Binomial expansion, we get

$$\frac{1}{R_{02}}(1-\alpha t) + \frac{1}{R_{01}}(1-\alpha t) = \frac{1}{R_{01}}(1-\alpha_1 t) + \frac{1}{R_{02}}(1-\alpha_2 t)$$

$$\Rightarrow \quad \alpha t\left(\frac{1}{R_{01}} + \frac{1}{R_{02}}\right) = \frac{\alpha_1}{R_{01}}t + \frac{\alpha_2}{R_{02}}t \quad \Rightarrow \quad \alpha = \frac{\alpha_1 R_{02} + \alpha_2 R_{01}}{R_{01} + R_{02}}$$

# Battery & E.M.F

**Battery**

Battery is a device which maintains a potential difference across its two terminals.

**Electromotive force (e.m.f)**

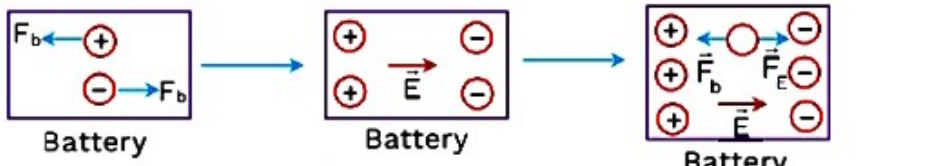
**Definition 1:** Electromotive force is the capacity of the system battery to make the charge flow.

**Definition 2:** It is the work done by the battery for the flow of 1 coulomb charge from lower potential terminal to higher potential terminal inside the battery.

**Mechanism of Battery**

**Condition 1 :**

When battery is not connected to external load.




Battery | Battery | Battery

$\vec{F}_b$ = Non-electrostatic force developed by Battery

$\vec{E}$ = Electric field generated due to charge separation

At full charge condition of battery net force on charge becomes zero

Initially, non-Electrostatic force $\vec{F}_b$ developed by battery mechanism, come in picture and starts pulling positive charge on one side and -ve charge on another side of battery. Due to this, electric field develop in the battery. At some instant.

$$\vec{F}_b = \vec{F}_E = q\vec{E}$$

At this condition battery is fully charged.

When battery is connected to external load 'R' electrons move from negative plate of battery through wire and reach at positive plate of battery so charge on the plates decreases and so does the $\vec{E}$.

$$\left(\vec{F}_b > \vec{F}_\epsilon\right)$$

Due to this, $\vec{F}_b$ again pull some charge to positive plate and negative plate, again $\vec{F}_b = q\vec{E}$ and this process going on. Here work done by $F_b$ come from chemical energy of battery.

**Important Point :**

(1) $\vec{F}_b$ does not decrease with time.

(2) $\vec{E}$ due to charge separation.

(3) Charges are not created only separated.

(4) Battery is source of energy not charge.

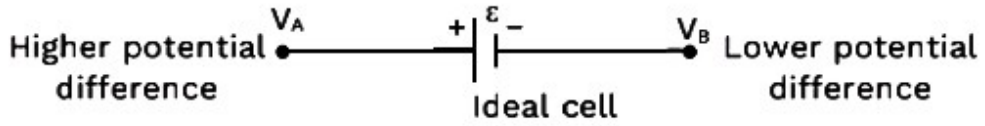
(5) $F_b$ depends on chemical energy.

(6) Inside the battery direction of 'I' and $\vec{E}$ is different.

(7) Electro motive force is measured in volt.

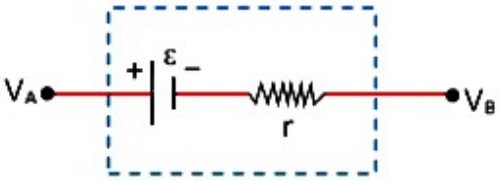
**Representation for Battery**

**(1) Ideal cell :** Cell in which there is no heating effect.




**(2) Non ideal cell :**

Internal resistance (r) : The potential of battery (real source in a circuit) is not equal to the emf of the cell. The reason is that charge moving through the material of battery (electrolyte) of the cell encounters collision causing resistance. We call this the internal resistance of the source.




**Case 1: Battery acting as a source (or battery is discharging)**

$(V_A - V_B) = \varepsilon - ir \Rightarrow V = \varepsilon - ir$

$V_A - V_B = V$ = called terminal voltage.

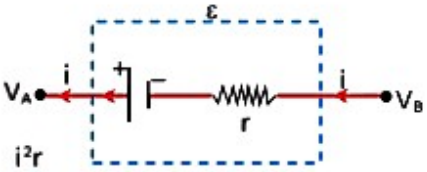
Charge flow in time 'dt' $\Rightarrow dq = idt$

Energy supply to load

$\Delta w = (\varepsilon - ir) \times (idt)$

$\Delta w = \varepsilon i dt - i^2 r dt$

$\frac{dw}{dt} = \varepsilon i - i^2 r$




$(\text{Power})_{\text{Load}} = \varepsilon i - i^2 r$

- $(\text{Power})_{\text{Load}}$ : Power supplied to load
- $\varepsilon i$ : Power developed battery
- $i^2 r$ : Internal power dissipation

**Case 2: Battery acting as a load (or battery charging)**

$V_A - V_B = \varepsilon + ir$

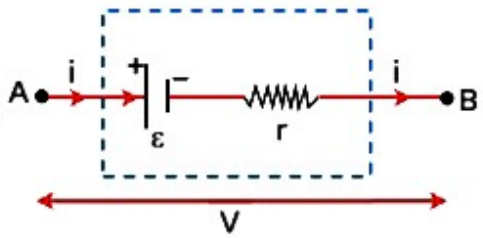
The rate at which chemical energy is stored in the cell = $\varepsilon i$

Thermal power inside the cell = $i^2 r$

Electrical power input = $\varepsilon i + i^2 r$

$= (\varepsilon + ir)i$

$= (V_A - V_B)i$



**For Electrolyte Cell :**

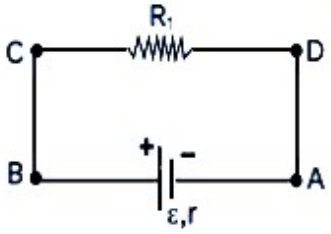
$r \propto d$ (distance between electrode)

$r \propto \frac{1}{s}$ (area of electrode)

$r \propto c$ (concentration)

$r \propto \frac{1}{\text{temperature}}$

**Q.** **In the given electric circuit calculate the following parts**




**(a)** **Current flow in the circuit.**

**(b)** **Power output delivered to load R**

**(c)** **Relation between internal resistance r and R so that the electric power output (that means power given to R) is maximum.**

**(d)** **Find value of maximum power output.**

**(e)** **Plot graph between output power and resistance load**

**(f)** **From graph we see that for a given power output there exists two values of external resistance, prove that the product of these resistance equals $r^2$.**

**(g)** **What is the efficiency of the cell when it is used to supply maximum power.**

**Sol.** (a) In the circuit shown if we assume that potential at A is zero then potential at B is $\varepsilon - ir$. Now since the connecting wires are of zero resistance.

$\therefore\ V_D = V_A = 0 \Rightarrow V_C = V_B = \varepsilon - ir$

Now current through CD is also I ($\because$ it is in series with the cell)

$$\therefore\ i = \frac{V_C - V_D}{R} = \frac{(\varepsilon - ir) - 0}{R}$$

$$\text{Current } i = \frac{\varepsilon}{r + R}$$

**Note :** After learning the concept of series combination we will able to calculate the current directly.

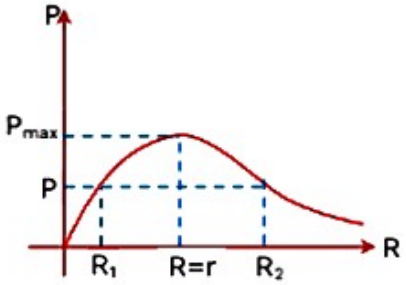
(b) Power output $P = i^2R = \frac{\varepsilon^2}{(r+R)^2}R$

(c) $\frac{dP}{dR} = \frac{\varepsilon^2}{(r+R)^2} - \frac{2\varepsilon^2 R}{(r+R)^3} = \frac{\varepsilon^2}{(R+r)^3}[R+r-2R]$

For maximum power supply $\frac{dP}{dR} = 0 \quad \Rightarrow r + R - 2R = 0 \Rightarrow r = R$

Here for maximum power output, outer resistance should be equal to internal resistance.

(d) $P_{max} = \frac{\varepsilon^2}{4r}$




(e) Graph between 'P' and R, where maximum power output at $R = r$, i.e., $P_{max} = \frac{\varepsilon^2}{4r}$

Relation between power output & resistance load is given as,

$\Rightarrow P = \frac{\varepsilon^2 R}{(r+R)^2}$

(f) Power output $P = \frac{\varepsilon^2 R}{(r+R)^2}$

$P(r^2 + 2rR + R^2) = \varepsilon^2 R \quad \Rightarrow \quad R^2 + \left(2r - \frac{\varepsilon^2}{P}\right)R + r^2 = 0$

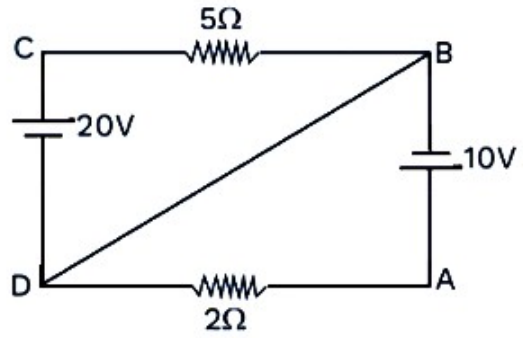
Above quadratic equation in R has two roots $R_1$ and $R_2$ for given values of $\varepsilon$, P and r such that

$\therefore \quad R_1 R_2 = r^2$ (product of roots), $\Rightarrow r^2 = R_1 R_2$

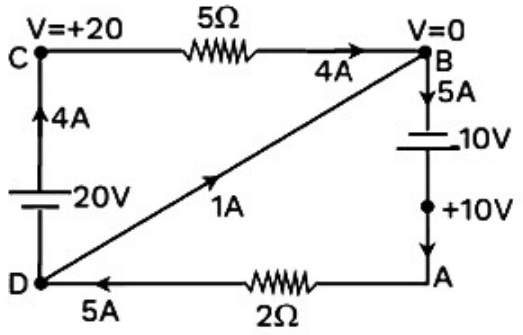
(g) Power of battery spent $= \frac{\varepsilon^2}{(r+r)^2}.2r = \frac{\varepsilon^2}{2r}$. Power (output) $= \left(\frac{\varepsilon}{r+r}\right)^2 \times r = \frac{\varepsilon^2}{4r}$

$$\text{Efficiency} = \frac{\text{power output}}{\text{total power spent by cell}} = \frac{\frac{\varepsilon^2}{4r} \times 100}{\varepsilon^2 / 2r} = \frac{1}{2} \times 100 = 50\%$$

**Q.** **In the figure given below, find out the current in the wire BD.**




**Sol.**




Let at point D, potential = 0 and write the potential of other points, Then current in wire $AD = \frac{10}{2} = 5\,A$ from A to D

Current in wire $CB = \frac{20}{5} = 4\,A$ from C to F

$\therefore$ Current in wire BD = 1 A from D to B

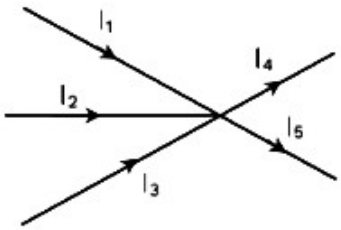
# Kirchhoff's Laws

**Kirchhoff's current law:** This law is based on law of conservation of charge. It state that **"The algebraic sum of the currents meeting at a point of the circuit is zero"** or total currents entering a junction equals total current leaving the junction.

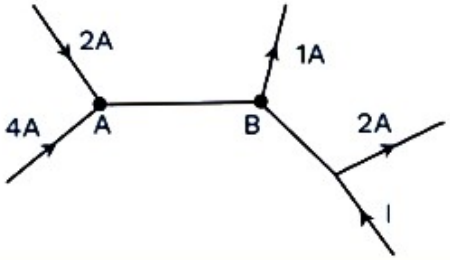
Incoming current = outgoing current

$I_1 + I_2 + I_3 = I_4 + I_5$

$\sum I_{in} = \sum I_{out}$




**Q.** **Calculate the value of current I in the circuit shown in figure.**



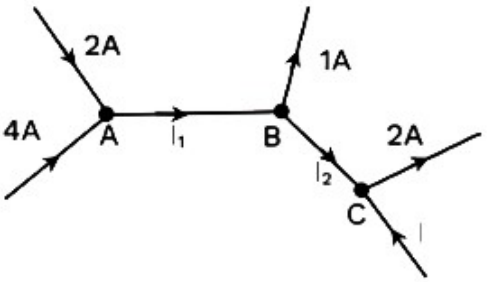

**Sol.** Apply KCL at A : $2 + 4 = I_1 \Rightarrow I_1 = 6A$

Apply KCL at B: $I_1 = 1 + I_2 \Rightarrow 6 = 1 + I_2 \Rightarrow I_2 = 5A$

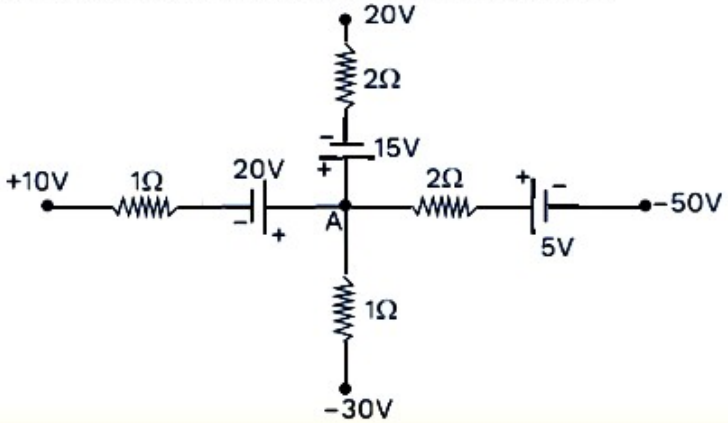
Apply KCL at C: $I_2 + I = 2 \Rightarrow 5A + I = 2$

Answer is $I = -3A$

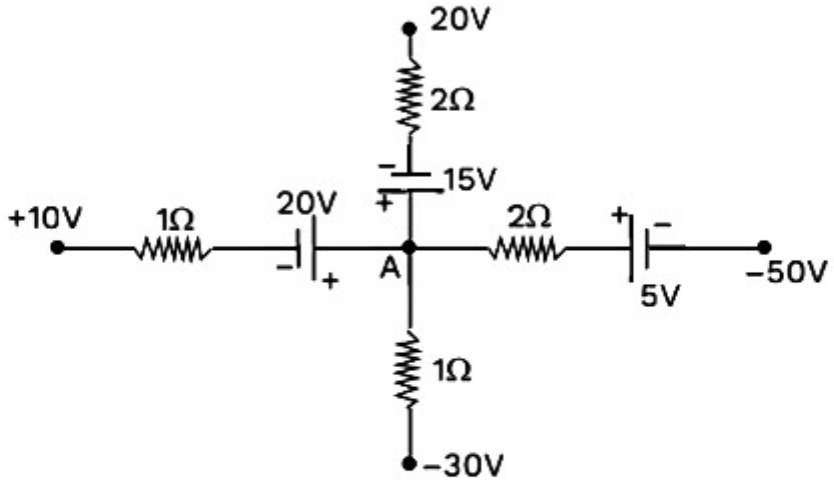
i.e., current I is 3A and is leaving the point C.




**Q.** **Find the potential at point 'A' in the given part of circuit.**



**Sol.**

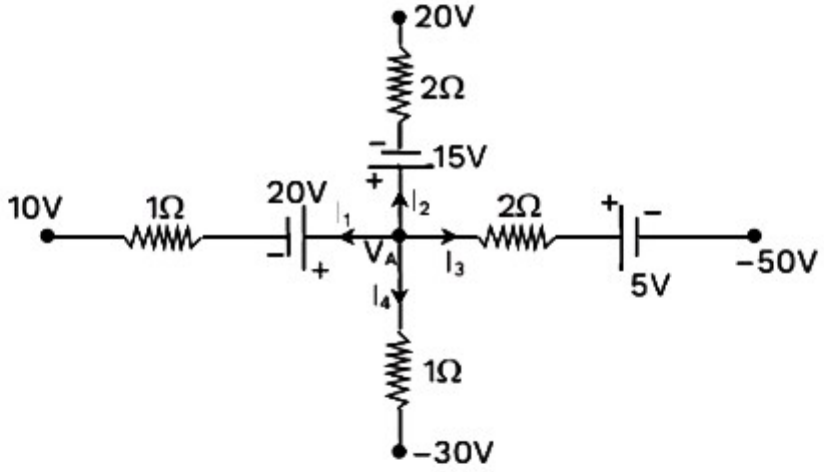



Steps for problem solving :

Step 1 : Assume outgoing current in all the branches.

Step 2: let potential of point A is $V_A$.

Step 3: Apply KCL at junction 'A'.




Apply KCL at junction 'A'

Outgoing current = incoming current

$$I_1 + I_2 + I_3 + I_4 = 0$$

$$\frac{V_A - 20 - 10}{1\Omega} + \frac{V_A - 15 - 20}{2\Omega} + \frac{V_A - 5 + 50}{2\Omega} + \frac{V_A + 30}{1\Omega} = 0$$

$$V_A\left[\frac{1}{1} + \frac{1}{2} + \frac{1}{2} + \frac{1}{1}\right] - \frac{30}{1} - \frac{35}{2} + \frac{45}{2} + 30 = 0$$

$$V_A[3] = \frac{30}{1} + \frac{35}{2} - \frac{45}{2} - \frac{30}{1} = \frac{60 + 35 - 45 - 60}{2} = \frac{-10}{2}$$

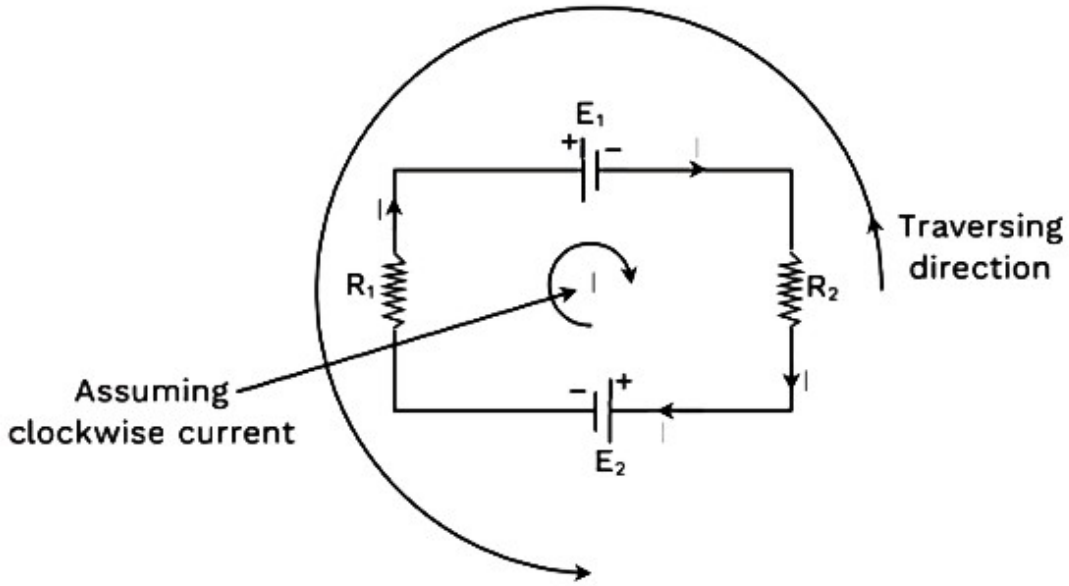
$$V_A[3] = -5 \quad \Rightarrow V_A = -\frac{5}{3}\text{volt}$$

**Kirchhoff's voltage law**

The algebraic sum of all the potential drop across each element along a closed loop is zero.

$$\sum_{\text{closed loop}} \Delta V = 0$$

"It is based on the law of conservation of energy".




$+ IR_2 + E_1 + I R_1 + E_2 = 0$

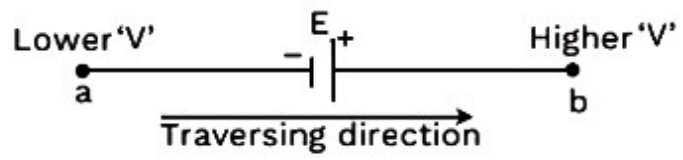
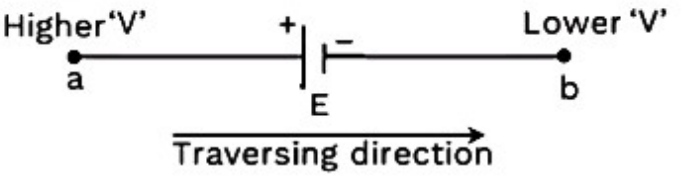
For 'R' and emf we can write as $\sum IR + \sum EMF = 0$

**The Rule for determining ΔV (Potential drop) across a resistor and a battery**

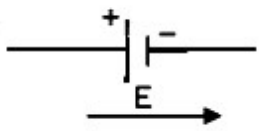
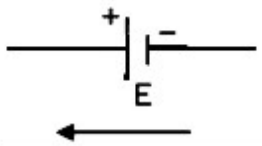
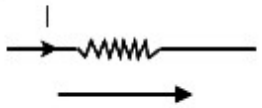
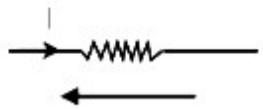
For Resistors :

| | |
|---|---|
| (i) | Higher I R Lower<br>a b<br>Traversing direction<br>$\Delta V = V_b - V_a = -IR$ |
| (ii) | Lower 'V' R I Higher 'V'<br>a b<br>Traversing direction<br>$\Delta V = V_b - V_a = +IR$ |

For Batteries :

| | |
|---|---|
| (i) |  $\Delta V = V_b - V_a = +E$ |
| (ii) |  $\Delta V = V_b - V_a = -E$ |

Summary of sign conventions :

| | |
|---|---|
| (i) |  $-E$ |
| (ii) |  $+E$ |
| (iii) |  $-IR$ |
| (iv) |  $+IR$ |

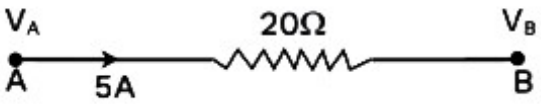
**Steps to solve numerical problems using KVL:**

Step 1: Assign direction of current in the loop/closed circuit (if not given).
Step 2: Choose the traversing direction.
Step 3: Apply KVL in the loop with sign convention given above.

**Q.** **What is the potential difference between A and B.**




**Sol.** Let potential of A is $V_A$ and for 'B' is $V_B$.




Choose traversing direction :

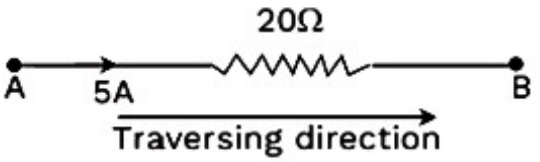
Case 1: Move from left to right

$V_A - I \times R - V_B = 0$

$V_A - 5 \times 20 - V_B = 0$

$V_A - V_B = 20 \times 5$

$V_A - V_B = 100V$



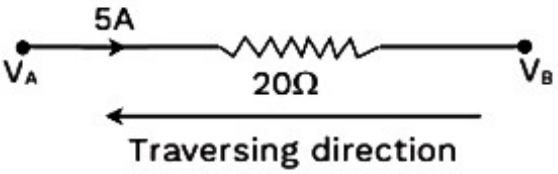

Case 2: Move from right to left :

$V_B + IR - V_A = 0$

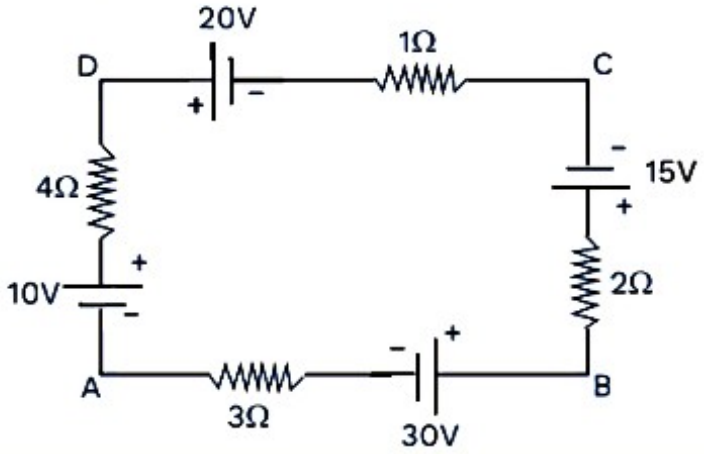
$V_B + 5 \times 20 - V_A = 0$

$V_B - V_A = -100$ Volt

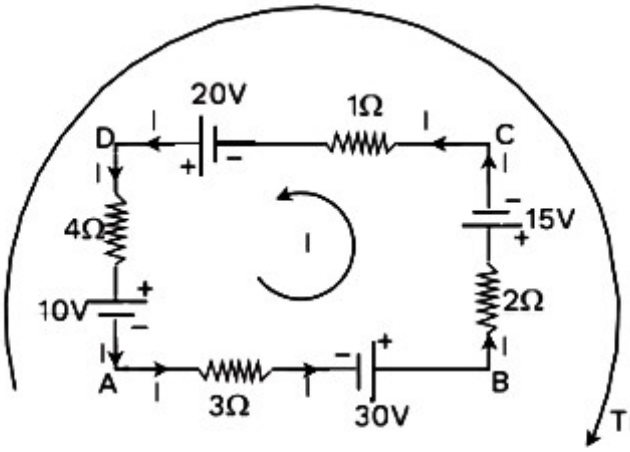
$V_A - V_B = 100$ volt




**Q.** **Find current in the circuit.**



**Sol.**



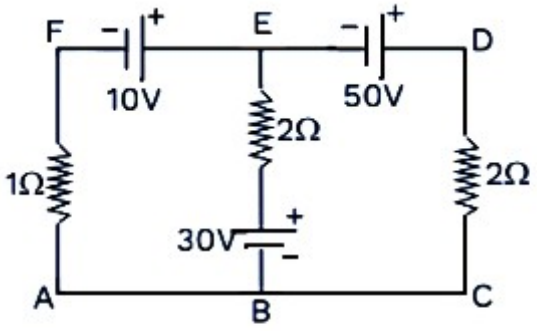

∵ All the element are connected in series. Apply Kirchhoff's voltage law in ABCDA loop

$10 + 4I - 20 + I + 15 + 2I - 30 + 3I = 0$

$10I = 25$

$I = 2.5A$

**Q.** **Using only Kirchhoff voltage law find the current in each branch.**




**Sol.** Step 1: Assign direction of current in the loop (you can choose any direction either clockwise or anti clockwise).

Step 2: Choose direction of traversing:

Let us traverse the circuit in clockwise direction.

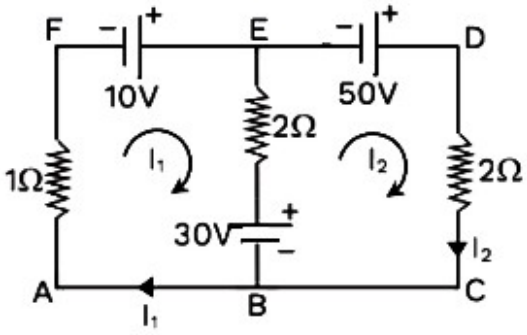
Step 3: Apply K.V.L. in the loops.

Applying Kirchhoff voltage law in the loop A F E B A: (For clockwise traversing)

$-1 \times I_1 + 10 - 2(I_1 - I_2) - 30 = 0$

$-I_1 + 10 - 2I_1 + 2I_2 - 30 = 0$

$2I_2 - 3I_1 = 20$ ...(1)




Applying KVL in the loop B E D C B :
(for clockwise traversing)

$+30 + 2(I_1 - I_2) + 50 - 2I_2 = 0$

$30 + 2I_1 - 2I_2 + 50 - 2I_2 = 0$

$2I_1 - 4I_2 = -80$

$I_1 - 2I_2 = -40$ ...(2)

After solving equation (1) and (2)

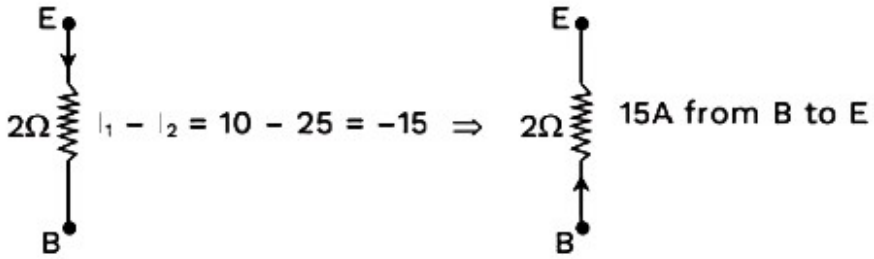
$2I_2 - 3I_1 = 20$

$I_1 - 2I_2 = -40, \quad -2I_1 = -20$

$I_1 = 10A$ ...(3)

From equation (3) and equation (2)

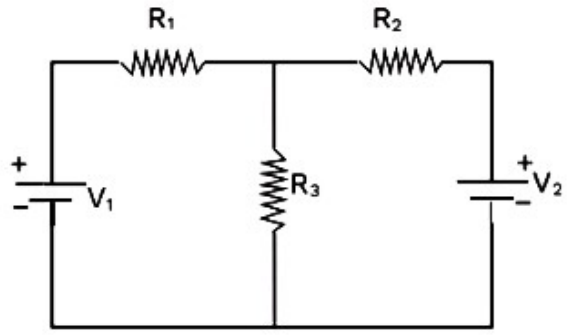
$10 - 2I_2 = -40 \Rightarrow I_2 = 25A$




Current in wire AF = 10 A from A to F.
Current in wire EB = 15A from B to E.
Current in wire DE = 25A from E to D.

# Nodal Analysis

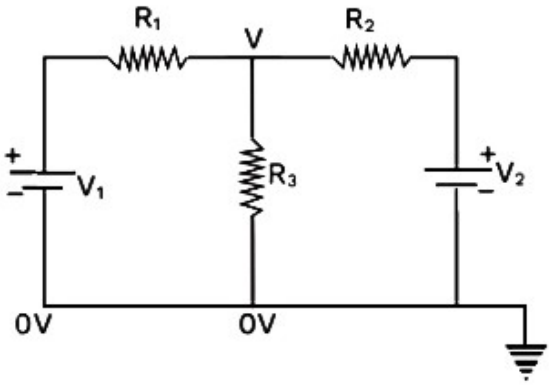
It is also called Node-voltage analysis or branch current method.
It is based on Kirchhoff's current law.



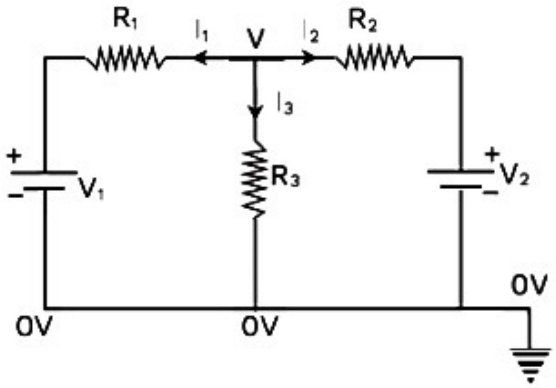

Suppose we want to calculate current in all branches or voltage at nodes.

**Steps for Problem Solving**

Step 1: Assign ground Node and mark voltages at all nodes.




Step 2: Assign current direction at Nodes (always take outward current for easy simplification).



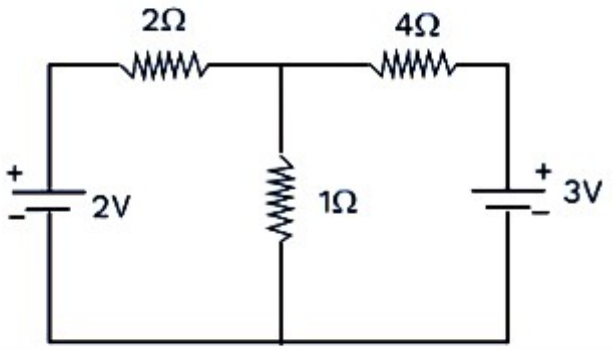
Step 3: Write KCL equation at assigned Node.

$$i_1 + i_2 + i_3 = 0$$

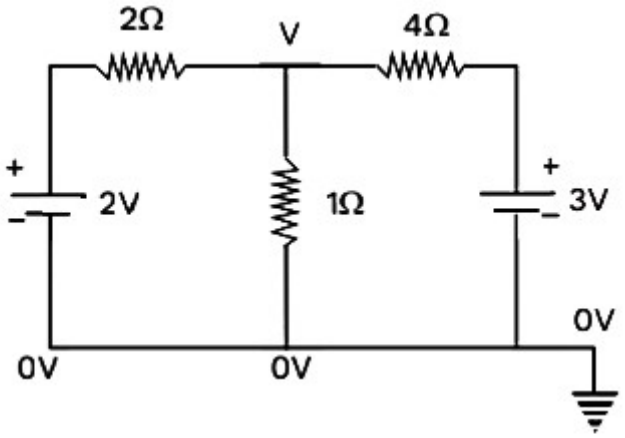
$$\frac{V - V_1}{R_1} + \frac{V - V_2}{R_2} + \frac{V - 0}{R_3} = 0$$

Step 4: Solve the above equation and find unknown voltage.

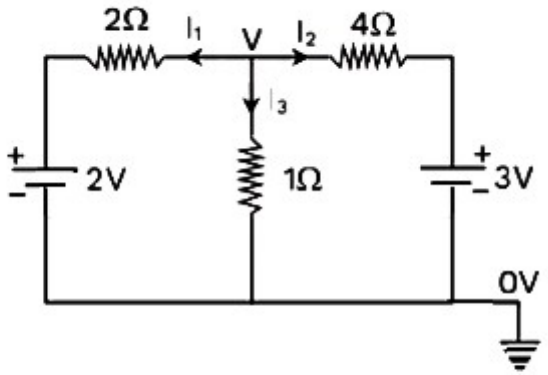
**Q.** **Find current in all branches.**




**Sol.** Step 1:




Step 2:



Step 3:

$$i_1 + i_2 + i_3 = 0$$

$$\frac{V-2}{2} + \frac{V-3}{4} + \frac{V-0}{1} = 0$$

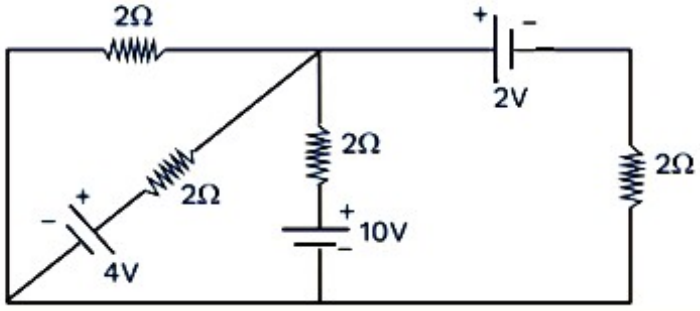
$$V\left\{\frac{1}{2} + \frac{1}{4} + \frac{1}{1}\right\} - \frac{2}{2} - \frac{3}{4} - \frac{0}{1} = 0$$

$$V\left\{\frac{2+1+4}{4}\right\} - 1 - \frac{3}{4} = 0 \quad \Rightarrow \frac{7}{4}V = \left(\frac{7}{4}\right) \quad \Rightarrow V = 1 \text{ volt}$$

$$I_1 = \frac{1-2}{2} = -\frac{1}{2};\ I_2 = \frac{1-3}{4} = -\frac{1}{2};\ I_3 = \frac{1}{1} = 1$$

Answer is $i_1 = \frac{1}{2}A,\ i_2 = \frac{1}{2}A,\ i_3 = 1A$

**Q.** **Find current in all branches.**




**Sol.** Apply KCL at Node A :

$$I_1 + I_2 + I_3 + I_4 = 0$$

$$\frac{V-0}{2} + \frac{V-4}{2} + \frac{V-10}{2} + \frac{V-2}{2} = 0$$

$$V\left\{\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2}\right\} - 0 - 2 - 5 - 1 = 0$$

$2V = 8$

$V = 4$ Volt

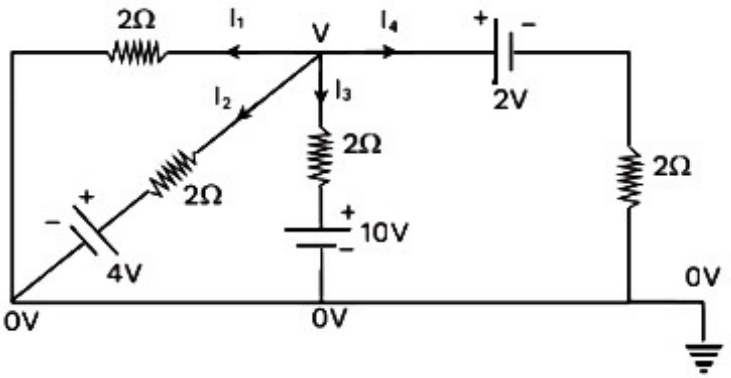
$$I_1 = \frac{4-0}{2} = 2A, \qquad I_3 = \frac{4-10}{2} = -3A$$

$$I_2 = 0,\ I_4 = \frac{4-2}{2} = 1\ A$$

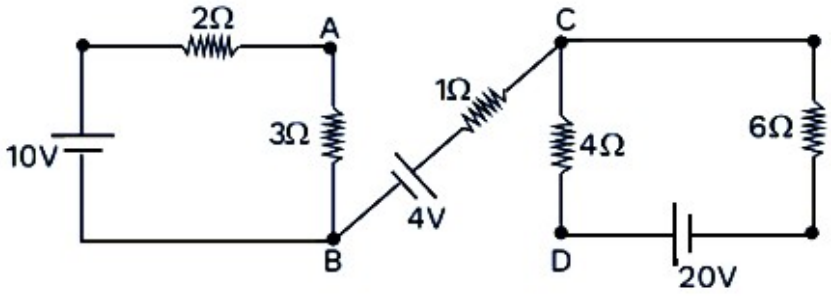
Answer is $I_1 = 2A,\ I_2 = 0,\ I_3 = -3A,\ I_4 = 1A$

Note : Verify the answer

$$\sum 1 = 2 + 0 - 3 + 1 = 0$$

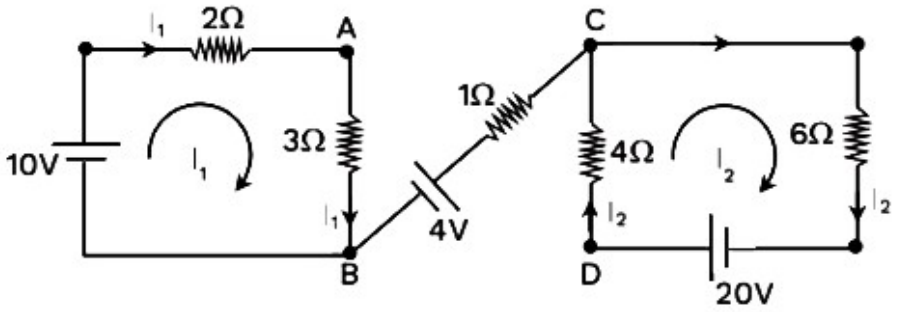


**Q.** **In the circuit shown in figure determine the voltage drop between A and D.**




**Sol.** We need to divide this circuit into three parts. We have left loop right loop, and the central part. To find out voltage drop between points A and D we have to apply Kirchhoff's second law to these circuits one by one.

Also let us assume that current $I_1$ flows in the left circuit in the clockwise direction and current $I_2$ flows in the right circuit in the anticlockwise direction. Direction of both the currents is decided by the battery present in the circuit as current flows from positive to negative.




Applying Kirchhoff's second law to left loop, we get,

$$-2I_1 - 3I_1 + 10 = 0$$

$$\Rightarrow \quad -5I_1 = -10$$

$$\Rightarrow \quad I_1 = 2A$$

Now applying Kirchhoff's second law to right loop, we get

$$-4I_2 - 6I_2 + 20 = 0$$

$$\Rightarrow \quad 10I_2 = 20$$

$$\Rightarrow \quad I_2 = 2A$$

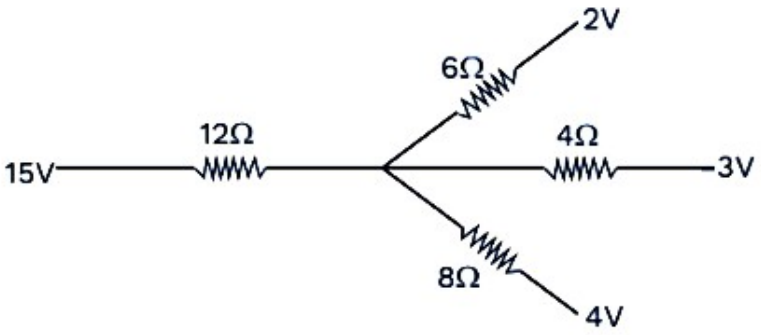
Applying between the points A and D we get

$$V_A - 3I_1 + 4 + 4I_2 - V_D = 0$$

Putting values of $I_1$ and $I_2$ in the above expression, we get

$$V_A - V_D = 3 \times 2 - 4 \times 2 - 4 = 6 - 12 = -6\ V$$

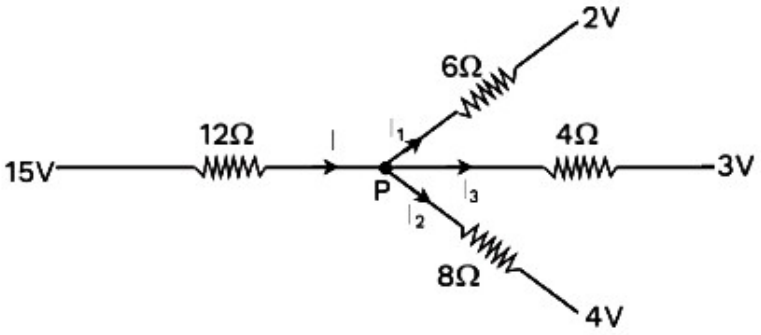
**Q.** **Find the current through 12Ω resistor in figure.**




**Sol.** Let V be the potential at P, then applying KCL at junction P, we get

$$I = I_1 + I_2 + I_3 \Rightarrow \frac{15 - V}{12} = \frac{V - 2}{6} + \frac{V - 3}{4} + \frac{V - 4}{8}$$

$$\text{or} \quad V = \frac{68}{15} V \text{ and } I = \frac{15 - (68/15)}{12} = \frac{157}{180} A$$

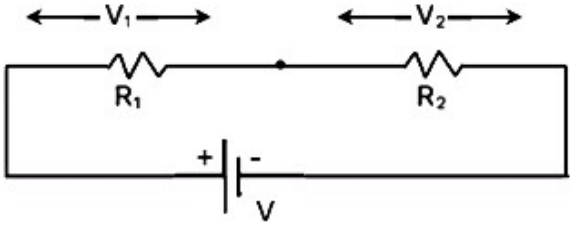


# Voltage & Current Division Rules

**Voltage Division Rule**

Voltage division rule is applicable only in case of series connection.

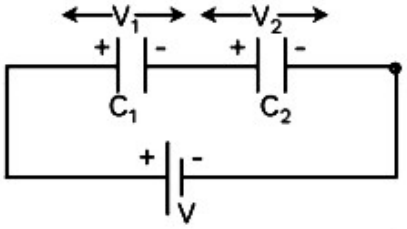
**(a) In case of Resistor :**




Potential drop across resistor $R_1$, $V_1 = \left(\frac{R_1}{R_1 + R_2}\right)(V)$

Potential drop across resistor $R_2$, $V_2 = \frac{R_2}{(R_1 + R_2)}(V)$

**(b) In Case of Capacitor :**

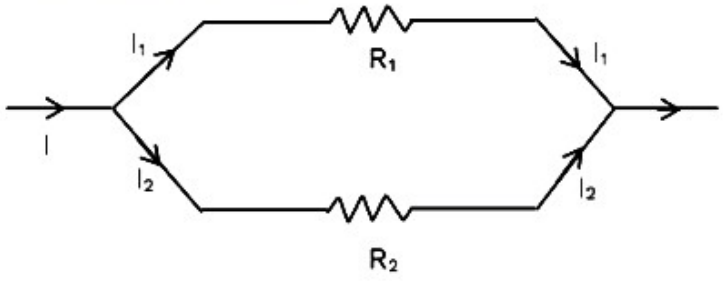



Potential difference across plates of capacitor $C_1$, $V_1 = \left(\frac{C_2}{C_1 + C_2}\right)(V)$

Potential difference across plates of capacitor $C_2$, $V_2 = \left(\frac{C_1}{C_1 + C_2}\right)(V)$

**Current Division Rules :**

**Case-1 :** Two resistors connected in Parallel



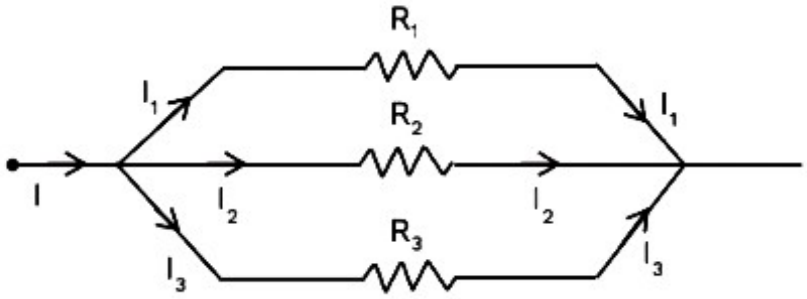

Use : $I = I_1 + I_2$

$I_1 R_1 = I_1 R_2$ = Voltages Across each resistor are same.

On solving equation (1) and (2) we get

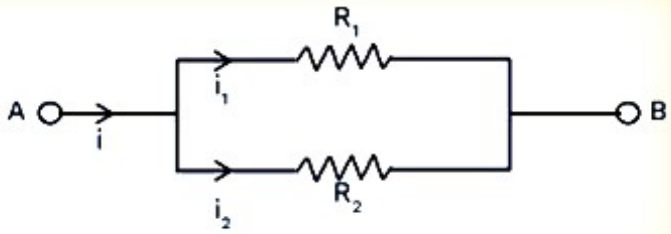
$I_1 = \frac{R_2}{(R_1 + R_2)}(I)$, $I_2 = \frac{R_1}{R_1 + R_2}(I)$

**Case-2 :**




$$I_1 = \left[\frac{\left(\frac{1}{R_1}\right)}{\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}}\right](I) \quad ; \quad I_2 = \left[\frac{\left(\frac{1}{R_2}\right)}{\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}}\right][I] \quad ; \quad I_3 = \left[\frac{\left(\frac{1}{R_3}\right)}{\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}}\right](I)$$

**Q.** **For a given figure determine $i_1$ and $i_2$ in terms of i.**

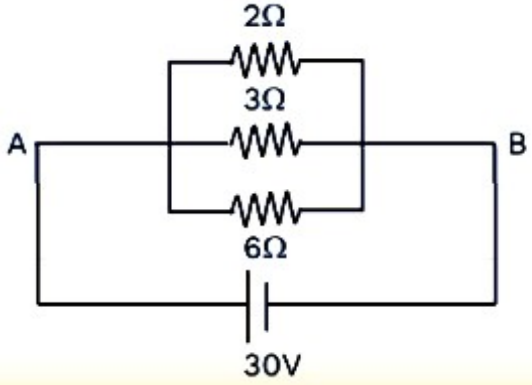



**Sol.** $\therefore \quad i_1R_2 = i_2R_2 \quad \text{or} \quad \frac{i_1}{i_2} = \frac{R_2}{R_1}$

$\Rightarrow \quad i_1 = \frac{R_2 i}{(R_1 + R_2)}$

Note : Remember this law of $i \propto \frac{1}{R}$ in the resistors connected in parallel.

**Q.** **Find out current passing through the battery and each resistor.**



**Sol.** Method (1)

Since all resistors connected in parallel so potential difference across each resistor is 30 V.

$\therefore$ Current is each resistors are $\frac{30}{2}=15A, \frac{30}{3}=10A$ and $\frac{30}{6}=5A$

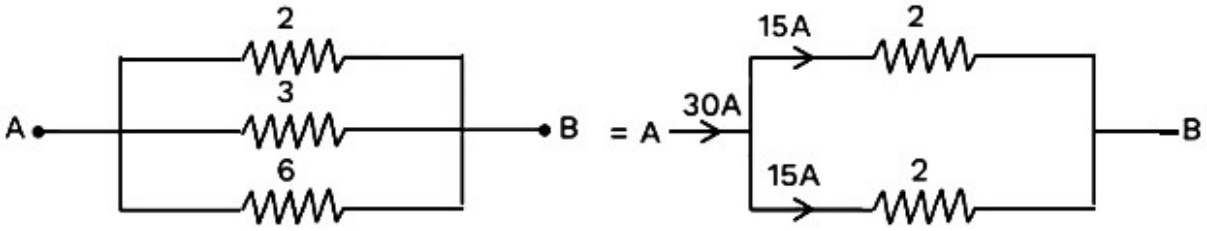
$\therefore$ Current through battery is equal to = 15 + 10 + 5 = 30 A.

Method (2)

By ohm's law $i=\frac{V}{R_{eq}}$

$$\frac{1}{R_{eq}}=\frac{1}{2}+\frac{1}{3}+\frac{1}{6}=1\Omega, \quad R_{eq}=1\Omega \quad \Rightarrow \quad i=\frac{30}{1}=30A$$

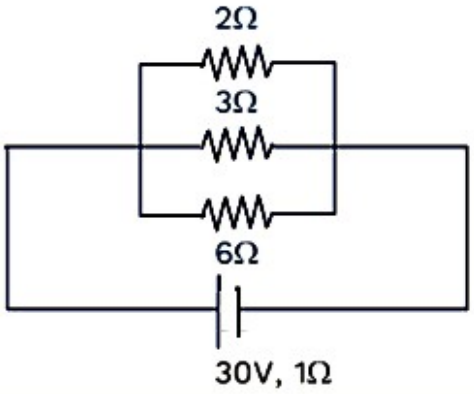
Now distribute this current in the resistors in their inverse ratio.




Current total in 3Ω and 6Ω is 15 A it will be divided as 10 A and 5A.

Note : The method (I) is better. But you will not find such an easy case everywhere.

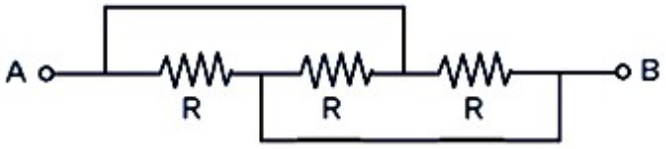
**Q.** **Find the magnitude of current which is passing through battery.**



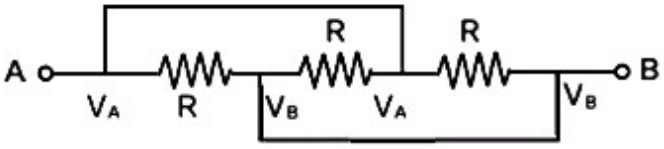
**Sol.** Here potential difference across each resistor is not 30 V
∵ battery has internal resistance here the concept of combination of resistors is useful.

$$\text{Req} = 1 + 1 = 2\ \Omega \quad \Rightarrow \quad i = \frac{30}{2} = 15\ \text{A}.$$

**Q.** **Find out the equivalent Resistance.**

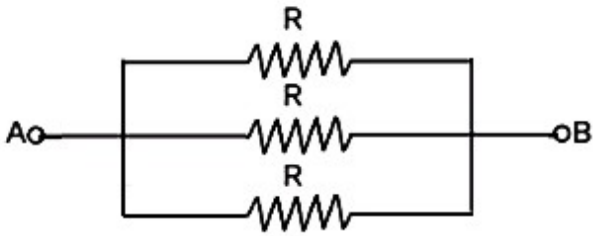



**Sol.**

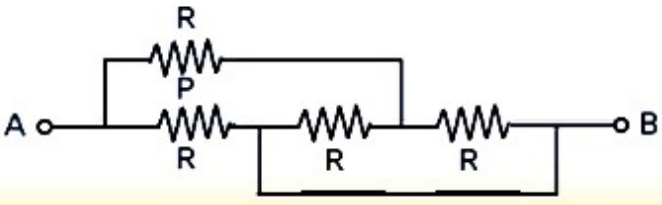



Here all the resistances are connected between the terminals A and B Modified circuit is

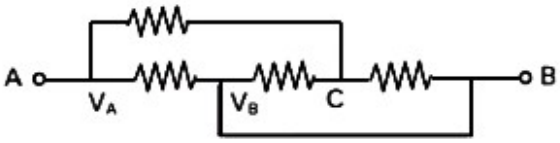
So $R_{eq} = \frac{R}{3}$



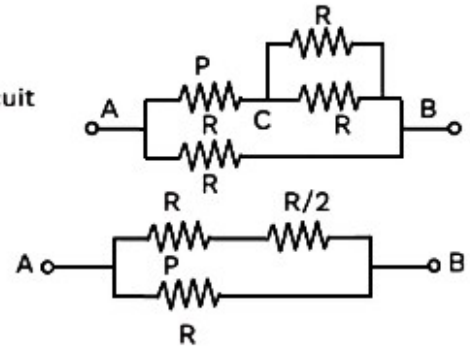

**Q.** **Find the current flow through the Resistance P if voltage supply between A and B is V volts.**



**Sol.**



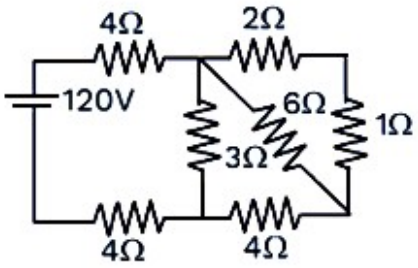

Modified circuit




$$R_{eq} = \frac{3R}{5} \qquad \Rightarrow \quad I = \frac{5V}{3R}$$

$$\text{Current in P} = \frac{R \times \frac{5V}{3R}}{1.5R + R} = \frac{2V}{3R}$$

**Q.** **Find the current in 2Ω resistance.**




**Sol.** 2Ω, 1Ω in series = 3Ω

3Ω, 6Ω in parallel = $\frac{18}{9}$ = 2Ω

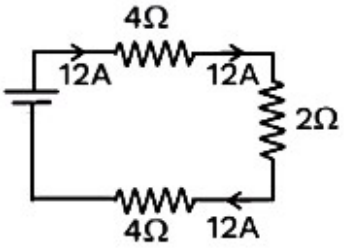
2Ω, 4Ω in series = 6Ω

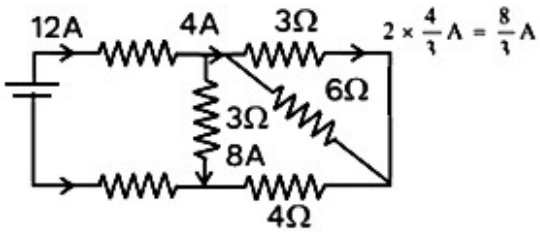
6Ω, 3Ω in parallel = 2Ω

$R_{eq}$ = 4 + 4 + 2 = 10Ω

$$i = \frac{120}{10} = 12A$$

So current in 2Ω Resistance = $\frac{8}{3}$A



**Calculation of Effective Resistance**

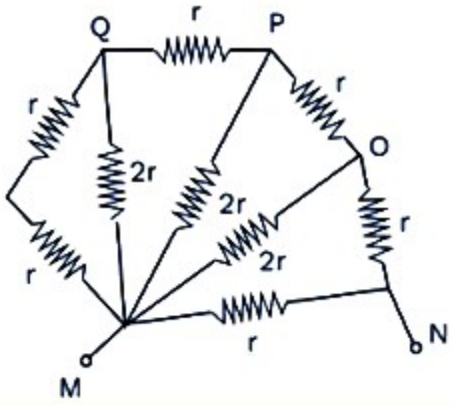
In some cases, resistors are neither in series nor in parallel. Such type of circuits can be solved by following methods.

1. Successive Reduction Method.
2. Balanced Wheat Stone Bridge.
3. Unbalanced Wheat Stone Bridge.
4. Symmetrical Circuit.
5. Ladder Problems.
6. Cube Problems.

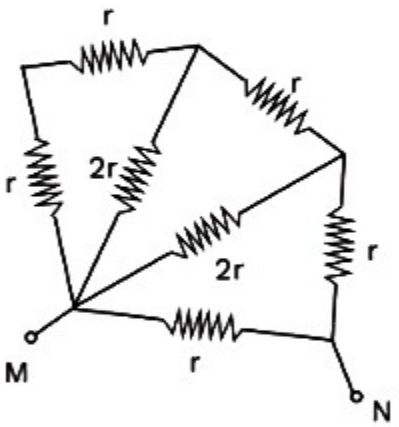
**Successive Reduction Method**

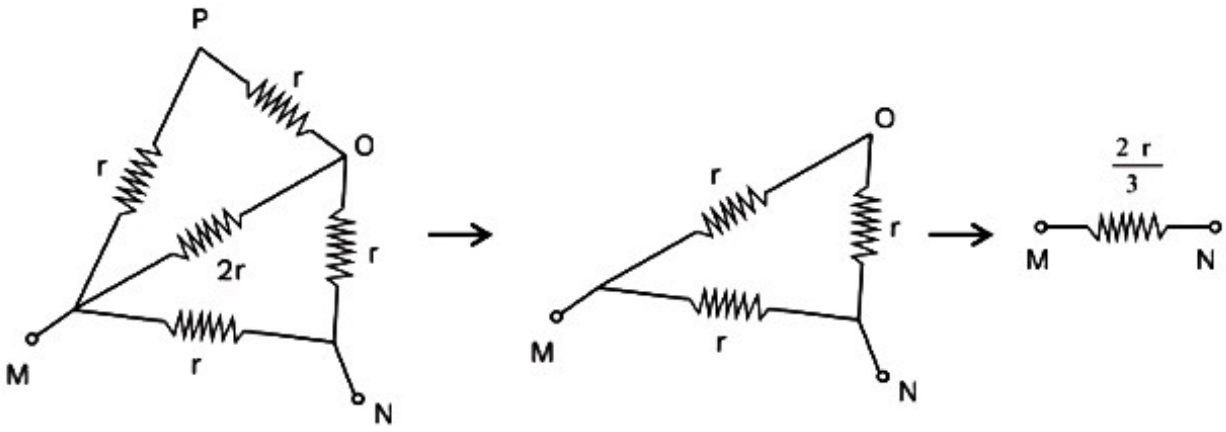
This method is applicable only when we are able to identity resistance in series or in parallel. The method is based on the simplification of the circuit by successive reduction of the series and parallel combination.

**Q.** **Calculate the equivalent resistance between points method M and N for the network shown in fig.**

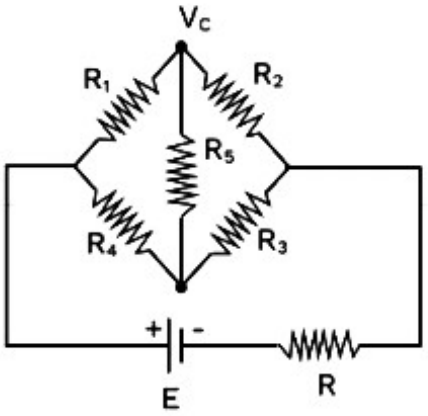



**Sol.**






**Balanced Wheat Stone Bridge**

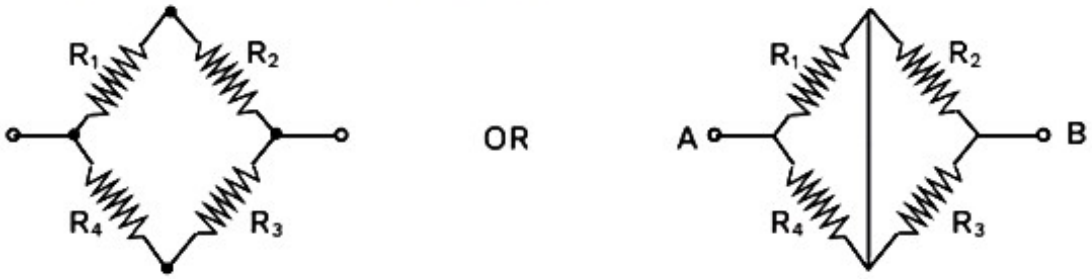



The arrangement as shown in figure, is known as wheat stone bridge. Here there are four terminals in which except two all are connected to each other through resistive elements. In this circuit if:

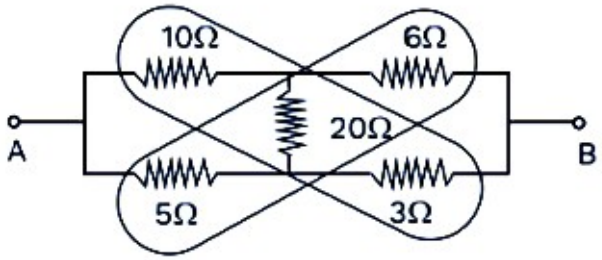
$\boxed{R_1 R_3 = R_2 R_4}$ then $V_C = V_D$ and current in $R_5 = 0$, this is called balance point or null point.

Here in this case products of opposite arms are equal. Potential difference between C and D at null point is zero. The null point is not affected by resistance $R_5$, E and R. It is not affected even if the positions of $R_5$ and E are interchanged.

Hence, here the circuit can be assumed to be :



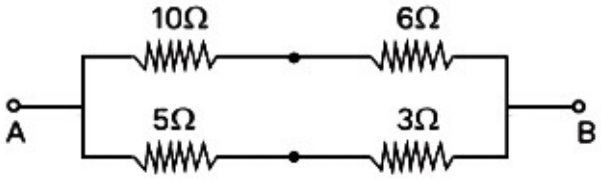
**Q.** **Find equivalent resistance of the circuit between the terminal A and B.**




**Sol.** Since the given circuit in wheat stone bridge and it is in balance condition.

$\therefore \quad 10 \times 3 = 6 \times 5$

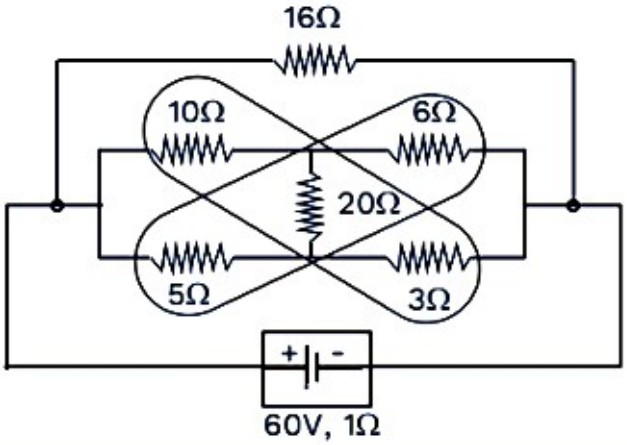
Hence this is equivalent to, $Req = \frac{16 \times 8}{16 + 8} = \frac{16}{3}\Omega$



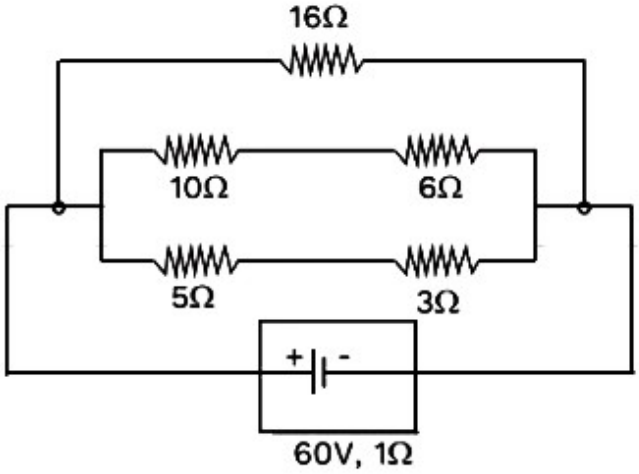

**Q.** **Find :**

**(a) Equivalent resistance.**

**(b) Current in each resistance in the given circuit.**



**Sol.** Redraw the circuit.



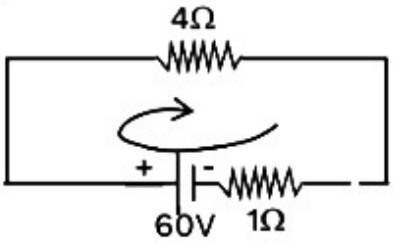

Since the given circuit is wheat stone bridge and which is in balance condition.

$\therefore \quad 10 \times 3 = 5 \times 6 = 30$

So, no current will flow through 20Ω resistance so we can remove 20Ω resistance.

$$\frac{1}{Req} = \frac{1}{16} + \frac{1}{16} + \frac{1}{8} = \frac{1+1+2}{16} = \frac{4}{16}$$

Current flow through battery $\Rightarrow I = \frac{60}{4+1} = \frac{60}{5} = 12A$



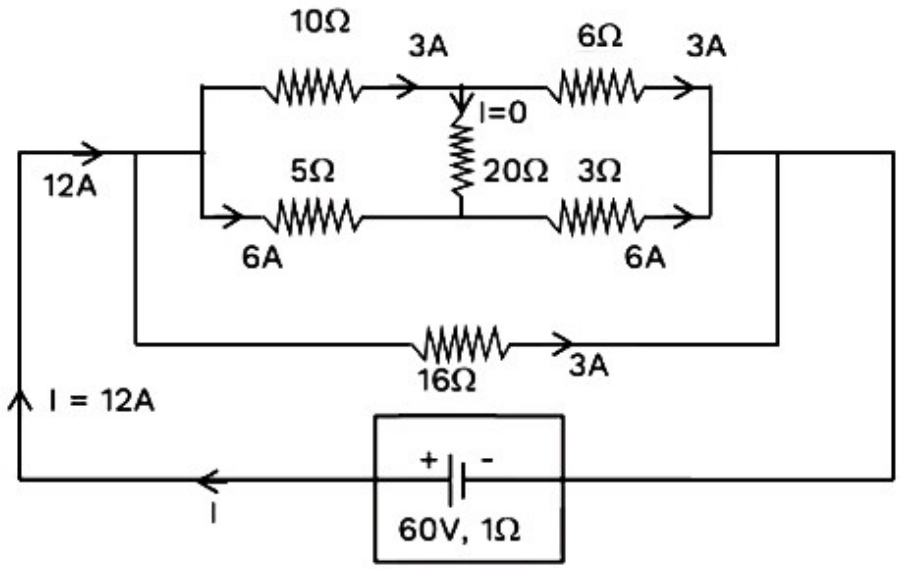

By using current distribution law

Current in resistance 10Ω and 6Ω = 3A

Current in resistance 5Ω and 3Ω = 6A

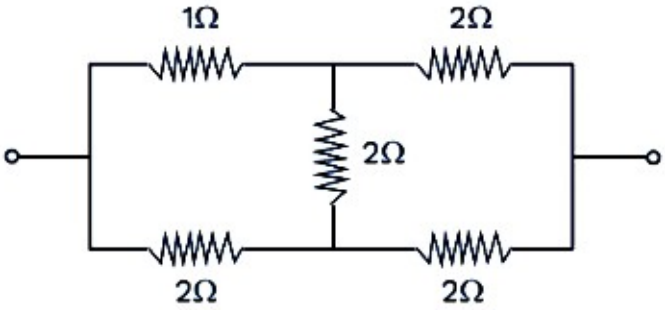
Current in resistance 20Ω = 0

Current in resistance 16Ω = 3A

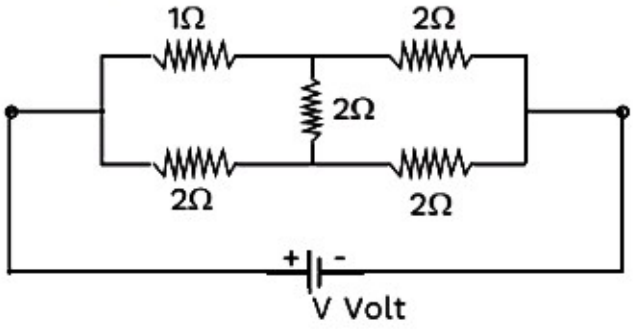


**Unbalanced Wheat-Stone-Bridge :**
In some circuits, resistances are neither connected in series nor in parallel and also wheat stone bridge is not balanced. We use V-I method to solve such type of circuit.

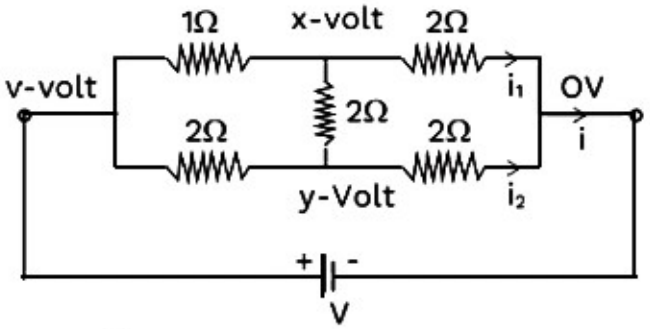
**Q.** **Find equivalent resistance.**




**Sol.** Step 1 : Connect a battery across the terminal.




Step 2: Mark the voltages of nodes.




Step 3: Calculate. $R_{eq} = \frac{V}{I}$

Apply KCL at node 'x'

$$\Rightarrow \quad \frac{x-V}{1}+\frac{x-0}{2}+\frac{x-y}{2}=0$$

$\Rightarrow \quad 2x - 2V + x + x-y = 0$

$\Rightarrow \quad 4x - 2V = y \qquad ...(1)$

Apply KCL at node 'y'

$$\frac{y-x}{2}+\frac{y-V}{2}+\frac{y}{2}=0$$

$\Rightarrow \quad 3y - v = x \qquad ...(2)$

Solve Eq. (1) and Eq. (2)

$$x = \frac{7V}{11}, \quad y = \frac{6v}{11}$$

Now calculate :

$$I_1 = \frac{x-0}{2} = \frac{7V}{11\times 2} = \frac{7V}{22} \quad \Rightarrow \quad I_2 = \frac{y}{2} = \frac{6V}{11\times 2} = \frac{6V}{22}$$

So, equivalent resistance

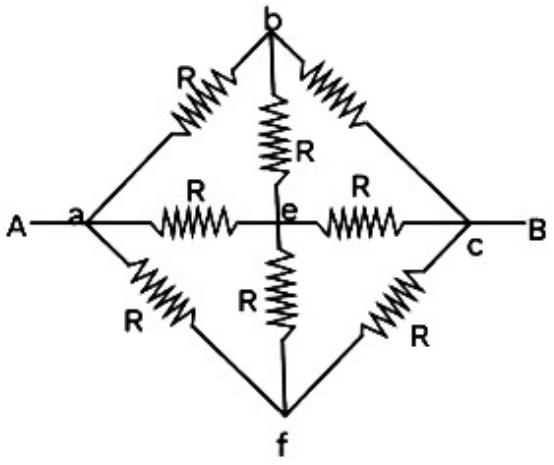
$$Req = \frac{V}{i_1 + i_2} = \frac{V}{\frac{7V}{22}+\frac{6V}{22}} = \frac{V}{\frac{13V}{22}} \Rightarrow \boxed{Req = \frac{22}{13}\Omega}$$

**Symmetrical Circuits**

There are two type of Symmetry : (1) Mirror symmetry, (2) Folding Symmetry

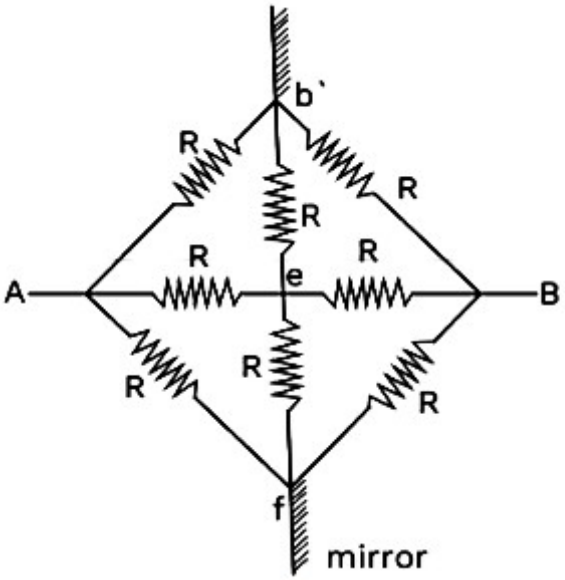
**Mirror Symmetry**

When a circuit is symmetrical about a line (by symmetry, we mean that two parts are mirror image of each other), then the potential and current must also be symmetrical.

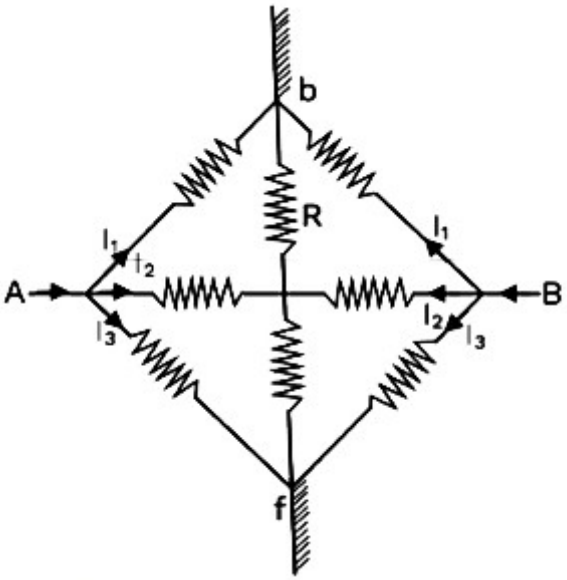



Therefore, current in ab and af are same. Currents in fc and bc are same, potential of the points b, e and f are same

**There are following steps to Solve Symmetrical Circuit**

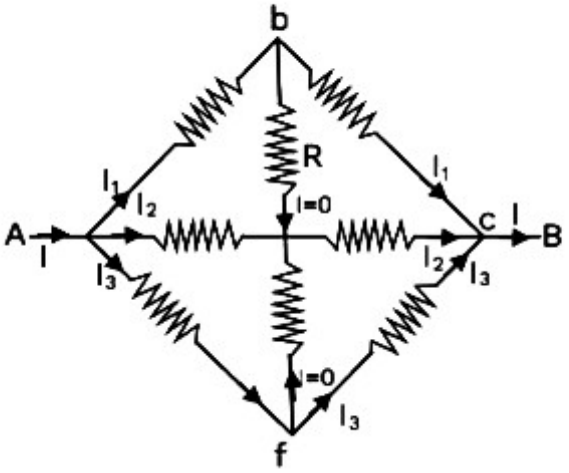



Step–1 : Draw a mirror




Step–2 : Mark the Current

Step-3 : Reverse the direction of Current

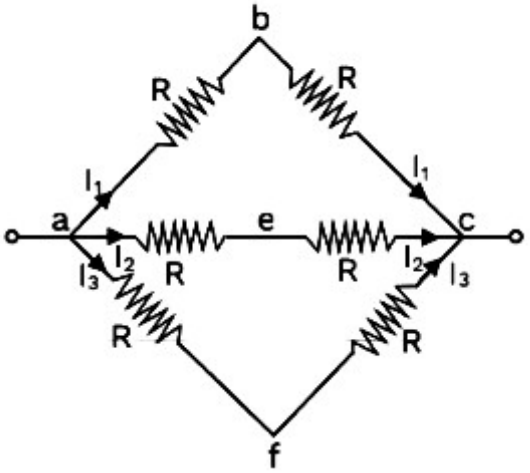



**Important points :**

Due to symmetry we observe that current $I_1$ is not divide at junction b, $I_2$ is not divide at junction e and $I_3$ is not divide at junction 'f'

So, No current flow through the Resistor connected between be and ef.

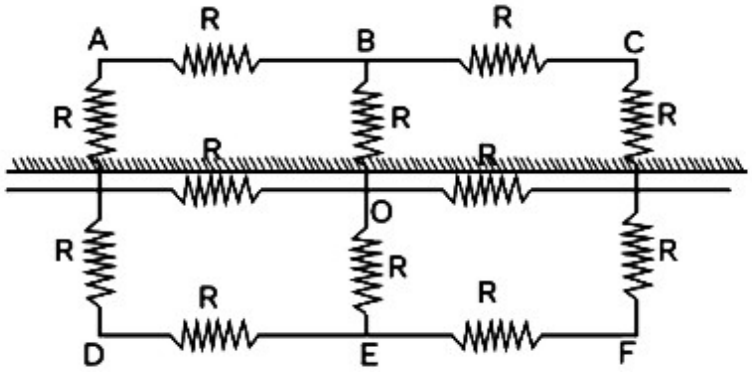
⇒ Now Re-draw the circuit




Now equivalent Resistance : $\frac{1}{R_{ab}} = \frac{1}{2R} + \frac{1}{2R} + \frac{1}{2R} = \frac{2R}{3}$

is Answer.

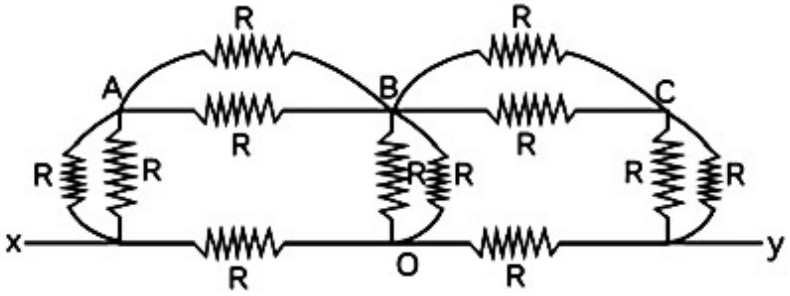
**Folding Symmetry**



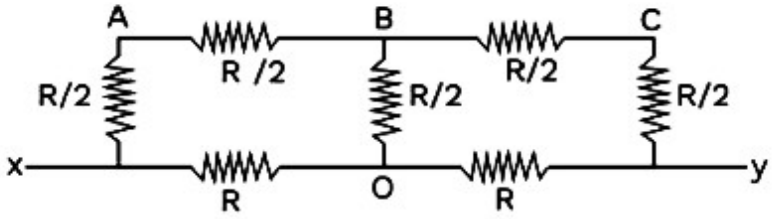

⇒ In this case the circuit has symmetry

⇒ With the help of symmetry we can write, $V_A = V_D$ ; $V_B = V_E$ ; $V_C = V_F$ and we know that same potential points can be connected.

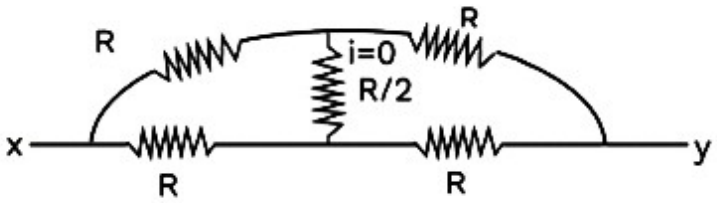
⇒



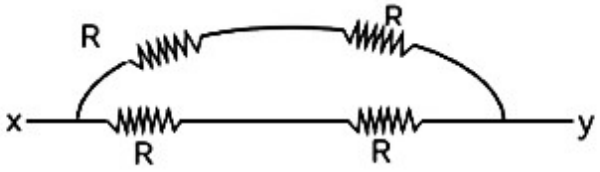

⇒




⇒

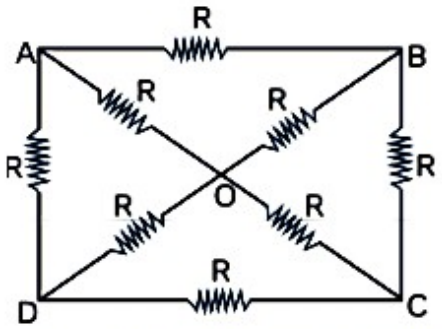



⇒



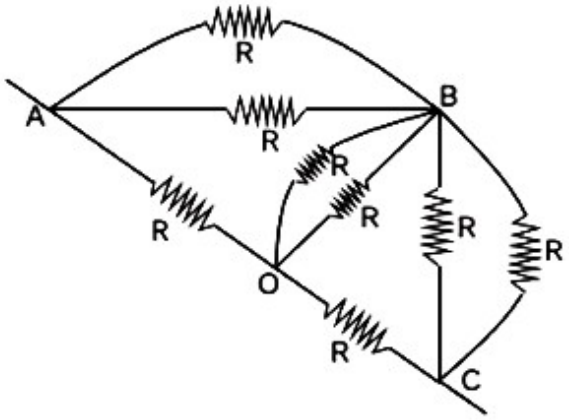

$R_{eq}$ = R. Answer is R

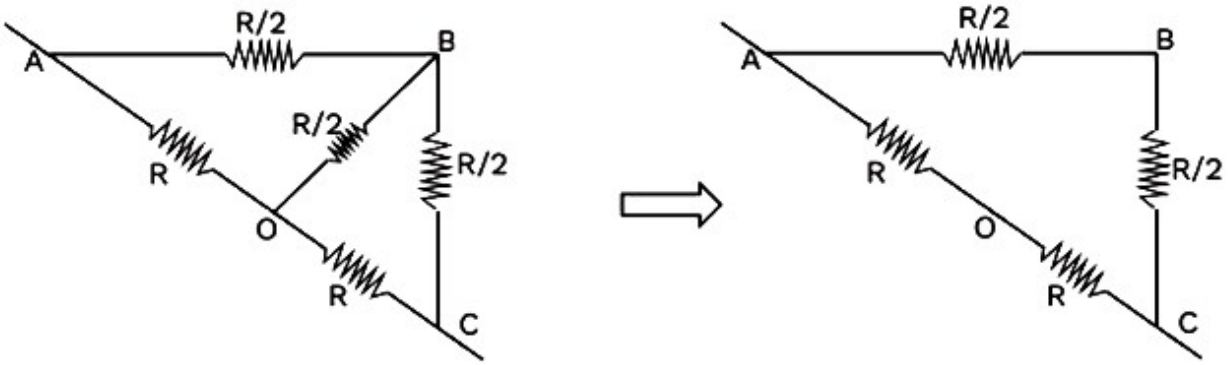
**Q.** **What is effective resistance between points A and C for the networks shown.**




With the help of symmetry about line AC we can write, $V_D = V_B = V_o$

**Sol.** Now fold the circuit about line AC.



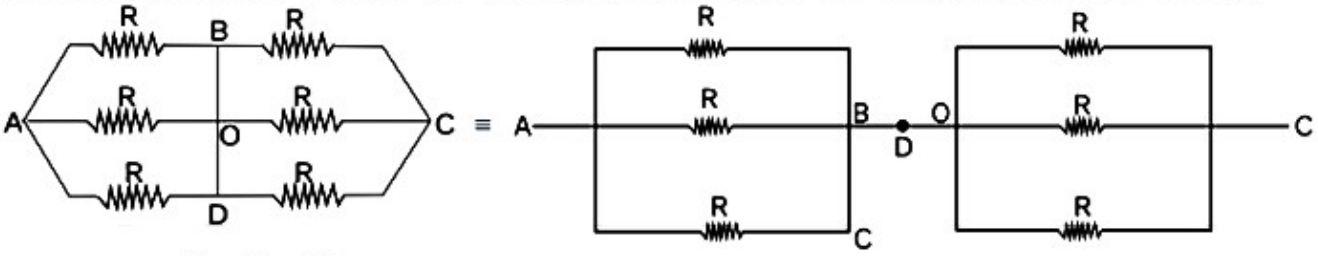




Thus, between points A and C

$\Rightarrow R_{eq} = \frac{2R}{3}$. Answer is 2R/3.

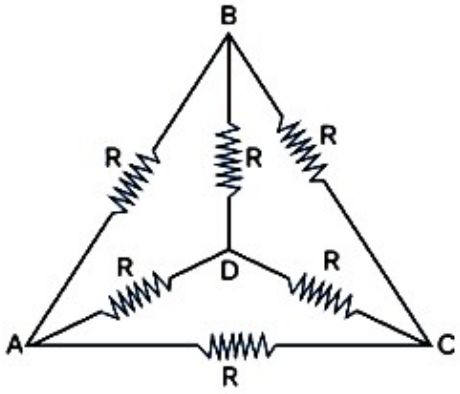
**Alternatively,**

Since the points B, O and D are at same potential, we can treat them shorte circuit.

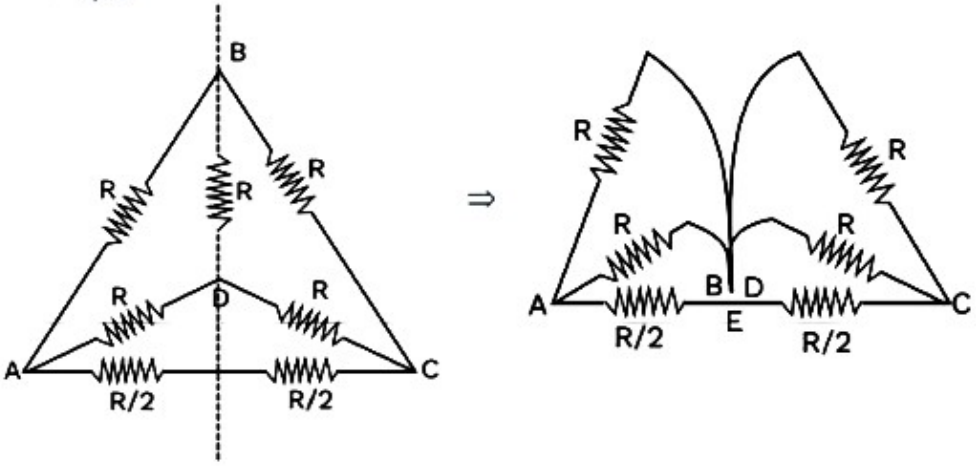



$\Rightarrow R_{eq} = \frac{R}{3} + \frac{R}{3} = \frac{2R}{3}$ is Answer.

**Q.** **What is the equivalent resistance between points A and C, by using symmetry concept.**




**Sol.** Break the branch AC int two resistors in series, each R/2, and consider the dotted line passing through B, D and E. The network on the two sides of this line is symmetrical. Hence, one can short- circuit the points B, D and E and calculate $(R_{eq})_{AE}$



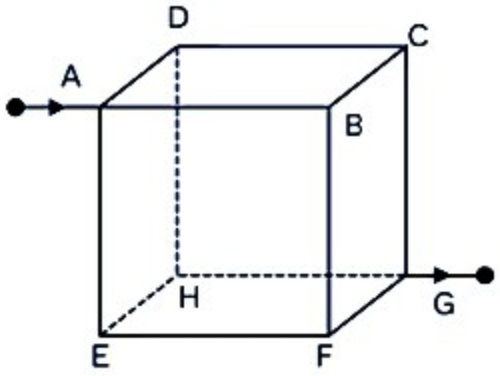
Now, we find that resistances R, R and R/2 are in parallel across A and E. The parallel combination of R and R gives R/2. This R/2 in parallel with R/2 gives R/4. Hence,

$$(R_{eq})_{AE} = \frac{R}{4} \quad \Rightarrow \quad (R_{eq})_{AC} = 2(R_{eq})_{AE} = 2\left(\frac{R}{4}\right) = \frac{R}{2}.$$

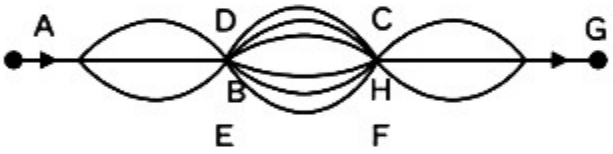
**Cube Problems:**

In some of the networks, we may not find symmetry on the two sides of a line. But, we may find that the network contains some set of points having equal potential. These points can be joined together to make the network simple and easy simplification.

**Q.** **Twelve equation wires of resistances R each are joined up to form the edges of a cube, as shown in figure. The cube is connected into a circuit across the diagonal AG. Find the equivalent resistance of the network.**




**Sol.** Let us search the points of identical potential. since the three edges of the cube from AB, viz., AB, AD and AE are identical , the points B, D and E are the equal potential . Same way , for the point G, the sides GC, GH and GF are symmetrical in all aspect and the points C, H and F are at the equal potential difference.




Next, to simplify the circuit, we bring together the points, B, D and E and also C, H & F.

Now the resistance between A and D = R/3
And the resistance between points D and C = R/6
Also the resistance between C and G is equal to = R/3
Thus, the resistance between C and G = R/3
Thus, the circuit is equivalent to three resistance of value. R/3, R/6 and R/3, in series, and hence the total resistance between A and G is

$$R_{AG} = \frac{R}{3} + \frac{R}{6} + \frac{R}{3} = \frac{5R}{6}$$

**Summary :**

Equivalent Resistance Between :

| Edge ($R_{AB}$) | Face diagonal ($R_{AF}$) | Body diagonal($R_{AG}$) |
|---|---|---|
| $\frac{7R}{12}$ | $\frac{3R}{4}$ | $\frac{5R}{6}$ |

**Infinite Ladder Problems**

To solve ladder problems we use the following steps.

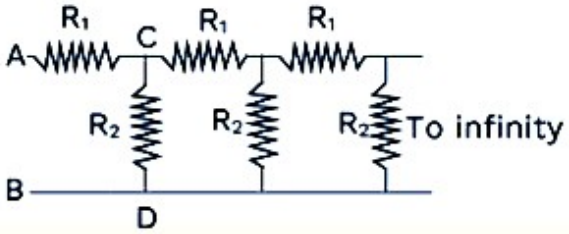
**Step-1:** Let us assume the total resistance of the infinite network to be X (say).

**Step-2:** Now just retain one basic repetitive unit and we observe the remaining circuit to be the same as the original circuit. So, resistance of this left out circuit must be X.

**Step-3:** Now the equivalent circuit, is the combination of basic unit and original repetitive circuit of resistance X, such that the net resistance of the entire circuit is X.

The following illustrations are done on the basis of these three steps.

Q. **What is equivalent resistance between points A and B of an infinite chain of resistors joined as shown in figure.**

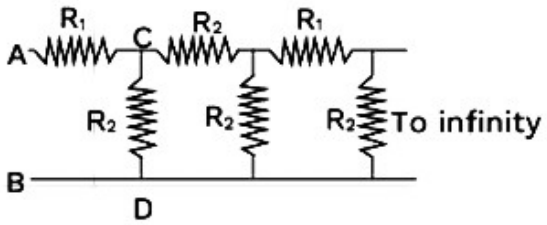


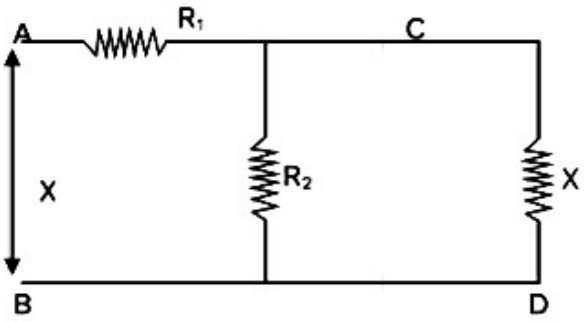
**Sol.** Suppose the equivalent resistance between A and B is X. Applying the steps discussed, we get

$$\Rightarrow X = R_1 + [R_2 \parallel X] = R_1 + \frac{R_2 X}{(R_2 + X)}$$

$$\Rightarrow X^2 - R_1 X - R_1 R_2 = 0$$

$$\Rightarrow X = \frac{1}{2}\left(R_1 \pm \sqrt{(R_1^2 + 4R_1 R_2)}\right)$$



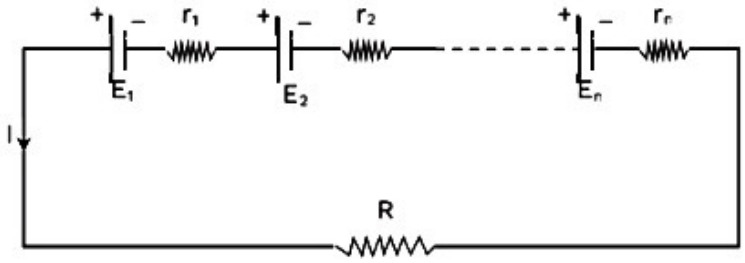

And as resistance cannot be negative, we have, $X = \frac{R_1}{2}\left(1 + \sqrt{1 + \frac{4R_2}{R_1}}\right)$

However, if $R_1 = R_2 = R$ we get, $X = \frac{R}{2}(1 + \sqrt{5}) = 1.6\ R$

# Combination of Cells

**Case 1: Series Combination**

When we connect cells are in series the total emf of the series combination is equal to the sum of the emf's of the individual cells and internal resistance of the cells also come in series.




Equivalent emf(E) = $E_1 + E_2 + E_3 \ldots E_n$

Equivalent internal resistance ($r_{eq}$) = $r_1 + r_2 + r_3 \ldots r_n$

Current, $I = \frac{E_{net}}{r_{net} + R}$ ; if all n cell are identical then $I = \frac{nE}{nr + R}$

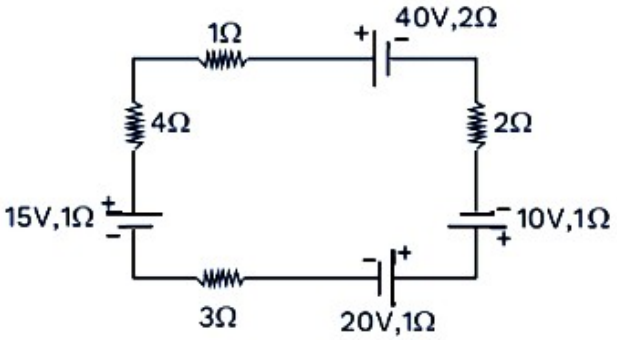
If nr >> R, $I = \frac{E}{r} \cong$ current from one cell.

If nr << R, $I = \frac{nE}{R} \approx n \times$ current from one cell.

*Note :* If polarity of m cells is reversed, then equivalent emf = (n – 2m)E while the equivalent resistance is still (nr + R), so current in R will be

$$I = \frac{(n-2m)E}{nr + R}$$

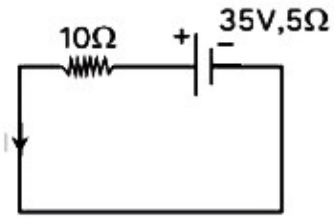
**Q.** **Find the current flow in the given circuit.**



**Sol.** The given circuit can be simplified as

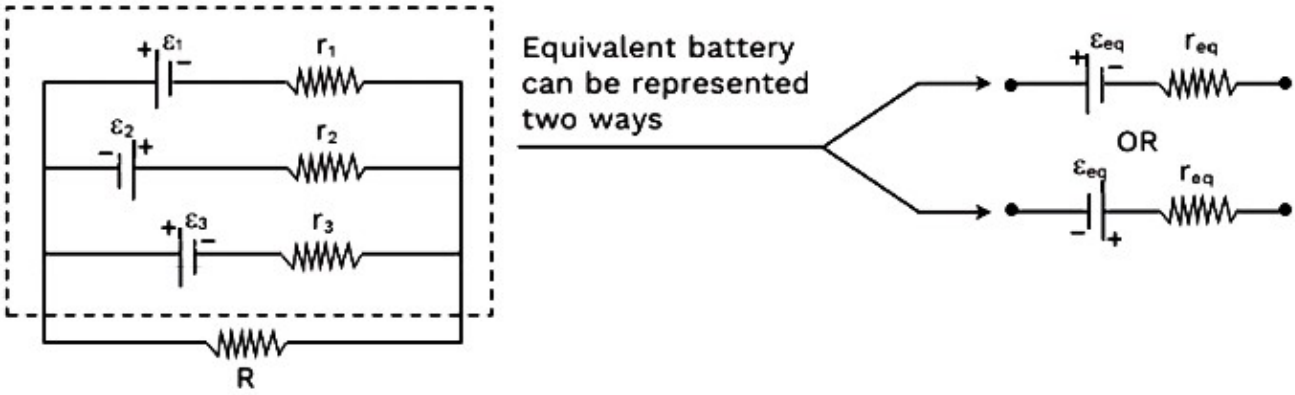
$$I = \frac{35}{10+5} = \frac{35}{15} = \frac{7}{3}A$$

$$I = \frac{7}{3}A$$




**Case 2 : Parallel Combination**

Let we have cells connected as follows:




Equivalent battery can be written in two ways:

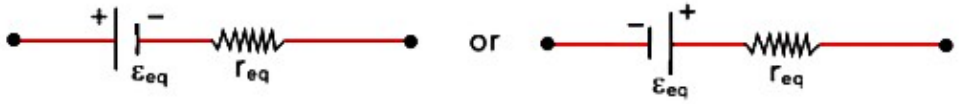



Figure-1 Figure-2

**Case – 1:** If we consider the polarity this (fig.1) way then value of $\varepsilon_{eq}$ can be written as

$$\varepsilon_{eq} = \frac{\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_3}{r_3} - \frac{\varepsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}} \qquad \Rightarrow \frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$



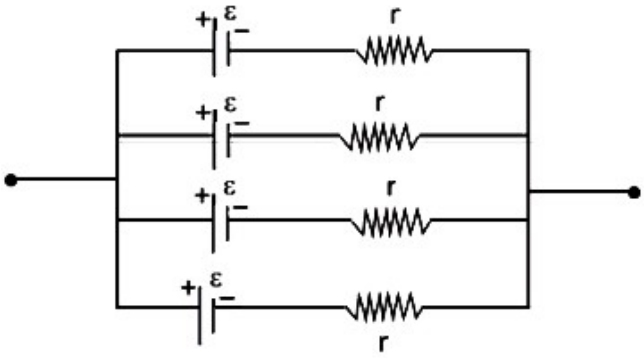

Figure-1

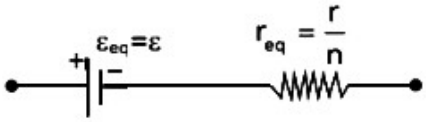
**Case -2:** If we consider the polarity of battery:

$$\text{Then, } \varepsilon_{eq} = \frac{\left[\frac{\varepsilon_2}{r_2} - \frac{\varepsilon_1}{r_1} - \frac{\varepsilon_3}{r_3}\right]}{\left[\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}\right]} \qquad \Rightarrow \frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$



**Special Case :** If m cells each of emf is equal to E and internal resistance r ohm be connected in parallel and if this combination is connected to an external resistance then equivalent emf of circuit is ε.



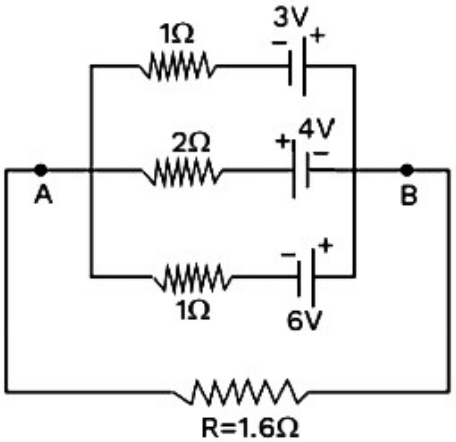



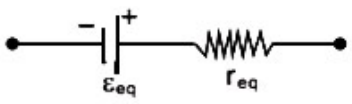

$$\varepsilon_q = \frac{\frac{\varepsilon}{r}+\frac{\varepsilon}{r}+\ldots\frac{\varepsilon}{r}\rightarrow n-\text{times}}{\frac{1}{r}+\frac{1}{r}+\ldots\frac{1}{r}\rightarrow n-\text{times}} = \frac{n\times\frac{\varepsilon}{r}}{n\times\frac{1}{r}} = \varepsilon \quad \Rightarrow \varepsilon_{eq} = \varepsilon$$

$$\frac{1}{r_{eq}} = \frac{1}{r}+\frac{1}{r}+\frac{1}{r}+\ldots n\text{ times}, \quad \frac{1}{r_{eq}} = \left(\frac{n}{r}\right) \Rightarrow r_{eq} = \frac{r}{n}$$

**Q.** **Three cells of emf 3V, 4V and 6V are connected in parallel. If their internal resistance are 1 Ω, 2 Ω and 1 Ω. Find the equivalent $r_{eq}$ and the current in the external load R = 1.6 Ω.**

**Sol.** Let's assume that polarity of equivalent battery is $\varepsilon_{eq}$.



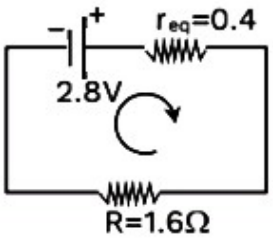
Now,

$$\varepsilon_{eq} = \left\{\frac{\frac{3}{1}+\frac{6}{1}-\frac{4}{2}}{\frac{1}{1}+\frac{1}{2}+\frac{1}{1}}\right\}$$

$$\varepsilon_{eq} = 2.8\,\text{volt}$$

$$\frac{1}{r_{eq}} = \frac{1}{1}+\frac{1}{2}+\frac{1}{1} = \frac{2+1+2}{2} = \frac{5}{2}$$

$$r_{eq} = \frac{2}{5} = 0.4\,\Omega \quad \Rightarrow r_{eq} = 0.4\ \Omega$$




Now redraw the circuit

Current flow in the circuit :

$$i = \frac{2.8V}{1.6+0.4} = \frac{2.8V}{2}$$

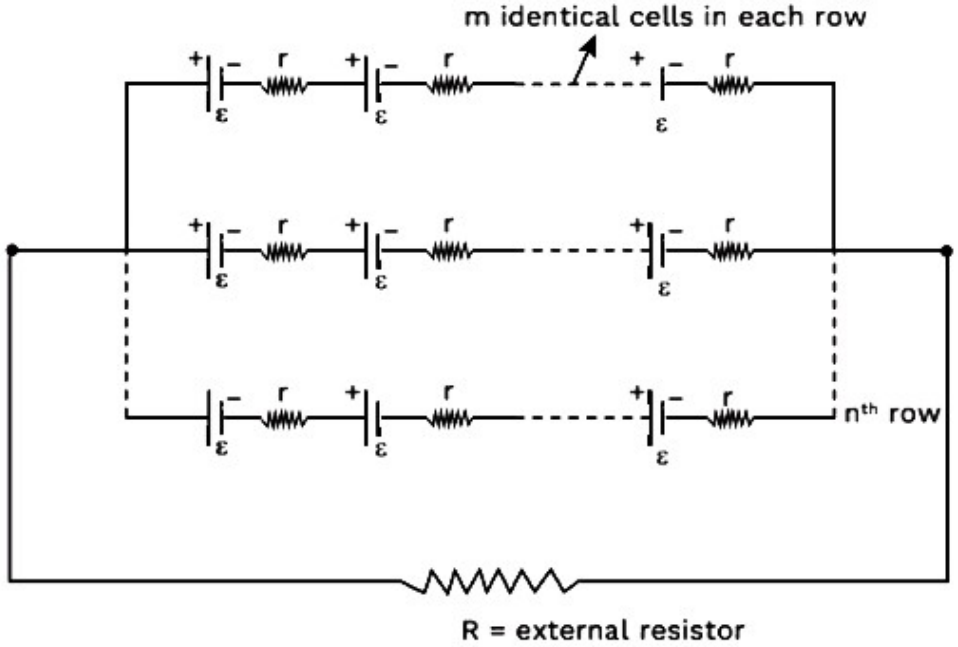
$$i = 1.4\,\text{amp}$$

Answer is $\varepsilon_{eq} = 2.8V$

$r_{eq} = 1.4\ \Omega$

$i = 1.4$ amp

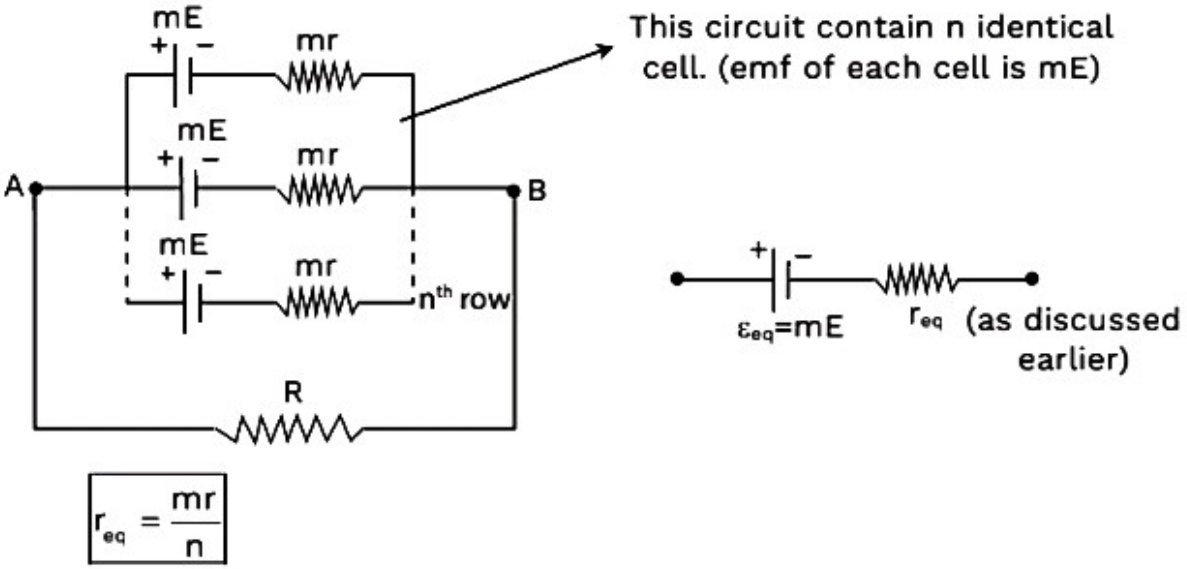
**Mixed Combination of Batteries**




R = external resistor

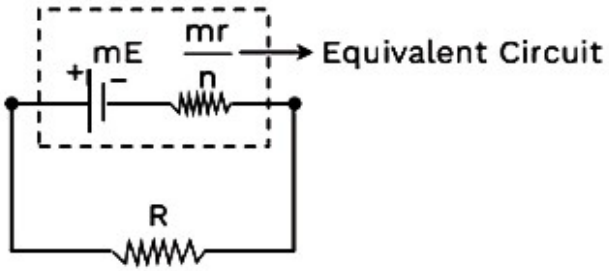
This situation is shown in figure. These are m identical cells in a row and number of rows in n. The emf of each cell is ε and internal resistance is 'r'.

**Now simplification of above circuit gives,**




$$r_{eq} = \frac{mr}{n}$$

Final circuit after reduction :




**Important Points Regarding this Circuit :**

(1) Total no of Battery = m × n

(2) Net emf = mE

(3) Total internal resistance = $\frac{mr}{n}$

(4) Total external resistor = R

(5) Current flow through the external resistor 'R' is, $i = \frac{mE}{R + \frac{mr}{n}}$

**Special Cases :**

(1) Condition for maximum power transfer to external resistor (load)

$\Rightarrow$ Internal resistance of circuit = External resistance of circuit

$\Rightarrow \frac{mr}{n} = R \Rightarrow mr = nR$

(2) In case of maximum power transfer to the load, current flow in the circuit:

$$i_{max} = \frac{mE}{2R} = \frac{mE}{2\left(\frac{mr}{n}\right)}$$

(3) In case of maximum power transfer to the load, formula for numerical solving in terms of total number of battery:

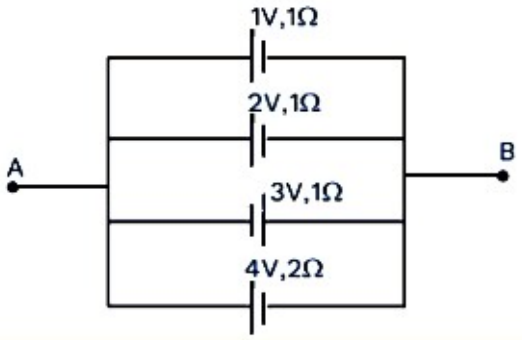
Total no. of battery, $N = m \times n$

From relation : $mr = nR$, $\frac{N}{n} \times r = n \times R$, $n^2 = N \times \frac{r}{R}$

$$n = \sqrt{\frac{r}{R} \times N} = \sqrt{\frac{\text{Internal resistance of battery}}{\text{External resistance}} \times \text{Total No. of Battery}}$$

$m \times n = N \Rightarrow m = \frac{N}{n}$

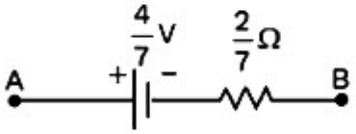
**Q.** **Calculate equivalent emf of four different cells connected in parallel combination as shown in figure.**



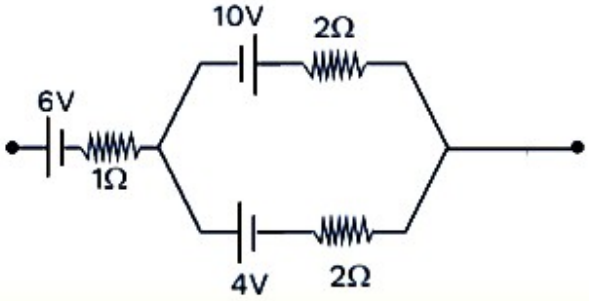

**Sol.**

Since, $E_{eq} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2} + \frac{E_3}{r_3} + \frac{E_4}{r_4}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r_4}}$ $\Rightarrow E_{eq} = \frac{\frac{1}{1} + \frac{2}{1} - \frac{3}{1} + \frac{4}{2}}{\frac{1}{1} + \frac{1}{1} + \frac{1}{1} + \frac{1}{2}} = \frac{2}{\left(\frac{7}{2}\right)} = \frac{4}{7}V$

The equivalent cell arrangement between A and B is shown for reference.



**Q.** **Find the emf and internal resistance of a single battery which is equivalent to a combination of three batteries as shown in figure.**

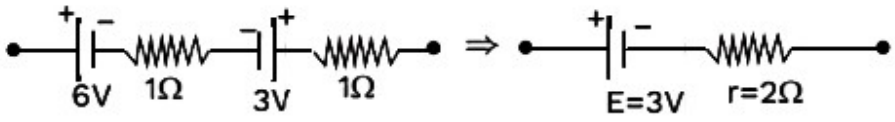



**Sol.** For parallel combination we can apply,

$$E_{eq} = \frac{\frac{E_1}{r_1} - \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\frac{10}{2} - \frac{4}{2}}{\frac{1}{2} + \frac{1}{2}} = 3\,V$$

Further, $\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} = \frac{1}{2} + \frac{1}{2} = 1 \quad \Rightarrow r_{eq} = 1\Omega$

The given combination consists of two batteries in parallel and resultant of these two in series with the third one.




**Numericals Based on mixed grouping of cells**

**Q.** **In a mixed grouping of identical cells, five rows are connected in parallel and each row contains 100 cells. This combination sends a current I through an external resistance of 20 Ω.**
**If the emf and internal resistance of each cell is 1.5V and 5Ω, respectively then find the value of i ?**

**Sol.** n = 5, m = 100, R = 20

$$i = \frac{mE}{R + \left(\frac{mr}{n}\right)} = \frac{100 \times 1.5}{20 + \frac{100 \times 5}{5}} = \frac{150}{20 + 100} = \frac{150}{120} = 1.25\ A$$

i = 1.25 Amp

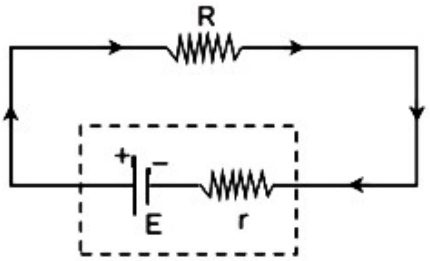
**Q.** **100 cells each of emf 5V and internal resistance 1 Ω are to be arranged to produce maximum current in a 25 Ω resistance.**
**Each row contains equal no of cells. Find number of rows and number of cell in each row.**

**Sol.** The relation $n = \sqrt{\frac{r}{R} \times N}$, $n = \sqrt{\left(\frac{1}{25}\right) \times (100)}$

$$n = 2$$
$$nm = N$$
$$m \times 2 = 100$$
$$m = 50$$

Two rows and each row contain 50 cells.

# Maximum Power Transfer Theorem




Suppose we want to find out for what value of external resistance the maximum power will be drawn from a battery.

For this, in the shown network let the power developed in resistance R be:

$$P = i^2 \times R \Rightarrow P = \left(\frac{\varepsilon}{R+r}\right)^2 \times R$$

$$P = \frac{\varepsilon^2 R}{(R+r)^2}$$

Now, for $\frac{dP}{dR} = 0$

Since P will be maximum if $\frac{dP}{dR} = 0$ (concept of maxima & minima)

$$\Rightarrow \frac{d}{dR}\left(\frac{E^2 \times R}{(R+r)^2}\right) = E^2 \times \frac{d}{dR}\left\{\frac{R}{(R+r)^2}\right\}$$

$$\Rightarrow E^2\left[\frac{(R+r)^2 \times 1 - (R)(2)(R+r)(1+0)}{\left[(R+r)^2\right]^2}\right] = \frac{dP}{dR}$$

$$\Rightarrow \frac{dP}{dR} = E^2\left[\frac{R^2 + r^2 + 2Rr - 2R^2 - 2Rr}{(R+r)^4}\right]$$

$$\Rightarrow \frac{dP}{dR} = E^2\left[\frac{r^2 - R^2}{(R+r)^4}\right] \qquad \ldots(1)$$

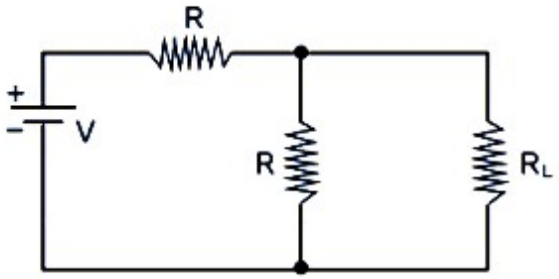
$$\Rightarrow \frac{dP}{dR} = 0 \qquad \Rightarrow \quad r^2 - R^2 = 0 \qquad \Rightarrow \quad r = R$$

Now we can find maximum power delivered to load:

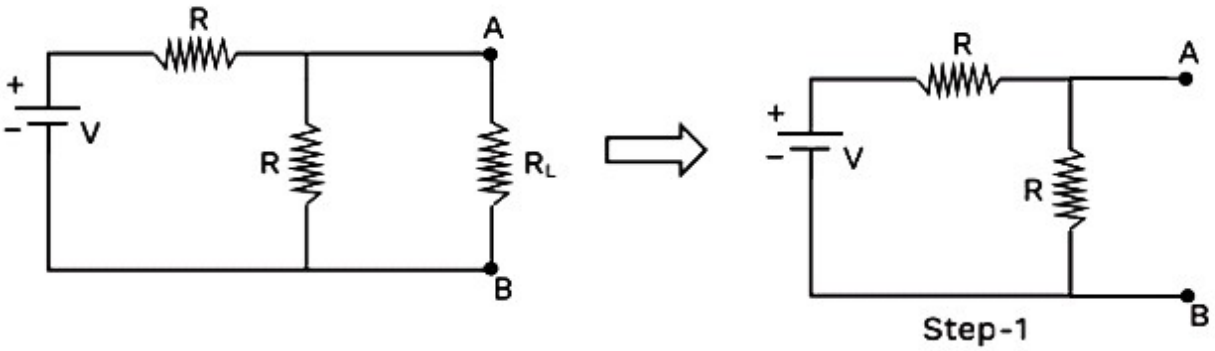
$$P_{max} = i^2 \times R_L \quad = \frac{V^2}{(R+r)^2} \times R_L = \frac{V^2}{(2R_L)^2} \times R_L \quad = \frac{V^2 R}{4R_L^2} = \frac{V}{4R_L} \quad \Rightarrow P_{max} = \frac{V}{4R_L}$$

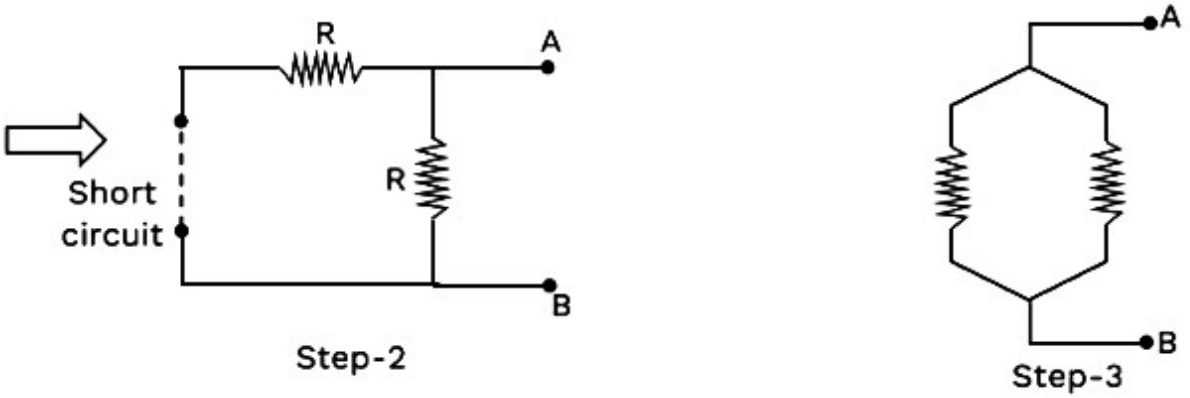
$R_L$ = resistance of load.

**Q.** **Find out value of $R_L$ such that it will receive maximum power.**




**Sol.** Steps to find out equivalent resistance across the load.
Step-1 : Remove load resistance
Step-2 : Short circuit the voltage source
Step-3 : Solve the given network of resistances.



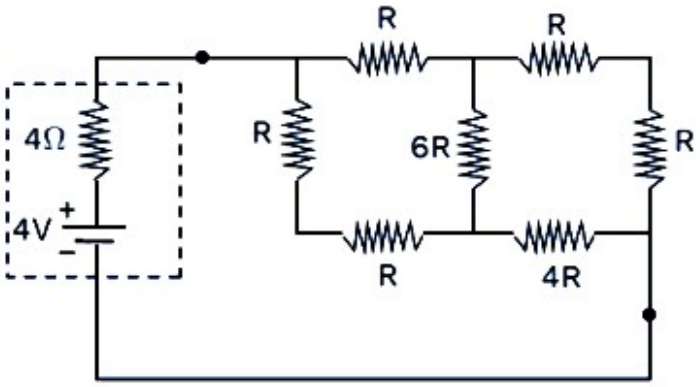




So, for maximum power received by load.

$$R_{eq} = R_L$$

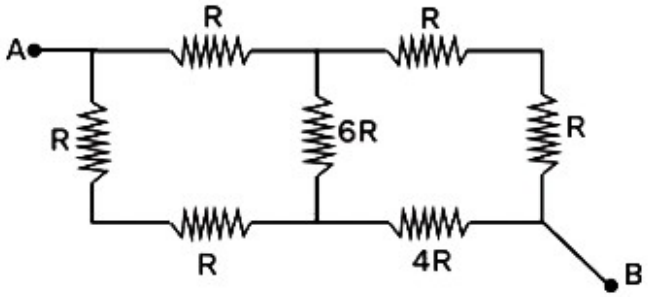
$$R_L = \frac{R}{2} \text{ Ans.}$$

**Q.** **A battery of internal resistance 4 Ω is connected to the network of resistance as shown. In order to given the maximum power to the network, the value of R should be**

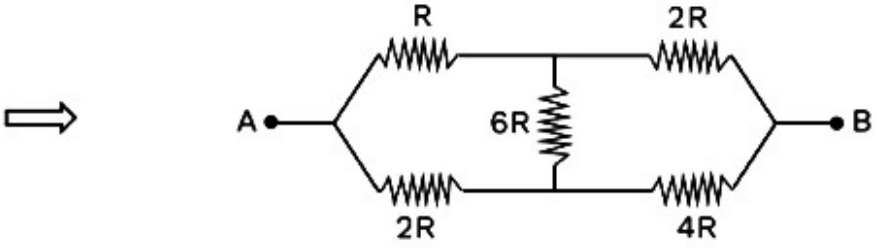


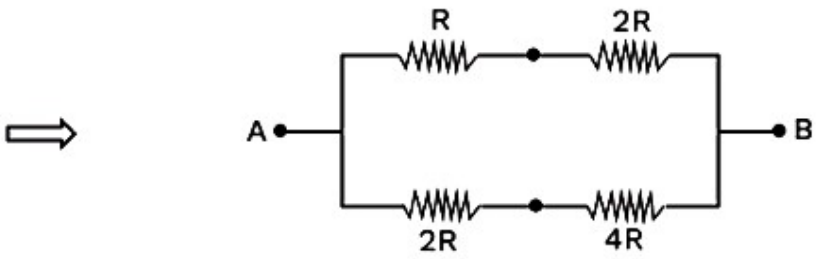

**(A)** $\frac{4}{9}\Omega$ **(B)** $\frac{8}{9}\Omega$ **(C) 2W** **(D)** $\frac{11}{7}\Omega$

**Sol.** Important Point : Load may be a single resistance or a group of resistances or may be a circuit which contains resistance as well as batteries.
Now, Given load network:




This is balanced wheat stone bridge.






$$R_{AB} = 3R \parallel 6R = \frac{3R \times 6R}{(3R + 6R)} = 2R$$

$\Rightarrow$ Now : In case maximum power transfer :

$2R = 4\Omega$

$R = 2\Omega$

Option C is correct.

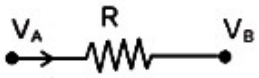
# Heating Effect of Current

When an electric current is passed through a conductor, it becomes hot and its temperature starts rising. This is known as 'Heating effect of current' or 'Joule's heating effect.

**What is Cause of Heating ?**

When we apply the potential difference applied across the two ends of conductor sets up electric field. Under the effect of electric-field, electron accelerated and as they start moving , they collide against the ions and atoms in the conductor, the energy of electrons transferred to the atoms and ions appears as heat.

**Heat-produced by an Electric Current**




The net amount of charge flow in time t

$q = I \times t$

Since we know that current always flows from higher potential to lower potential, so decrease in potential of this charge is given by:

$\Delta V = (V_A - V_B)$

$\Rightarrow$ Decrease in potential energy of charge 'q'

$\Delta U = q \times \Delta V$

$\Delta U = (V_A - V_B) \times q$

$\boxed{\Delta U = I \times t(V_A - V_B)}$ $\Rightarrow$ This decrease in the potential energy will appear in form of heat energy.

$\boxed{H = I \times t(V_A - V_B)}$

**Case-1:**

For ohmic resistance

$V_A - V_B = I \times R$

$H = I \times t \times IR$

$\boxed{\text{Heat energy} = I^2 \times R \times t}$

**Joule's Law of Heating**

Joule's law of heating states that the amount of heat produced in a conductor is directly proportional to:

$H \propto i^2$

$H \propto R$

$H \propto t$

$\boxed{H = i^2 R \times t}$

2nd form $V = i \times R \Rightarrow i = \frac{V}{R}$, $H = \left(\frac{V}{R}\right)^2 \times R \times t = \frac{V^2}{R} \times t$

3rd form $R = \frac{V}{I}$

$$H = i^2 \times R \times t \quad \Rightarrow \quad H = I^2 \times \frac{V}{I} \times t \qquad \Rightarrow \quad H = V \times I \times t$$

$$\text{So, energy} = i^2 \times R \times t = \frac{V^2}{R} \times t = V \times i \times t$$

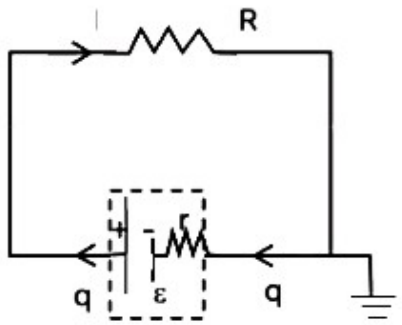
**Note :**

1. "Heat produced in conductor docs not depends upon the direction of current".
2. Heating effect of current is common to both DC as well as AC currents.

**Unit of Heat**

1. SI Unit : Joule
2. Practical unit : 1 Kilowatt hour (KWh)

Electrical power produced in the circuit :




The rate at which the energy is dissipated is called as electric power (P).
It is given as :

$$P = \frac{\text{Heat disspated}}{t} = \frac{I^2 \times R \times t}{t}$$

$P = I^2R$. So $P = I^2R = \frac{V^2}{R} = VI$

**Q.** **What is 1 kilo watt Hour ?**

**Sol.** It is unit of energy. 1 k.Wh is the energy consumed by an appliance of Power 1 kilowatt when it runs for 1 hour.

1 k.Wh = 1000 × 60 × 60
1 k.Wh = $3.6 \times 10^6$ Joule.

**Summary :**

1) Electrical power (P) = $i^2R = \frac{V^2}{R} = VI$

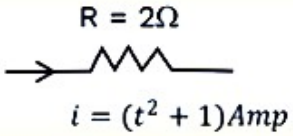
2) Electrical power 'P' = $\frac{d(\text{Energy})}{dt}$

$$\boxed{\int dE = \int P \cdot dt}$$

Energy produced in form of heat.

Heat energy = $= \int_{t_1}^{t_2} i^2 \times R dt$

**Q.** **Find heat dissipation through resistor in 2 sec.**




**Sol.** Heat $= \int i^2 R dt$

Given $i = (t^2 + 1)$

$$= \int_0^2 (t^2 + 1)^2 \times 2 \times dt$$

$$= \int_0^2 (t^4 + 1 + 2t^2) 2dt$$

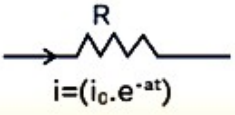
$$= 2 \times \left(\frac{t^5}{5}\right)_0^2 + 2(t)_0^2 + 4\left(\frac{t^3}{3}\right)_0^2$$

$$= \frac{2}{5}[2^5 - 0] + 2(2 - 0) + \frac{4}{3}(8 - 0)$$

$$= \frac{32 \times 2}{5} + 4 + \frac{32}{3}$$

$$= 27.3 \text{ J}$$

**Q.** **Find-out heat dissipation through 'R' in infinite time.**



**Sol.** Heat Energy $= \int_0^{\infty} i^2 \times R \times dt \quad = \int \left(i_0.e^{-at}\right)^2 \times R dt$

$$= i_0^{\,2} \times R \int_0^{\infty} e^{-2at}.dt \qquad = i_0^{\,2} \times R \times \frac{\left|e^{-2at}\right|_0^{\infty}}{-2a}$$

$$= \frac{i_0^{\,2} \times R}{-2a}\left\{e^{-\infty} - e^{-0}\right\} \qquad = \frac{i_0^2 R}{-2a}(0-1)$$

Ans. Heat Energy $= \boxed{\frac{i_0^2 R}{2a} \text{ Joule}}$

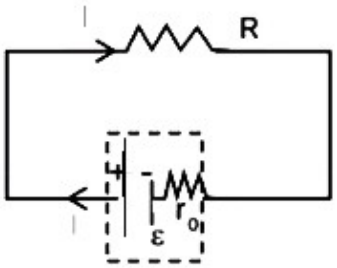
**Discussion on power of battery**

(1) Power delivered by battery :

$P_{Battery}$ = Rate of work done by Battery

$$= \frac{d}{dt}(WD) = \frac{d}{dt}(q \times \text{potential diff.}) = \left(\frac{dq}{dt}\right) \times (\text{Emf})$$

$$P_B = \varepsilon I, \quad P_B = \varepsilon \times \frac{\varepsilon}{R+r}, \quad P_B = \frac{\varepsilon^2}{R+r}$$




$$I = \frac{\varepsilon}{R + r_0}$$

(2) Power consumed in external resistance :

$$P_R = i^2 \times R, \quad P_R = \left(\frac{\varepsilon}{R+r}\right)^2 \times R, \quad P_R = \frac{\varepsilon^2 R}{(R+r)^2}$$

(3) Power consumed in internal resistance of Battery.

$$P_r = i^2 \times r_0 = \left(\frac{\varepsilon}{R+r_0}\right)^2 \times r_0, \; P_r = \frac{\varepsilon^2 r}{(R+r)^2}$$

**Power-Calculation in different cases**

(1) Bulb problems : series and parallel combination.

(2) Water – Heater problems.

**Important Points :**

a) Resistance of Bulb :



Rating of Bulb

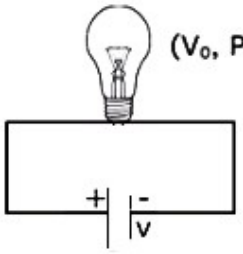
$(V_0, P_0)$

We know, $P = \frac{V^2}{R}$

Resistance of bulb $R = \frac{V_0^2}{P_0}$

This 'R' is constant through-out problem.

b) Power dissipation for a given bulb (given rating) when connected to different voltage.

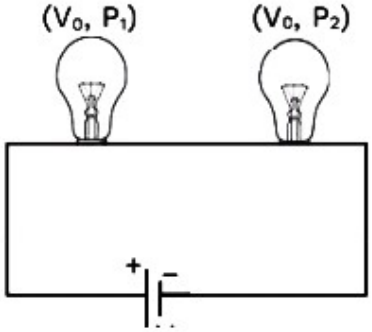



$$P = \frac{V^2}{R} = \frac{V^2}{\left(\frac{V_0^2}{P_0}\right)}$$

$$P = \left(\frac{V}{V_0}\right)^2 \times P_0$$

Power calculation in case of two bulbs.

a) Series combination:




Resistance of bulb $B_1$, $R_1 = \frac{V_0^2}{P_1}$

Resistance of bulb $B_2$, $R_2 = \frac{V_0^2}{P_2}$

Derivation

Total power dissipated, $P_T = P_1 + P_2$

$$P_T = \frac{V^2}{R_{eq}} = \frac{V^2}{R_1 + R_2} = \frac{V^2}{\frac{V_0^2}{P_1} + \frac{V_0^2}{P_2}} = \frac{V^2}{V_0^2\left[\frac{P_2 + P_1}{P_1 P_2}\right]} \Rightarrow P_T = \left(\frac{V}{V_0}\right)^2 \times \frac{P_1 P_2}{(P_1 + P_2)}$$

When connected voltage and rated voltage are same then total power dissipated for n number of bulbs in series,

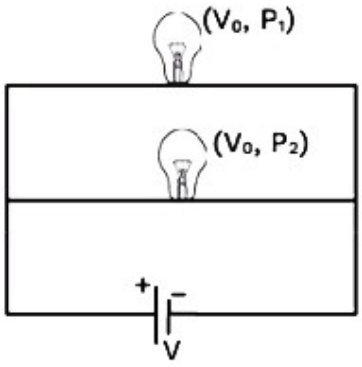
$$\frac{1}{P_T} = \frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3} + ..... + \frac{1}{P_n}$$

b) Parallel combination

$$P_T = P_1 + P_2$$

$$= \frac{V^2}{R_1} + \frac{V^2}{R_2} = \frac{V^2}{\frac{V_0^2}{P_1}} + \frac{V^2}{\frac{V_0^2}{P_2}} = \frac{V^2}{V_0^2}(P_1 + P_2)$$

$$P_T = \left(\frac{V}{V_0}\right)^2 (P_1 + P_2)$$



When connected voltage and rated voltage are same then total power dissipated for n number of bulbs in parallel,

$P_T = P_1 + P_2 + P_3 + ..... + P_n$

**Water Heater Problem**

**Q.** **It is given that an Electric Kettle has two coils of same power when one-coil is switched on, it takes 15 minutes to boil water, and when the second coil is switched on, it takes 30 min. How much time will it take to boil water when both the coil are used in.**

**a) series** **b) parallel**

**Sol.** Heat produced in resistance 'R' in time 't' is

$= P \times t = \frac{V^2}{R} \times t$

For coil-1 $H_1 = \frac{V^2}{R_1}(15 \times 60)$

For coil-2, $H_2 = \frac{V^2}{R_2}(30 \times 60)$

But according to the given problem

$H_1 = H_2$

i.e. $\frac{15}{R_1} = \frac{30}{R_2} \Rightarrow R_2 = 2R_1$

(a) When both the coils are used in series, we have

$H_s = \frac{V^2}{(R_1 + R_2)} \times t_s = \frac{V^2}{3R_1} \times t_s$ [as $R_2 = 2R_1$]

Here $H_s = H_1 = H_2$

So $\frac{V^2}{R_1}(15 \times 60) = \frac{V^2}{3R_1} \times t_s \Rightarrow t_s = (45 \times 60)\,sec$

$t_s$ = 45 minute

(b) When both the coils are used in parallel

We have $H_p = \left[\frac{V^2}{R_1} + \frac{V^2}{R_2}\right] \times t_p = \frac{3V^2}{2R_1} \times t_p$ [as $R_2 = 2R_1$]

According to the given problems, $H_p = H_1$

or $\frac{3V^2}{2R_1} \times t_p = \frac{V^2}{R_1} \times (15 \times 60)$ or $t_p = (10 \times 60)sec \Rightarrow t_p = 10$ min

Ans. (a) 45 Min. (b) 10 min

**Shortcut method :**

If a heater boils m kg water in time $T_1$ and another heater boils the same water in $T_2$, then both connected in series will boil the same water in time,

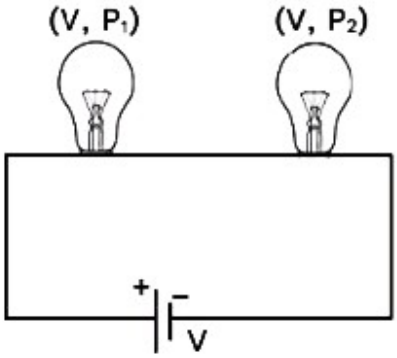
a) $Ts = T_1 + T_2 = 30 + 15$ min = 45 minute Ans.

b) For parallel connection

$$T_p = \frac{T_1 T_2}{T_1 + T_2} = \frac{30 \times 15}{30 + 15} \quad \Rightarrow \quad T_p = 10 \text{ minute.}$$

**Important Concepts**

a) In series combination of Resistances (bulbs) :

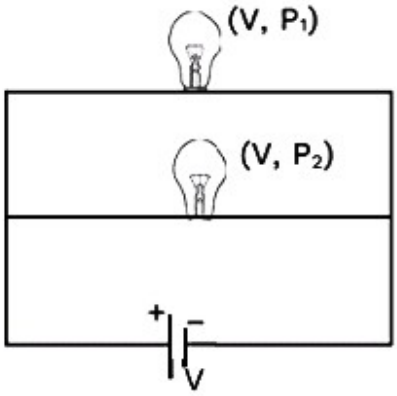



Brightness $\propto$ power consumed by the bulb $\propto t^2 \propto R \propto \frac{1}{P_{Rated}}$

So, bulb of lesser wattage (rated power) will shine more.

For same current $P = I^2 \times R$; $P \propto R$; $R \uparrow \Rightarrow P \uparrow$

b) Parallel combination :




Brightness $\propto$ power consumed by Bulb

$\propto V^2 \propto \frac{1}{R}$ Bulb of greater wattage (rated power) will shine more.

For same 'V', more power will be consumed in smaller resistance $P \propto \frac{1}{R}$.

**Characteristics of a Fuse**

Fuse is used with the main electrical circuit for the safety of electrical appliances.

A fuse wire should have high resistance and low melting point. Hence, generally it is made of tin-lead alloy.

Let R be the resistance, $\rho_0$ resistivity, l length, 'a' cross-sectional area and I ampere be its current carrying capacity.

When the fuse is safe, then for its steady state temperature, heat produced per second must be equal to heat radiated by it per second. Heat produced in fuse wire per second.

$$H = \frac{Q}{t} = I^2R = I^2\left(\frac{\rho l}{A}\right)$$

$$H = \frac{I^2\rho l}{\pi r^2} \text{ joule sec}^{-1} \qquad ...(1)$$

If e is emissivity of fuse material of radius r and T is the excess safe temperature of wire above surroundings then according to Newton's Law of cooling, the energy radiated per second.

$$H = e(2\pi rl)T \qquad ...(2)$$

For steady state

$$e(2\pi rl)T = I^2\frac{\rho l}{\pi r^2} \quad \Rightarrow \quad T = \frac{I^2\rho}{2\pi^2 er^3}$$

So, we observe that the steady state temperature of a fuse is independent of its length.

Hence length is immaterial for an electric fuse.

For a given material of a fuse wire.

$I^2 \propto r^3$

**Q.** **A fuse which have a circular cross-sectional having radius of 0.15mm blows when a 15 amp current pass through it. Find out is the radius of a fuse, made of the same material which will blow at 120A ?**

**Sol.** For fuse wire $I \propto r^{3/2}$

$$\text{So } \frac{r_2}{r_1} = \left(\frac{I_2}{I_1}\right)^{2/3} = \left(\frac{120}{15}\right)^{2/3} = (8)^{2/3} = 4 \quad \Rightarrow 4r_1 = 0.60\,\text{mm}.$$

**Q.** **Calculate amount of heat will be generated in a coil of resistance R due to a charge q passing through it if the current in the coil decreases down to zero.**

**(a) uniformly during a time interval $t_0$ ?**

**(b) halving its value every $t_0$ seconds ?**

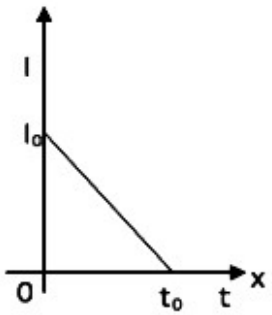
**Sol.** Heat generated in a resistance is given by

$H = I^2Rt$

We can directly use this formula provided I is constant. Here, I is varying. So, first we will calculate I at any time t, then find a small heat dH in a short interval of time dt. Then by integrating it with proper limits we can obtain the total heat produced.

(a) The corresponding I-t graph will be a straight line with I decreasing from a peak value (say $I_0$) to zero in time $t_0$. I-t equation will be as

$$I = I_0 - \left(\frac{I_0}{t_0}\right)t \quad (y = -mx + c) \qquad ...(1)$$




Here, $I_0$ is unknown, which can be obtained by using the fact that area under I-t graph gives the flow of charge. Hence

$$q = \frac{1}{2}(t_0)(I_0) \Rightarrow I_0 = \frac{2q}{t_0}$$

Substituting in (1), we get

$$I = \frac{2q}{t_0}\left(1 - \frac{t}{t_0}\right) \Rightarrow I = \left(\frac{2q}{t_0} - \frac{2qt}{t_0^2}\right) \Rightarrow I = \left(\frac{2q}{t_0} - \frac{2qt}{t_0^2}\right)$$

Now at time t, heat produced in a short interval dt is, $dH = I^2Rdt$

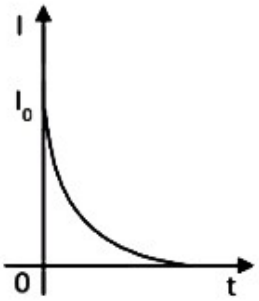
$$\Rightarrow dH = \left(\frac{2q}{t_0} - \frac{2qt}{t_0^2}\right)^2 Rdt$$

So, total heat produced is $H = \int_0^{t_0} dH$

$$\Rightarrow H = \int_0^{t_0}\left(\frac{2q}{t_0} - \frac{2qt}{t_0^2}\right)^2 Rdt \Rightarrow H = \frac{4}{3}\frac{q^2R}{t_0}$$

(b) Here, current decreases from some peak value (say $I_0$) to zero exponentially with half life $t_0$. I-t equation in this case will be

$I = I_0 e^{-\lambda t}$




Here, $\lambda = \frac{\log_e(2)}{t_0}$

Now, $q = \int_0^\infty I dt = \int_0^\infty I_0 e^{-\lambda t} dt = \left(\frac{I_0}{\lambda}\right)$

$\Rightarrow I_0 = \lambda q$

$\Rightarrow I = (\lambda q) e^{-\lambda t}$

$\Rightarrow dH = I^2 R dt = \lambda^2 q^2 e^{-2\lambda t} R dt$

$\Rightarrow H = \int_0^\infty dH = \lambda^2 q^2 R \int_0^\infty e^{-2\lambda t} dt = \frac{q^2 \lambda R}{2}$

Substituting $\lambda = \frac{\log_e(2)}{t_0}$, we have $H = \frac{q^2 R \log_e(2)}{2t_0}$

**Q.** **An electric heater and an electric bulb are rated 500W, 220V and 100W, 220V respectively. Both are connected in series to a 220V a.c. mains. Calculate power consumed by (i) heater (ii) bulb.**

**Sol.** $\because P = \frac{V^2}{R}$ or $R = \frac{V^2}{P}$, (for heater).

Resistance $R_h = \frac{(220)^2}{500} = 96.8\Omega$,

For bulb, resistance $R_L = \frac{(220)^2}{100} = 484\Omega$

Current in the circuit when both are connected in series

$I = \frac{V}{R_L + R_h} = \frac{220}{484 + 96.8} = 0.38\,A$

(i) Power consumed by heater = $I^2 R_h = (0.38)^2 \times 96.8 = 13.98$ W

(ii) Power consumed by bulb = $I^2 R_L = (0.38)^2 \times 484 = 69.89$ W

**Q.** **A heater coil is rated 100 W, 200 V. It is cut into two identical parts. Both parts are connected together in parallel, to the same source of 200V. Calculate the energy liberated per second in the new combination.**

**Sol.** $\because\ P = \frac{V^2}{R} \qquad \therefore\ R = \frac{V^2}{P} = \frac{(200)^2}{100} = 400\,\Omega$

Resistance of half piece $= \frac{400}{2} = 200\,\Omega$

Resistance of pieces connected in parallel $= \frac{200}{2} = 100\,\Omega$

Energy liberated/second $P = \frac{V^2}{R} = \frac{(200)^2}{100} = 400\ W$

**Q.** **Given that the power of a heater is 500W at 800°C. What will be its power at 200°C. If $\alpha = 4 \times 10^{-4}$ per°C ?**

**Sol.** $P = \frac{V^2}{R} \therefore \quad \frac{P_{200}}{P_{800}} = \frac{R_{800}}{R_{200}} = \frac{R_0(1 + 4 \times 10^{-4} \times 800)}{R_0(1 + 4 \times 10^{-4} \times 200)}$

$$\frac{P_{200}}{500} = \frac{(1 + 3200 \times 10^{-4})}{1 + 800 \times 10^{-4}}$$

$$\frac{P_{200}}{500} = \frac{(1 + 0.32)}{(1 + 0.08)}$$

$$\frac{P_{200}}{500} = \frac{1.32}{(1.08)}$$

$$P_{200} = \frac{1.32}{1.08} \times 500$$

$$\boxed{P_{200} = 611.11 \text{ watt}}$$

**Q.** **When a battery sends current through a resistance $R_1$ for time t, the heat produced in the resistor is equal to Q. When the same battery sends current through another resistance $R_2$ for time t, the heat produced in $R_2$ is again Q. What is the internal resistance of battery.**

**Sol.** $$\left[\frac{E}{R_1+r}\right]^2 R_1 = \left[\frac{E}{R_2+r}\right]^2 R_2 \quad \Rightarrow \quad r = \sqrt{R_1 R_2}$$

**Q.** **If bulb rating is 100 watt and 220V, for this given bulb determine**

**(a) Resistance of filament.**

**(b) Current through filament.**

**(c) If bulb operate at 110 volt power supply then find power consumed by bulb.**

**Sol.** Bulb rating is 100W and 220 V bulb means when 220 V potential difference is applied between the two ends then the power consumed is 100W

Here V = 220 Volt, P = 100W

$\frac{V^2}{R} = 100$. So R = 484Ω

Since resistance depends only on material hence it is constant for bulb.

$$I = \frac{V}{R} = \frac{220}{22 \times 22} = \frac{5}{11} \text{ Amp}$$

Power consumed at 110V

$$P = \left(\frac{V}{V_0}\right)^2 \times P_0 = \left(\frac{110}{220}\right)^2 \times 100 = \left(\frac{1}{2}\right)^2 \times 100 = \frac{100}{4} = 25 \text{ W}$$

Answer is R = 484Ω

$$I = \frac{5}{11} \text{Amp}, \quad P = 25 \text{watt}$$

# Measurement & Electrical Instruments

Measurement is a process of comparison between a standard and an unknown, resulting in knowing the magnitude of unknown, in terms of standard value. Instrument is a device that allows us to make this comparison.

**Here the list of electrical instruments**

(1) Galvanometer
(2) Ammeter
(3) Voltmeter
(4) Meter Bridge
(5) Potentiometer

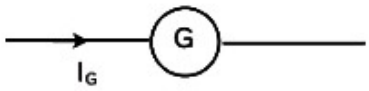
**Galvanometer**

The instrument used to measure the current of low strength by measuring the deflection of the coil due to torque produced by a magnetic field is known as galvanometer.

If we pass a current through a galvanometer more than the specified amount, it may get damaged. The specified current through a galvanometer is known as the full scale deflection current ($I_G$).

The resistance of a galvanometer is very small.

**Symbol :**




**Construction of Galvanometer :**

It consists of a pivoted coil placed in the magnetic field of a permanent magnet. Attached to the coil is a spring. In the equilibrium position, with no current in the coil, the pointer is at zero and spring is relaxed. When there is a current in the coil, the magnetic field exerts a torque on the coil that is proportional to current. As the coil turns, the spring exerts a restoring torque that is proportional to the angular displacement. Thus the angular deflection of the coil and pointer is directly proportional to the coil current and the device can be calibrated to measure current.

When coil rotates the spring is twisted and it exerts an opposing torque on the coil.

There is a resistive torque also against motion to damp the motion. Finally in equilibrium

$$\tau_{magnetic} = \tau_{spring} \Rightarrow BINA\sin\theta = C\phi$$

But by making the magnetic field radial, $\theta = 90°$

$\therefore$ $BINA = C\phi$

Here, B = magnetic field
A = Area of the coil
I = Current
C = torsional constant
N = Number of turns
$\phi$ = angle rotate by coil

**Current sensitivity :**

The ratio of deflection to the current flowing through galvanometer i.e., deflection per unit current is called sensitivity (C.S.) of the galvanometer $CS = \frac{\phi}{I} = \frac{BNA}{C}$

**Voltage Sensitivity :**

The ratio of deflection to the voltage across galvanometer is called. Voltage sensitivity VS = $\frac{\phi}{V}$

**Note :** Before study of ammeter and voltmeter, we should know about shunt.

**Shunt :** The small resistance connected in parallel to galvanometer coil in order to control current flowing through the galvanometer is knows as shunt.

**Advantage :**

(1) To protect the galvanometer coil from burning.
(2) Any galvanometer can be converted into ammeter of desired range with the help of shunt.
(3) The range of an ammeter can be changed by using shunt resistance of difference values.

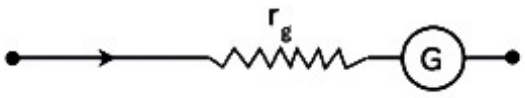
**Disadvantage :** Shunt resistance decreases the sensitivity of galvanometer.

**Ammeter**

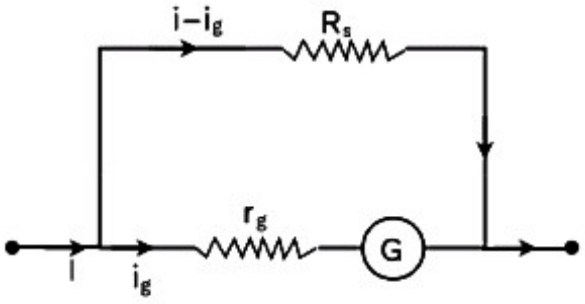
A galvanometer can not be used directly to measure large current. To measure large current we have to carry out some modification in the galvanometer.
"An ammeter is a measuring instrument used to measure the current in circuit."

**Conversion of Galvanometer into Ammeter :**

Let a galvanometer is given:




Its operating range is between (1μA – 1mA). If we want to measure the current in Ampere range we have to connect by pass resistor.



'A galvanometer can be converted into ammeter by connecting low resistance in parallel to its coil.

The value of shunt resistance to be connected in parallel to galvanometer coil is given by :

$$R_s = \frac{R_g i_g}{i - i_g}$$

Where :

$i$ = range of ammeter

$i_g$ = current required for full scale deflection of galvanometer

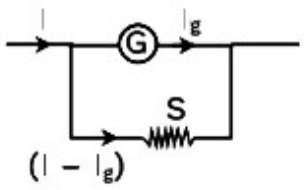
$R_g$ = resistance of galvanometer coil

**Q.** **What is the value of shunt which passes 10% of the main current through a galvanometer of 99 ohm?**

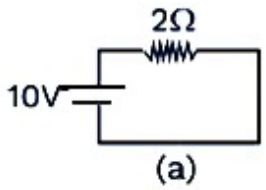
**Sol.** As in figure $R_g I_g = (I - I_g)S$

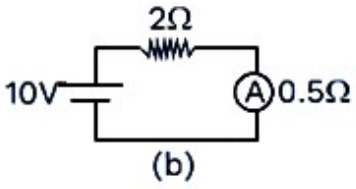
$$\Rightarrow 99 \times \frac{I}{10} = \left(I - \frac{I}{10}\right) \times S$$

$$\Rightarrow S = 11\ \Omega$$




**Q.** **Find the current of the circuit (a) & (b) and also determine percentage error in measuring the current through an ammeter.**




**Sol.** In figure a, $I = \frac{10}{2} = 5\,A$

In figure b, $I = \frac{10}{2.5} = 4\,A$

Percentage error is $= \frac{i - i'}{i} \times 100 = 20\%$

Here we see that due to ammeter, the current has reduced. A good ammeter has very low resistance as compared with other resistors, so that due to its presence in the circuit, the current is not affected.

**Voltmeter**

A voltmeter is an instrument used to find the potential difference across any two points in a circuit.

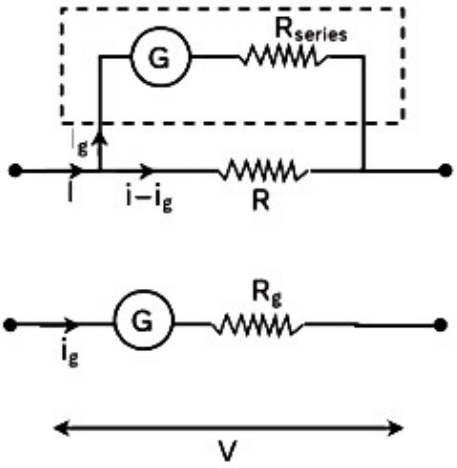
**Conversion of Galvanometer into Ammeter :**

The galvanometer can be converted into ammeter by connected high resistance in series with its coil.

The high resistance to be connected in series with galvanometer coil is given by

$$R = \frac{V}{i_g} - R_g$$

$$R = \frac{V}{i_g} - R_g$$




$$V = i_g (R_s + r_g)$$

$i_g$ = current through galvanometer, $R_s$ = Series resistance,
$r_g$ = resistance of galvanometer

**Q.** **How can we make a galvanometer with G = 20Ω and $I_g$ = 1 mA into a voltmeter with a maximum range of 10 V ?**

**Sol.** Using $R = \frac{V}{I_g} - G$. We have, $R = \frac{10}{10^{-3}} - 20 = 9980\,\Omega$

Thus, a resistance of 9980 Ω is to be connected is series with the galvanometer to convert it into the voltmeter of desired range.

Please note that at full scale deflection current through the galvanometer, the voltage drop across the galvanometer

$$V_g = I_g G = 20 \times 10^{-3} V = 0.02V$$

and the voltage drop across the series resistance R is,

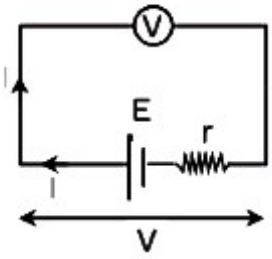
$$V = I_g R = 9980 \times 10^{-3} V = 9.98V$$

**Important Points :**

(a) An ideal ammeter should have zero resistance, but a non-ideal ammeter has small but non-zero resistance.

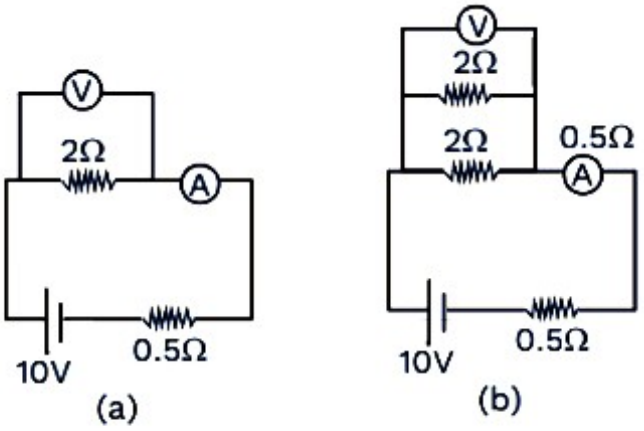
(b) An ideal voltmeter should have infinite resistance, but a non-ideal voltmeter has larger but not infinite resistance.

(c) Non-ideal voltmeter (i.e., a practical voltmeter) always draws some current from the circuit and there is always a potential drop across a non-ideal ammeter (i.e., a practical ammeter).




Therefore, when it is used to measure the emf of a cell, it will read $V = E - Ir$. As $I \neq 0$ therefore, reading of a voltmeter will not be equal to emf of the cell.

Q. **In the electrical circuits shown in figure (a) and figure (b), the instruments are ideal in figure (a) while their resistance are marked in figure (b). Determine their readings.**



**Sol.** For figure (a), we have

$$I = \frac{10}{2 + 0.5} = 4\,A$$

So, reading of ammeter is 4A and reading of voltmeter is IR = 8V
For figure (b), we have

$$I' = \frac{10}{R' + 0.5 + 0.5}, \text{ where } R' = \frac{2 \times 2}{2 + 2} = 1\Omega \quad \Rightarrow \quad I' = \frac{10}{2} = 5A$$

So, reading of ammeter is 5A and reading of voltmeter is $I'R' = 5 \times (1) = 5V$

**Q.** **A galvanometer with a scale divide into 100 equal division, has a current sensitivity of 10 division per mA and voltage sensitivity of 2 per mV. What adoptions are required to use it (a) to read 5A full scale and (b) 1 division per volt?**

**Sol.** Full scale deflection current $i_g = \frac{\phi}{CS} = \frac{100}{10}mA = 10mA$

Full scale deflection voltage $V_g = \frac{\phi}{V_s} = 50\ mV$

So, galvanometer resistance $G = \frac{V_g}{i_g} = \frac{50\,mV}{10\,mA} = 5\Omega$

(a) To convert the galvanometer into an ammeter of range 5A, a resistance of value SΩ is connected in parallel with it such that

$(i - i_g)\,S = i_g\,G$

$(5 - 0.01)\,S = 0.01 \times 5$

$$S = \frac{5}{499} \cong 0.01\Omega$$

(b) To convert the galvanometer into a voltmeter which reads 1 division per volt, i.e., of range 100V.

$V = i_g\,(R + G)$

$100 = 10 \times 10^{-3}(R + 5)$

$R = 10000 - 5$

$R = 9995\ \Omega = 9.995\ k\Omega$

**Q.** **The scale of galvanometer is divided into 150 equal divisions. The galvanometer has the current sensitivity of 10 divisions per mA and the voltage sensitivity of 2 divisions per mV. How can you convert this galvanometer into**

**(a) ammeter of 6A per division**

**(b) voltmeter of 1V per division**

**Calculate the galvanometer resistance G, value of shunt S to be connected to convert into ammeter and value of series resistance R to be connected to convert into voltmeter.**

**Sol.** Since galvanometer resistance is

$$G = \frac{\text{Full scale voltage}}{\text{Full scale current}}$$

Also, full scale voltage $= \dfrac{\text{No. of divisions}}{\text{Voltage sensitivity}} = \dfrac{150}{2} = 75\ \text{mV}$

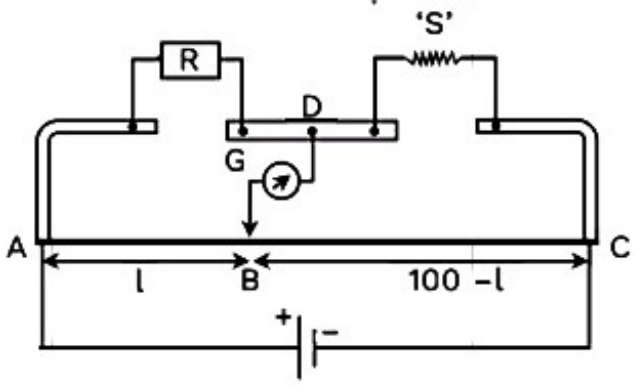
and full scale current $= \dfrac{\text{No. of divisions}}{\text{Current sensitivity}} = \dfrac{150}{10} = 15\,\text{mA} = 15 \times 10^{-3}\ \text{A}$

$$\Rightarrow \quad G = \frac{75 \times 10^{-3}}{15 \times 10^{-3}} = 5\,\Omega$$

$$\Rightarrow \quad S = \frac{I_g G}{I - I_g} = \frac{15 \times 10^{-3} \times 5}{150 \times 6 - 15 \times 10^{-3}} = 83\,\text{m}\Omega \Rightarrow R = \frac{V}{I_g} - G = \frac{150}{15 \times 10^{-3}} - 5 = 9995\ \Omega$$

## Meter Bridge

It is based on principle of balanced wheat stone bridge. Meter Bridge is used to find out unknown resistance of wire. AC is 1 meter long uniform wire, R is known resistance and S is unknown resistance. A cell is connected across 1 meter long wire and a galvanometer is connected to mid-point D. To calculate unknown resistance we touch jockey from 'A' to 'C' and find balanced condition. Let balance is a B point on wire.




Meter Bridge

AB = l cm

Resistance across wire AB, P = rl

BC = (100 – l) cm

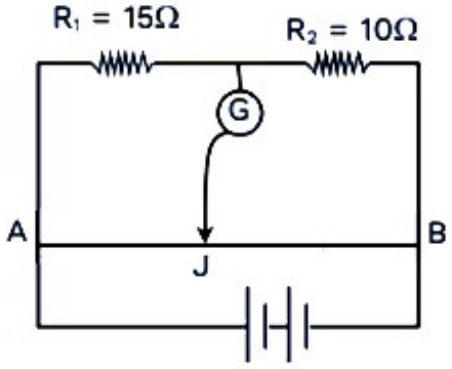
Resistance across wire BC, Q = r (100 – l)

Where r = resistance per unit length on wire

At balanced condition $\frac{P}{Q}=\frac{R}{S} \Rightarrow \frac{rl}{r(100-l)}=\frac{P}{S} \quad \Rightarrow \quad S=\frac{(100-l)R}{l}$

**Note :** The bridge is most sensitive when the resistance in all the four branches of the bridge is of same order.

**Q.** **In the meter bridge circuit shown in figure. Calculate the length AJ for null deflection in galvanometer.**



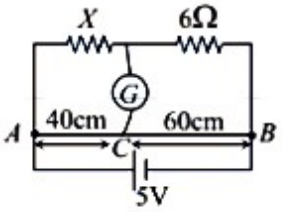
**Sol.** Let AJ = l cm, then JB = (100 – l) cm,
At zero deflection of galvanometer,

$$\frac{R_1}{R_2} = \frac{R_{AJ}}{R_{JB}} = \frac{l}{100-l} \quad \Rightarrow \quad \frac{15}{10} = \frac{l}{100-l}$$

$$\Rightarrow \quad \frac{3}{2} = \frac{l}{100-l} \quad \Rightarrow \quad 300 - 3l = 2l$$

$$\Rightarrow \quad 5l = 300 \quad \Rightarrow \quad l = 60 \text{ cm}$$

**Q.** **In the circuit shown, a meter bridge is in its balanced state. The meter bridge wire has a resistance 0.1 ohm/cm. The value of unknown resistance X and the current drawn from the battery of negligible resistance is**




**(A) 6Ω, 5 amp** **(B) 4Ω, 0.1 amp** **(C) 4Ω, 1.0 amp** **(D) 12Ω, 0.5 amp**

**Sol.** Resistance of the part AC

$R_{AC} = 0.1 \times 40 = 4\Omega$ and $R_{CB} = 0.1 \times 60 = 6\Omega$

In balanced condition $\frac{X}{6} = \frac{4}{6} \Rightarrow X = 4\Omega$

Equivalent resistance $R_{eq} = 5\Omega$ so current drawn from battery $i = \frac{5}{5} = 1A$

$\therefore$ (C)

**Q.** **If resistance $R_1$ in resistance box is 300 Ω, then the balanced length is found to be 75 cm from end A. It is given that the diameter of unknown wire is 1 mm and length of the unknown wire is 31.4 cm. Calculate the specific resistance of the unknown wire.**

**Sol.** Since, $\frac{R}{X} = \frac{l}{100 - l}$

$$\Rightarrow \quad X = \left(\frac{100 - l}{l}\right) r \Rightarrow \quad X = \left(\frac{100 - 75}{75}\right)(300) = 100\Omega$$

Where $X = \frac{\rho l}{A} = \frac{\rho l}{\left(\frac{\pi d^2}{4}\right)}$

$$\Rightarrow \quad \rho = \frac{\pi d^2 X}{4l} \quad \Rightarrow \quad \rho = \frac{\left(\frac{22}{7}\right)\left(10^{-3}\right)^2 (100)}{(4)(0.314)} \quad \Rightarrow \quad \rho = 2.5 \times 10^{-4}\ \Omega m$$

**Potentiometer**

**Necessity of Potentiometer :**

Practically voltmeter has a finite resistance (ideally it should be $\infty$). In other words, it draws some current from the circuit. To solve this problem potentiometer is used because at the instant of measurement it draws no current from the circuit. It means its effective resistance is infinite.

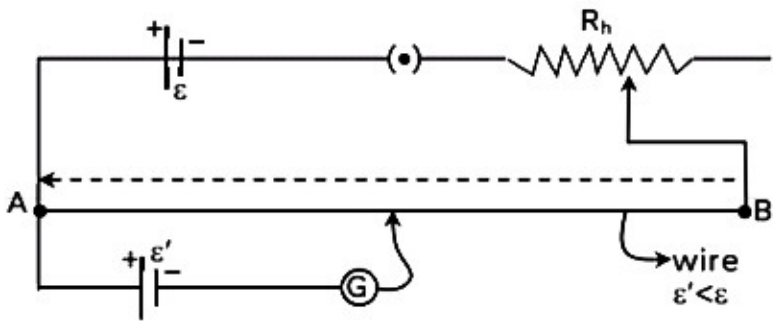
**Working Principle of Potentiometer :**

Any unknown potential difference is balanced on a known potential difference which is uniformly distributed over entire length of potentiometer wire. This process is named as zero deflection or null deflection method.

**Potentiometer wire :**

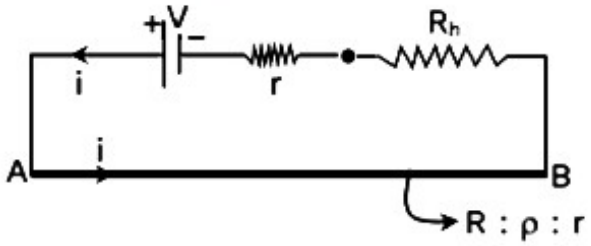
Made up of alloys of manganin, constantan, Eureka, specific properties of these alloys are high specific resistance, negligible temperature coefficient of resistance.

**Circuit of Potentiometer :**



**Primary Circuit :**

Primary circuit contains source of voltage, rheostat or resistance box.




$$i = \frac{V}{R + r + R_h} \quad \Rightarrow \quad V_{AB} = i \times R = \left(\frac{V}{R + r + R_h}\right) \times R$$

**Potential Gradient :**

The fall of potential per unit length of potentiometer wire is called potential gradient.

$$k = \frac{V_{AB}}{l} \Rightarrow k = \frac{i \times R}{l} = \frac{i}{l} \times \left(\rho \frac{l}{A}\right), \quad k = i\frac{\rho}{A}; \qquad \begin{Bmatrix} k \propto i \\ k \propto \rho \\ k \propto \frac{1}{A} \end{Bmatrix}$$

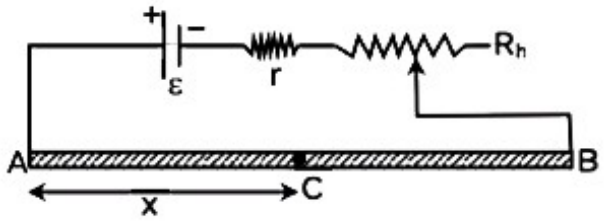
Where k = potential gradient

i = current flow in primary circuit

ρ = resistivity of wire

A = area of cross-section of wire

**Potential drop on potentiometer wire :**




Potential drop between points A and C = Potential gradient × length of wire,

$$V_{AC} = k \times x \Rightarrow V_{AC} = \frac{V_{AB}}{l} \times x$$

$V_{AC} \propto x \quad \Rightarrow$ "Principle of potentiometer"

"If a constant current is passed through a wire of uniform cross-section the potential difference across any segment of the wire is proportional to its length.

**Application of potentiometer :** It is used to

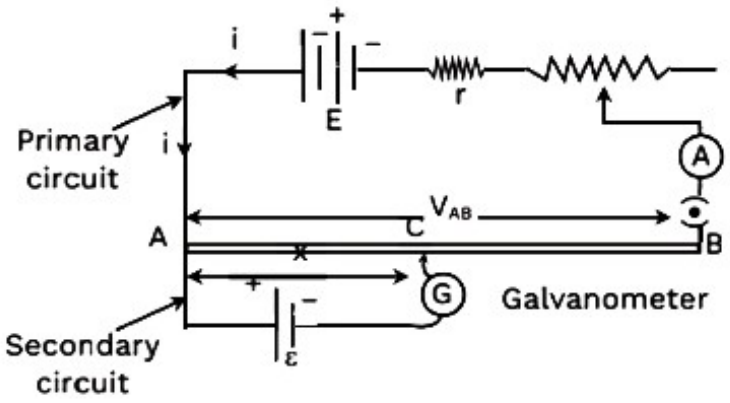
(1) Find emf of a cell

(2) Find internal resistance of a cell

(3) Compare emf's of two cells

(4) Compare two small resistances

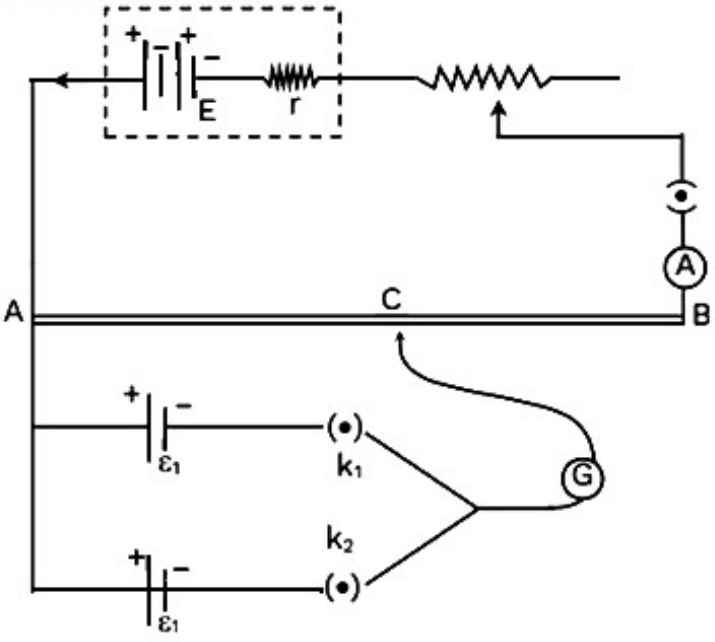
**(1) Emf of a cell:**




**At null deflection condition :** Emf = kx, where

K = potential gradient, X = balancing length

**(2) To compare Emf's of two cells:**




**Step 1:**

$k_1 \rightarrow$ closed, $k_2 \rightarrow$ open, $\varepsilon_1 = kl_1$ ...(1)

Where,

$\varepsilon_1$ = emf of first cell, k = potential gradient, $l_1$ = balancing length for cell $\varepsilon_1$

**Step 2:**

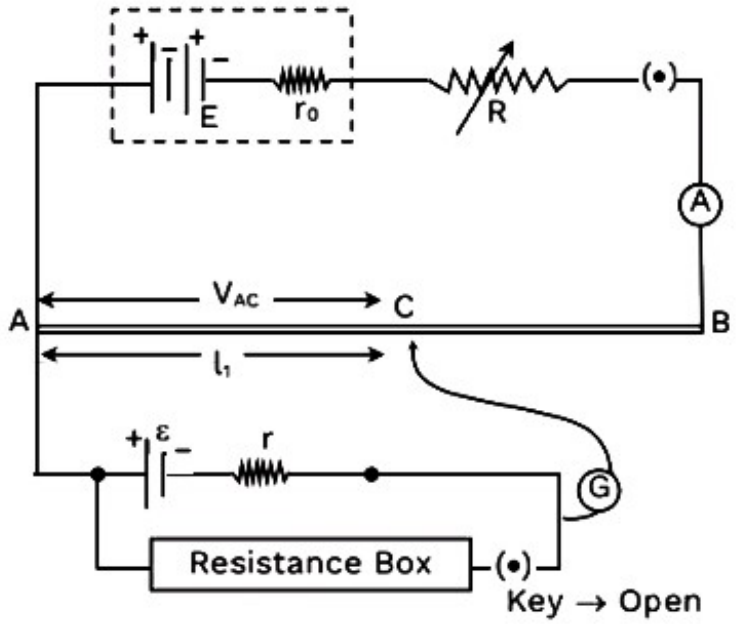
$k_1$ = open, $k_2$ = closed, $\varepsilon_2 = kl_2$ ...(2)

Where,

$\varepsilon_2$ = emf of second cell, k = potential gradient, $l_2$ = balancing length for cell $\varepsilon_2$

So, $\frac{\varepsilon_1}{\varepsilon_2} = \frac{kl_1}{kl_2} \Rightarrow \frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2}$

**(3) Internal resistance of a cell :**

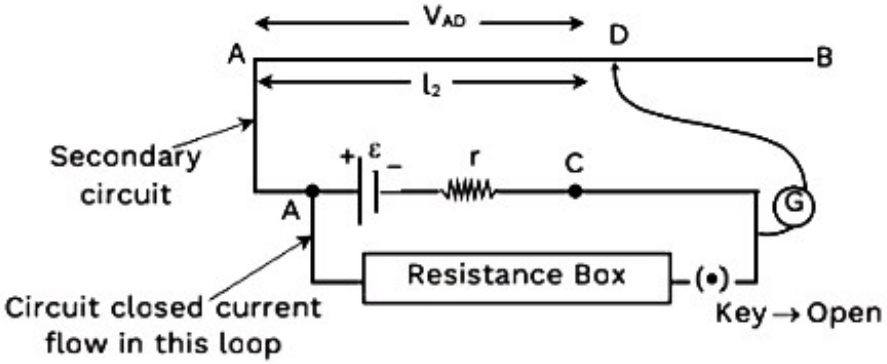



Step 1: Open the resistance box key and find the balancing length.

Emf of cell = $kl_1$ ...(1)

$l_1$ = Balancing length in step 1, k = potential gradient

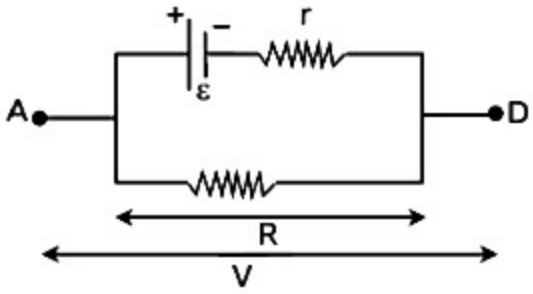
Step 2: Closed the key and plug-out 'R' resistance from 'Resistance Box' and find the length of new Balance point.




Potential difference V = $kl_2$

**Now see carefully closed circuit :**

$$V = \frac{\varepsilon \times R}{(R + r)} \qquad ...(2)$$




⇒ From equation (1) and (2)

$$\frac{V}{\varepsilon} = \frac{R}{R + r} \Rightarrow \frac{kl_2}{kl_1} = \frac{R}{(R + r)}$$

$$\Rightarrow \qquad r = \left(\frac{l_1 - l_2}{l_2}\right) \times R$$

Where, r = initial resistance of battery

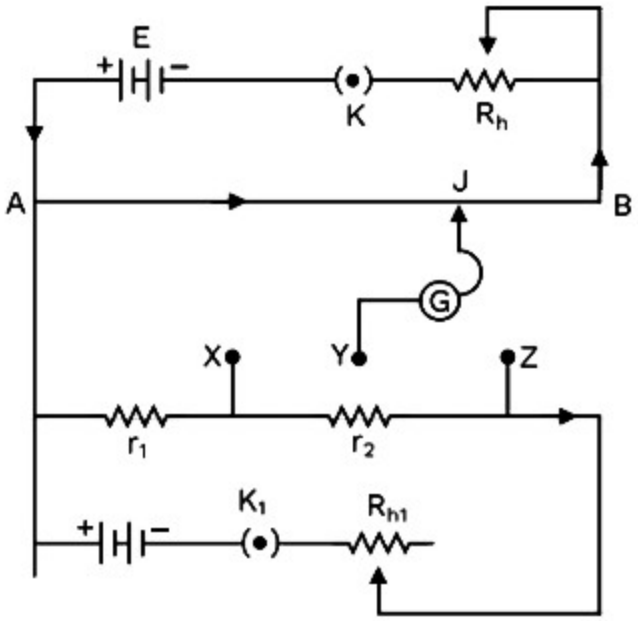
$l_1$ = Balancing length without 'R' (Step-1)

$l_2$ = Balancing length with 'R' (Step-2)

R = value of resistance plugged out from resistance box.

**Comparison of two small resistances :**

(a) For comparison of two small resistances with the help of potentiometer the following circuit is used :




If the potential difference across the ends of resistance $r_1$ is $V_1$ and balancing length for it is $l_1$ then

$V_1 = ir_1 = xl_1 \qquad ...(1)$

(b) The potential difference across the resistances $r_1$ and $r_2$ connected in series is $V_2$ and balancing length for them is $\lambda_2$ then

$V_2 = i(r_1 + r_2) = xl_2 \qquad ...(2)$

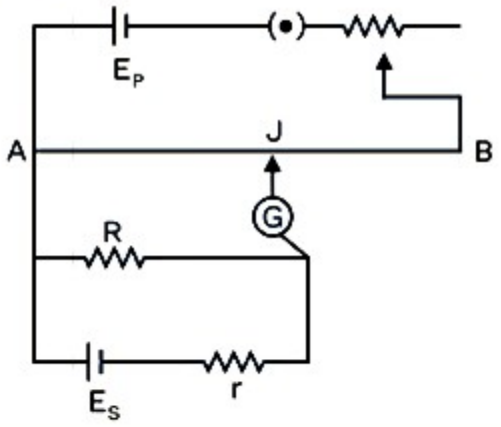
From equation (1) and (2), $\frac{i(r_1 + r_2)}{ir_1} = \frac{x\ell_2}{x\ell_1}$ or $\frac{r_1 + r_2}{r_1} = \frac{\ell_2}{\ell_1}$, $\frac{r_2}{r_1} = \frac{\ell_2 - \ell_1}{\ell_1}$

(c) If $r_1$ is known then $r_2$ will be, $r_2 = \left(\frac{\ell_2 - \ell_1}{\ell_1}\right) r_1$

(d) An unknown small resistance can also be determined with the help of potentiometer. The small resistance r is connected in series with a known resistance R and the balancing lengths for the potential differences across R and (R + r) are found out. If the balancing lengths are respectively $l_1$ and $l_2$ then

$iR = x\ell_1$ and $i(r + R) = x\ell_2$ hence $r = \frac{\ell_2 - \ell_1}{\ell_1} R$

**Q.** **One of the circuits for the measurement of resistance using a potentiometer is shown. The galvanometer is connected and zero deflection is observed at length AJ = 30 cm. In second case the secondary cell is changed.**
**Take $E_s$ = 10 V and r = 1Ω in 1st reading**
**And $E_s$ = 5 V and r = 2Ω in 2nd reading.**
**In second case, the zero deflection is observed at length AJ = 10 cm. Calculate the resistance R (in ohm).**




**Sol.** At zero deflection, we have

$$\frac{V}{l} = \frac{I_1 R}{30} = \frac{I_2 R}{10} \qquad \ldots(1)$$

Now $I_1 = \frac{E_{S_1}}{r_1 + R} \Rightarrow I_1 = \frac{10}{1+R}$, $I_2 = \frac{E_{S_2}}{r_2 + R} \Rightarrow I_2 = \frac{5}{2+R}$

Substituting in (1), we get

$$\frac{10R}{(1+R)\times 30} = \frac{5R}{(2+R)\times 10} \Rightarrow 4 + 2R = 3 + 3R \Rightarrow R = 1\Omega$$

**Q.** **In a potentiometer circuit to find the internal resistance r of the cell of emf E, when K is open, the balance point is obtained at 60 cm. When K is closed, the balance point is obtained at 50 cm. What is the value of r? If the cell is shorted by a 10Ω resistor, then calculate the internal resistance of the cell.**

**Sol.** $l_1$ = 60 cm, when K is open and
$l_2$ = 50 cm, when K is closed

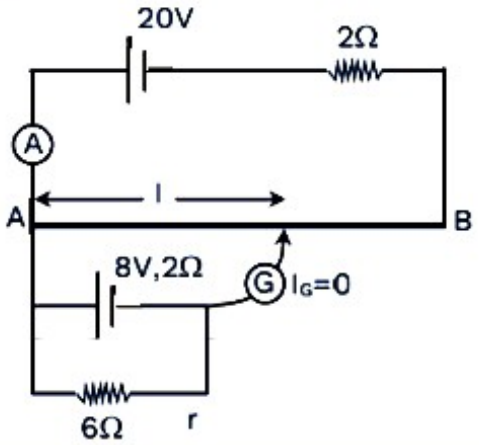
$$\text{Since } r = \left(\frac{l_{\text{key open}}}{l_{\text{key closed}}} - 1\right)R \quad \Rightarrow \quad r = \left(\frac{l_1}{l_2} - 1\right)R = \left(\frac{60}{50} - 1\right)10\Omega = \frac{1}{5} \times 10\Omega = 2\Omega$$

**Q.** **In a potentiometer arrangement a cell of emf 1.5 V gives a balance point at 30 cm length of wire. Now, when the cell is replaced by another cell, the balance point shifts to 50 cm. Calculate the emf of second cell?**

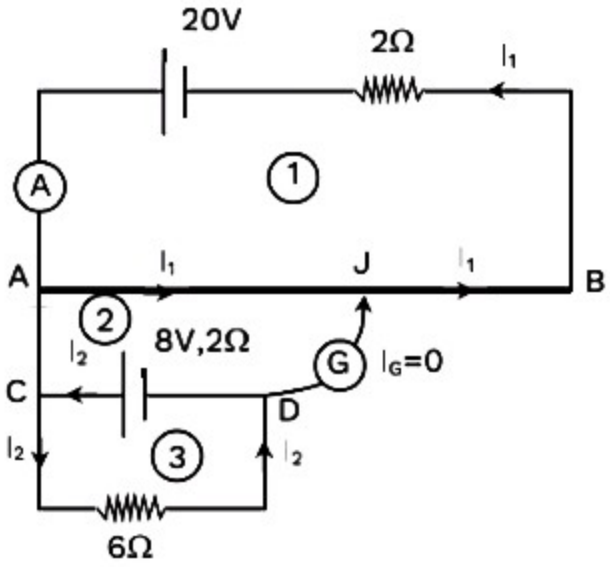
**Sol.** $E_1$ = 1.5V, $l_1$ = 30 cm
$E_2$ = ? , $l_2$ = 50 cm
Using the formula for comparison of emf of cells by potentiometer, we have

$$\frac{E_2}{E_1} = \frac{l_2}{l_1} \quad \Rightarrow \quad E_2 = \frac{l_2}{l_1} \times E_1 = \frac{50}{30} \times 1.5\text{V} = 2.5\text{V}$$

**Q.** **In the figure shown, wire AB has a length of 100 cm and resistance 8Ω. Find the balance point length l.**



**Sol.**




For condition of balance, we have $I_G = 0$.

So loops (1) and (3) are independent of each other. Hence we observe that $V_{AJ}$ = $V_{CD}$

$\Rightarrow \quad I_1 R_{AJ} = E - I_2 r \qquad ...(1)$

Where E = 8 V, r = 2 Ω

$$I_1 = \left(\frac{20}{2+8}\right) = 2A$$

$$R_{AJ} = \left(\frac{R_{AB}}{l_{AB}}\right) l_{AJ} = \left(\frac{8}{100}\right) l = \frac{2l}{25}$$

$$I_2 = \frac{8}{2+6} = 1A$$

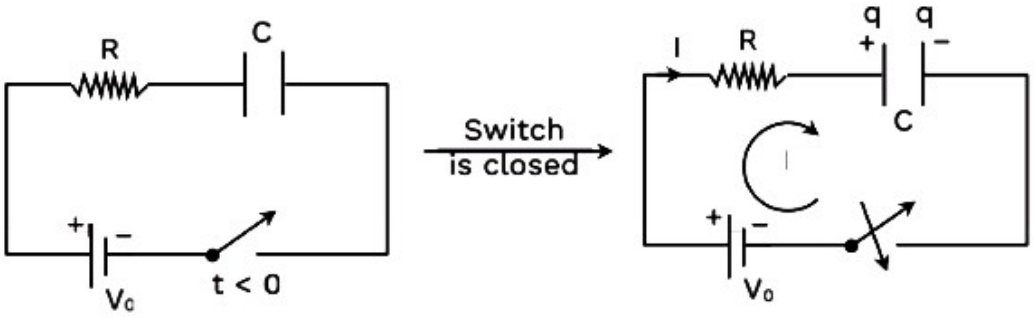
Substituting these values in equation (1), we get

$$2\left(\frac{2l}{25}\right) = 8 - (1)(2) \quad \Rightarrow \quad \frac{4l}{25} = 6$$

$$\Rightarrow \; 4l = 150 \quad \Rightarrow \quad l = \frac{150}{4} = 37.5\,cm$$

# Charging & Discharging of a Capacitor (R-C Circuit)

**Charging of a Capacitor**




It is given that initially capacitor is uncharged. Let at any time 't' charge on capacitor is 'q'. Apply Kirchhoff's voltage law

$$V_0 - IR - \frac{q}{C} = 0 \qquad \Rightarrow \quad IR = V_0 - \frac{q}{C}$$

$$\Rightarrow \quad IR = V_0 - \frac{q}{C} \qquad \Rightarrow \quad I = \frac{V_0C - q}{RC}$$

$$\Rightarrow \quad \frac{dq}{dt} = \frac{V_0C - q}{RC} \qquad \Rightarrow \quad \int_0^q \frac{dq}{V_0C - q} = \int_0^t \frac{dt}{RC}$$

$$\Rightarrow \quad -\left|\ln(V_0C - q)\right|_0^q = \frac{t}{RC} \qquad \Rightarrow \quad \ln\left(\frac{V_0C - q}{V_0C}\right) = -\frac{t}{RC}$$

Take antilog Both side :

$$\frac{V_0C - q}{V_0C} = e^{-\frac{t}{RC}}, \quad 1 - \frac{q}{CV_0} = e^{-\frac{t}{RC}}, \quad q = CV_0\left(1 - e^{-\frac{t}{RC}}\right)$$

Where,

Q = charge on capacitor at any time 't'

V = voltage across the capacitor at final state (Terminal voltage)

RC = τ (Time constant)

CV = $q_{max}$ = maximum charge on capacitor

$$q = q_{max}\left(1 - e^{-\frac{t}{RC}}\right)$$

**Time constant of the circuit (t) :** Time constant of circuit is defined as the time at which charge on the capacitor reach at 63% of maximum value of charge.

### q-t Graph

$$q = q_{max}\left(1 - e^{-\frac{t}{RC}}\right)$$

at $(t = \tau = RC)$;

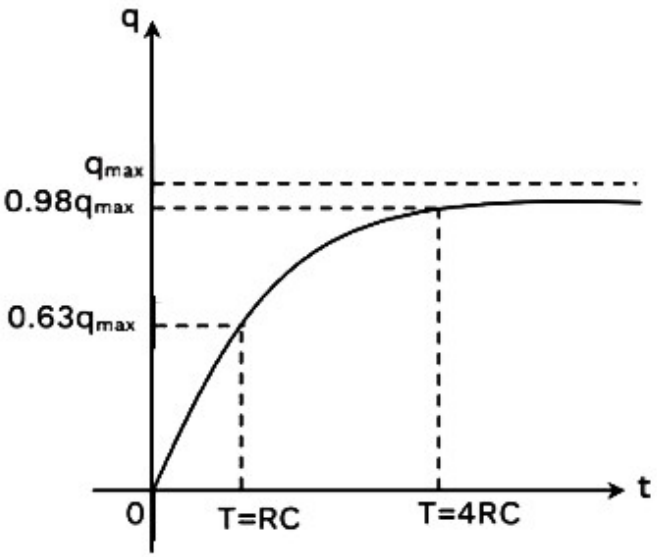
$$q = q_{max}\left(1 - e^{-\frac{RC}{RC}}\right)$$

$$q = q_{max}\left(1 - e^{-1}\right)$$

$$q = q_{max}\,(1 - 0.37)$$

$$q = 0.63\,q_{max}$$

$$q = 0.63\,VC$$




### Current at any Time 't'

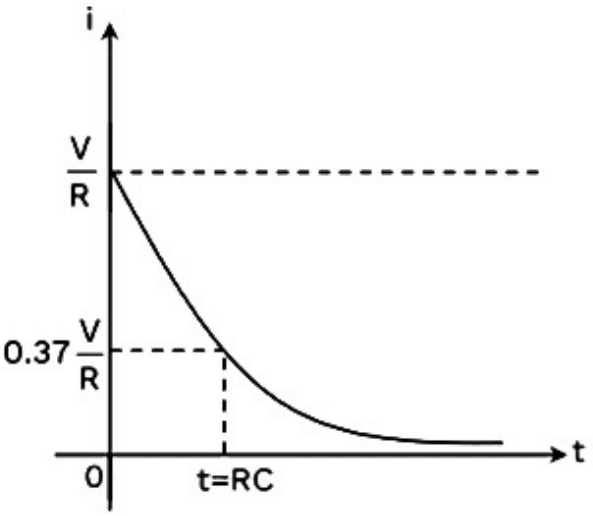
$$i = \frac{dq}{dt} = \frac{d}{dt}\left[q_{max}\left(1 - e^{-\frac{t}{RC}}\right)\right] = q_{max}\left[0 - e^{\frac{t}{Rc}}\left(-\frac{1}{RC}\right)\right]$$

$$i = \frac{q_{max}}{RC}.e^{-\frac{t}{RC}} \Rightarrow i = \left(\frac{q_{max}}{C}\right) \times \frac{1}{R}.e^{-\frac{t}{RC}} \Rightarrow i = \frac{V_0}{R}e^{-\frac{t}{RC}}$$

$\Rightarrow$ at $t = RC$ :

$$i = \frac{V_0}{R}.e^{-\frac{RC}{RC}} \Rightarrow i = \frac{V_0}{R}e^{-1} \Rightarrow i = 0.37\frac{V}{R}$$

### Graph :



Voltage across the capacitor at any Time 't' :

$q = CV_C(t)$

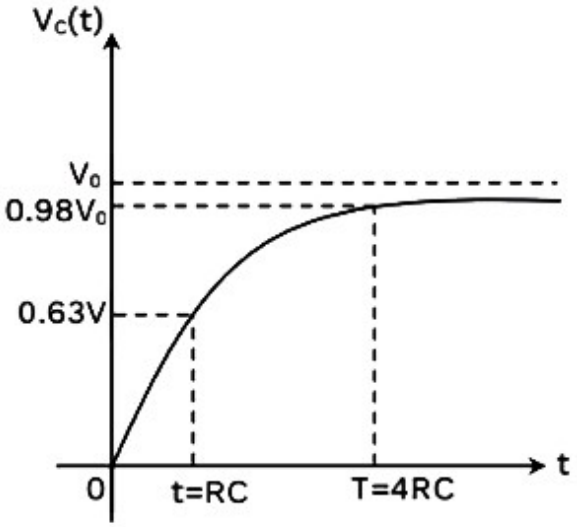
$$V_C(t) = \frac{q}{C} = \frac{1}{C} \times q_{max}\left(1 - e^{-\frac{t}{RC}}\right) = \frac{CV_0}{C}\left(1 - e^{-\frac{t}{RC}}\right) \Rightarrow V_C(t) = V_0\left(1 - e^{-\frac{t}{RC}}\right)$$

Where,

$V_C(t)$ = voltage across the capacitor at any Time 't'

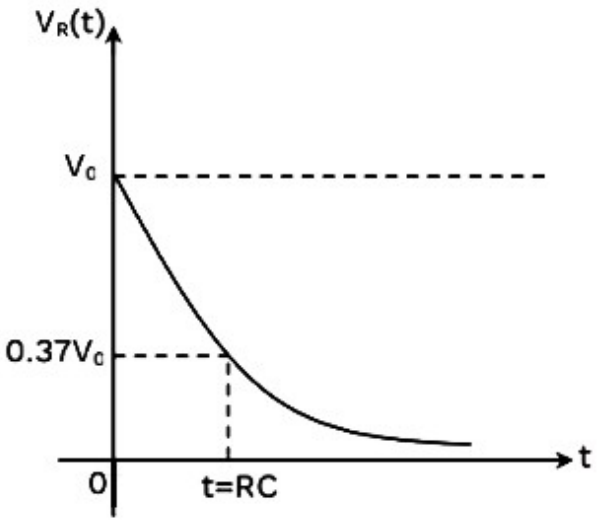
V = terminal voltage across the capacitor (at the condition when capacitor fully charge)

RC = Time constant.




**Voltage across the Resistor :** $V_R = IR$, $V_R = \left(\frac{V_0}{R}e^{-\frac{t}{RC}}\right) \times R$, $V_R(t) = V_0 e^{-\frac{t}{RC}}$

**Graph :**



**Heat Generated**

(A) First Method :

By energy conservation

H = work done by battery – $\Delta U_{capacitor}$

= charge flow through the battery – (energy stored in capacitor)

$= (q_{max})\,(V_0) - \frac{1}{2}CV_0^2 \quad = (CV_0) \times V_0 - \frac{1}{2}CV_0^2$

Heat generated = $\frac{1}{2}CV_0^2$

(B) Second Method :

Heat (H) = $\int_0^\infty i^2 R\,dt \;= \int_0^\infty \left(\frac{V_0}{R}\right)^2 \times e^{-\frac{2t}{RC}} \times R \times dt \;= \frac{V_0^2}{R}\int_0^\infty e^{-\frac{2t}{RC}}.dt$

Heat generated = $\frac{1}{2}CV_0^2$

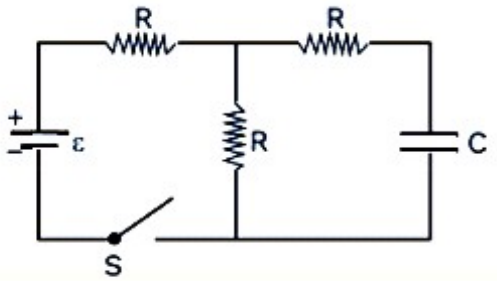
**Q.** **A capacitor is connected to a 12V battery through a resistance of 10Ω. It is found that the potential difference across the capacitor rises to 4.0 V in 1 μs. Find the capacitance of the capacitor.**

**Sol.** We know, $V = V_0\left(1 - e^{-\frac{t}{RC}}\right) \Rightarrow 4 = 12\left(1 - e^{-\frac{t}{RC}}\right) \Rightarrow 1 - e^{-\frac{t}{RC}} = \frac{1}{3}$

$\Rightarrow \quad e^{-\frac{t}{RC}} = \frac{2}{3} \quad \Rightarrow \quad \frac{t}{RC} = \ln\left(\frac{3}{2}\right) = 0.405 \quad \Rightarrow \quad RC = \frac{t}{0.405} = 2.469\,\mu s$

$\Rightarrow \qquad C = \frac{2.469\,\mu s}{10\Omega} = 0.25\,\mu F$

**Q.** **Initially the capacitor is uncharged. Find the charge on capacitor as a function of time, if switch is closed at t = 0.**



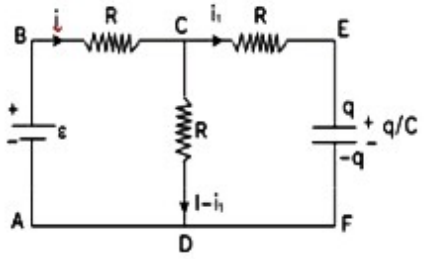
**Sol.** First Method :

Without using the formula of equivalent.

Applying KVL in loop ABCDA

$$\varepsilon - iR - (i - i_1)R = 0$$

$$\varepsilon - 2iR + i_1R = 0$$




Applying KVL in loop ABCEFDA

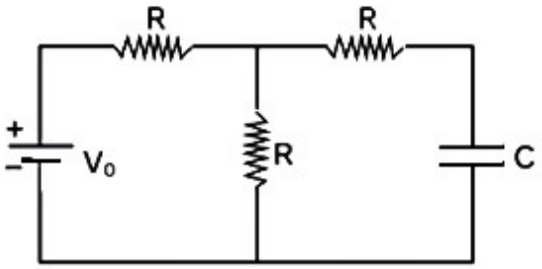
$$\varepsilon - iR - i_1R - \frac{q}{C} = 0$$

$$\frac{2\varepsilon - \varepsilon - i_1R - 2i_1R}{2} = \frac{q}{C}, \quad \varepsilon C - 3i_1RC = 2q, \quad \varepsilon C - 2q = 3\frac{dq}{dt}RC$$

$$\int_0^q \frac{dq}{\varepsilon C - 2q} = \int_0^t \frac{dt}{3RC}, \quad -\frac{1}{2}\ln\frac{\varepsilon C - 2q}{\varepsilon C} = \frac{t}{3RC}, \quad q = \frac{\varepsilon C}{2}\left(1 - e^{-\frac{2t}{3RC}}\right)$$

Second Method :

By using the formula of equivalent.




We know that charge on capacitor at any Time 't' us given by:

$$q = q_{max}\left(1 - e^{-\frac{t}{R_{eq}C}}\right) \qquad ...(1)$$

$R = R_{eq}$ = equivale resistance across the capacitor when battery is replaced by its internal resistance.

$q_{max}$ = maximum charge on capacitor.

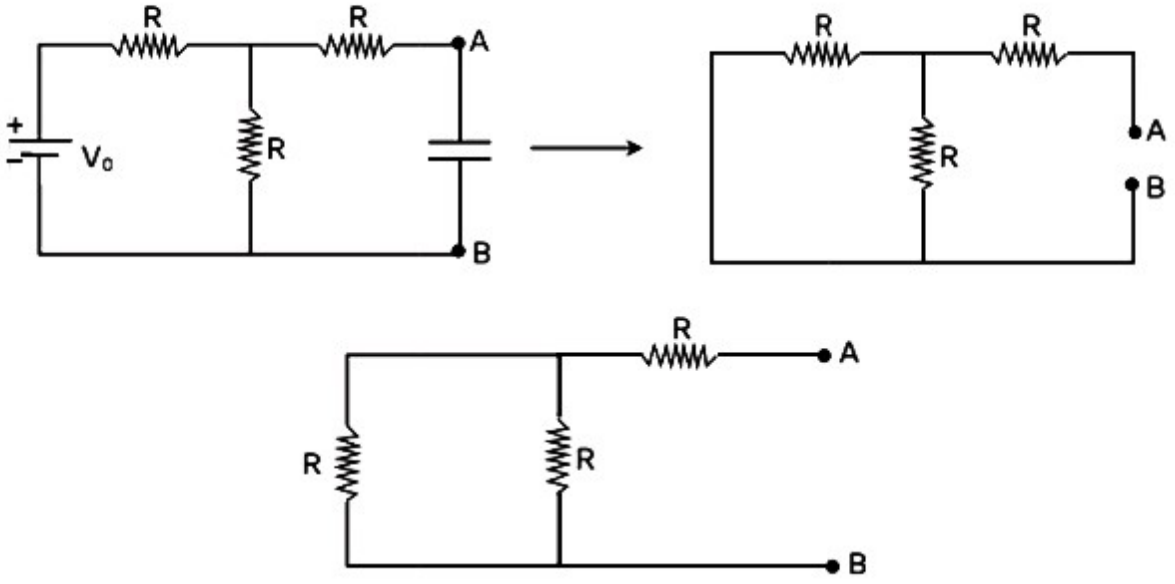
**Step 1:** Calculation for $R_{eq}$

Follow these steps for $R_{eq}$ calculation

(i) Remove capacitor

(ii) Voltage source by its internal resistance

(iii) Calculate equivalent resistance across the capacitor terminal.




$$\Rightarrow R_{AB} = R_{eq} = \frac{3R}{2}$$

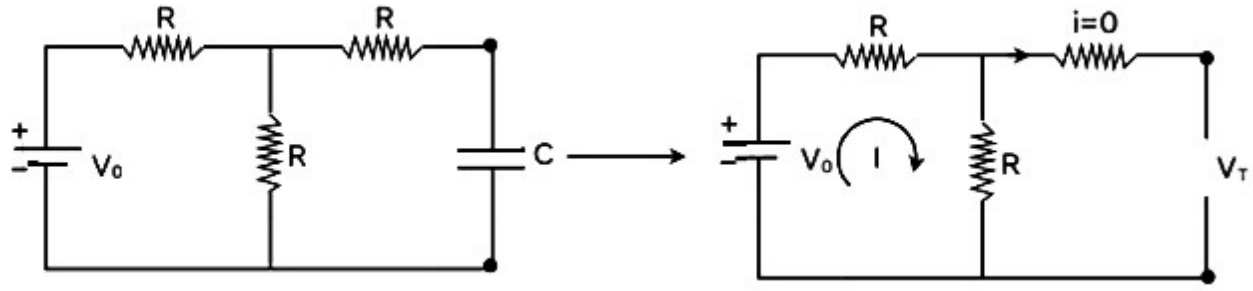
**Step – 2:** Calculate maximum charge on Capacitor

Follow these steps to calculate maximum charge on capacitor.

(i) Remove capacitor

(ii) calculate terminal voltage $V_T$ across the capacitor

(iii) $q_{max}$ = C × terminal voltage




$$I = \frac{V_0}{2R} \; ; \; V_T = I \times R = \frac{V_0}{2R} \times R = \frac{V_0}{2}$$

$V_T$ = terminal voltage across the capacitor

= it is voltage across the when capacitor is fully charged

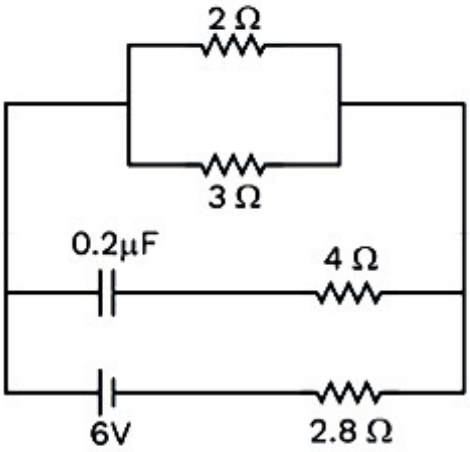
It is also called voltage across the capacitor at final state.

$$\Rightarrow \quad q_{max} = C \times V_T = C \times \frac{V_0}{2} \quad \Rightarrow \quad q_{max} = \frac{CV_0}{2}$$

Now from equation (1)

$$q = q_{max}\left(1 - e^{-\frac{t}{R_{eq} \times C}}\right), \quad q = \frac{CV_0}{2}\left(1 - e^{-\frac{2t}{3RC}}\right)$$

**Q.** **In the given circuit, the steady state current through the 2Ω resistor is**



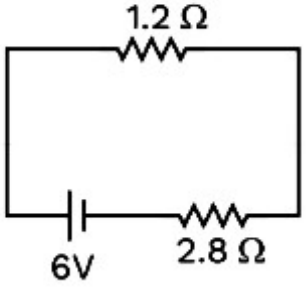

**(A) 0.6 A**

**(B) 0.9 A**

**(C) 1.2 A**

**(D) 1.5 A**

**Sol.** Since capacitor is a dc blocking element, so no current flows through the branch containing the capacitor. However, a voltage drop will exist across this branch.




$$R_{net} = 4\Omega \quad \Rightarrow \quad I_{total} = \frac{V}{R_{net}} = 1.5\,A$$

Voltage drop across 2.8 Ω is $V_1 = (2.8)\,(1.5) \Rightarrow V_1 = 4.2\ V$

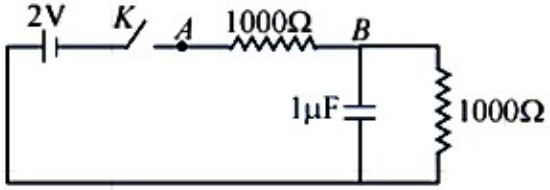
So, potential across parallel combination of 2Ω and 3Ω is

$$V_2 = 6 - 4.2 = 1.8\,V$$

If $I_2$ is the current in 2Ω resistor, then

$$I_2 = \frac{1.8}{2} = 0.9\,A$$

**Q.** **In the adjoining circuit, when the key K is pressed at time t = 0, which of the following statements about current I in the resistor AB is true?**




**(A)** **I = 2 mA at all t**
**(B)** **I oscillates between 1 mA and 2 mA**
**(C)** **I = 1 mA at all t**
**(D)** **At t = 0, I = 2 mA and with time it goes to 1 mA**

**Sol.** **Case 1:** At = 0 (Initial condition of capacitor), capacitor behaves as short circuit since it is uncharged.

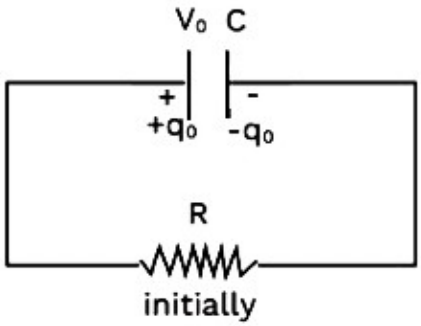
$$\text{So, } i = \frac{2}{1000\ \Omega} = 2\,\text{mA}$$

**Case 2:** At = $\infty$ (steady state condition), after long time when capacitor fully charged it behaves as open circuit then

$$i = \frac{2}{1000 + 1000} = \frac{2}{2000} \qquad \boxed{i = 1\ \text{mA}}$$

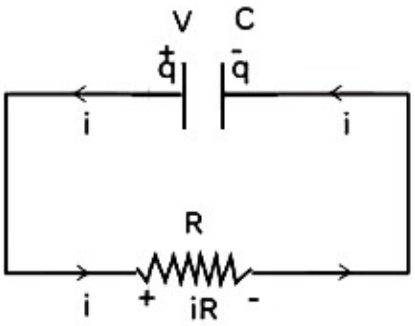
**Discharging of a Capacitor**

**Derivation of equation of discharging circuit :**




initially

At t = 0



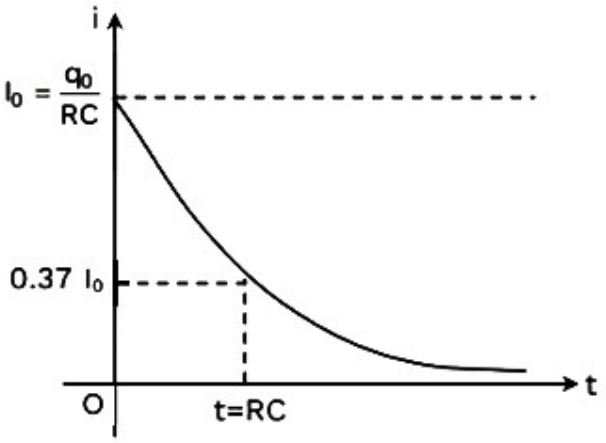

At any time t

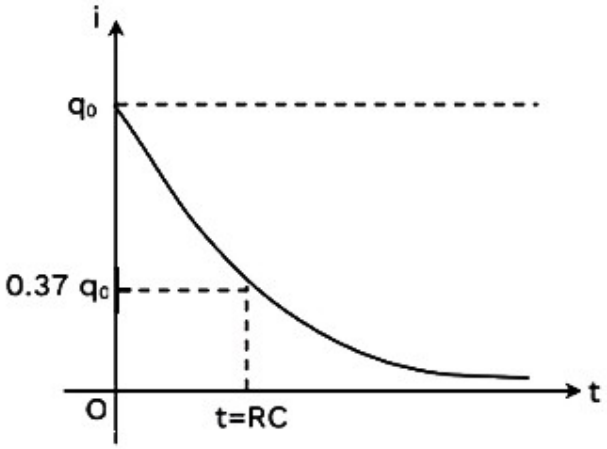
Applying K.V.L.

$$+\frac{q}{C}-iR=0\ ;\quad i=\frac{q}{CR}\Rightarrow -\frac{dq}{dt}=\frac{q}{CR},\quad \int_{q_0}^{q}\frac{-dq}{q}=\int_{0}^{t}\frac{dt}{CR}\quad\Rightarrow -\ln\frac{q}{q_0}=+\frac{t}{RC}$$

$$\boxed{q = q_0.e^{-t/RC}}$$

$$\boxed{i=-\frac{dq}{dt}=\frac{q_0}{RC}e^{-t/RC}}$$




**Q.** **The electric field between the plates of a parallel-plate capacitance 2.0 μF drops to one third of its initial value in 4.4 μs when the plates are connected by a thin wire. Find the resistance of the wire.**

**Sol.** The electric field between the plates is

$$E=\frac{Q}{A\varepsilon_0}=\frac{Q_0}{A\varepsilon_0}e^{-t/RC}\quad\Rightarrow\quad E=E_0e^{-t/RC}$$

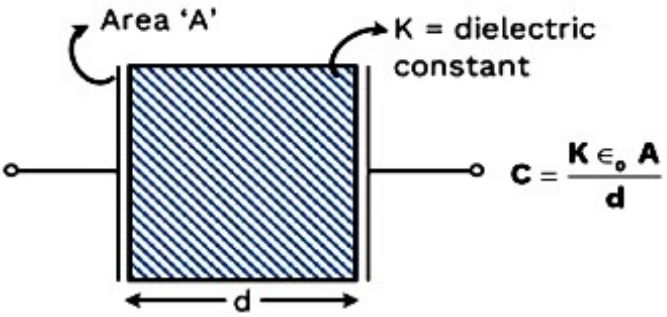
In the given problem, $E=\frac{1}{3}E_0$ at $t = 4.4\,\mu s$.

Thus, $\frac{1}{3}=e^{-\frac{4.4\mu s}{RC}}$, $\frac{4.4\mu s}{RC}=\ln 3=1.1$, $R=\frac{4.4\mu s}{1.1\times 2.0\mu F}=2.0\Omega$.

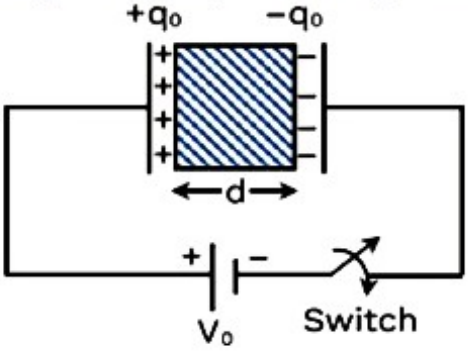
## Leakage Current through a Capacitor

Dielectric is not a perfect insulating material. So, when the space between the plates is completely filled with dielectric, there flows some charge through the dielectric medium. The flow of charge per unit time through a dielectric medium is called Leakage Current.

Suppose there is a capacitor of plate area 'A' and separation between plates is d and is filled with dielectric material of dielectric constant 'K'.




Now, connect this capacitor through a battery for a long time.




After long time, this capacitor gets fully charged.

Let : Charge on the capacitor = q0 at fully charged condition

Voltage of the capacitor = V0 = Battery Voltage.

Now, disconnect this battery, and observe the capacitor.

We observe when battery is disconnected, the charge on positive plate is going towards the negative plate vi dielectric material and capacitor starts discharging. Since dielectric materials have some resistance which is given by

$$R = \rho \times \frac{d}{A}$$

Where; R = Resistance of the dielectric material

$\rho$ = Resistivity of the dielectric material

So, it is a clear case of discharging of capacitor as we discussed in previous article :-

Charge on the capacitor at any time instant in a discharging RC circuit is given as,

$$q = q_0 e^{-\left(\frac{t}{RC}\right)}$$

$$R = \rho \times \frac{d}{A}$$

$$C = \frac{K \epsilon_0 A}{d}$$

$$RC = \text{Time constant} = \rho \times \frac{d}{A} \times \frac{K \epsilon_0 A}{d}$$

$$RC = \rho K \epsilon_0$$

Now, $q = q_0 e^{-\left(\frac{t}{K\epsilon_0 \rho}\right)}$

Leakage current $i = -\frac{dq}{dt}$

$$= -\frac{d}{dt}\left(q_0 e^{-\left(\frac{t}{K\epsilon_0 \rho}\right)}\right) = -q_0 \times e^{-\left(\frac{t}{K\epsilon_0 \rho}\right)} \left\{-\frac{1}{K \epsilon_0 \rho}\right\}$$

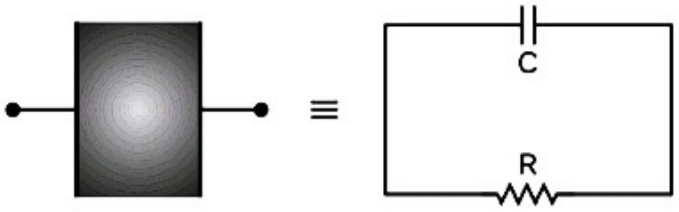
$$i = \frac{q_0}{K \epsilon_0 \rho} e^{-\left(\frac{t}{K\epsilon_0 \rho}\right)}$$

$$\boxed{i = \frac{q_0}{K \epsilon_0 \rho} \cdot e^{-\left(\frac{t}{K\epsilon_0 \rho}\right)}}$$

Where
i = Leakage current
$q_0$ = maximum charge at t = 0
K = Dielectric constant.
d = Separation between plates of the capacitor
$\varepsilon_0$ = Permittivity of material
$\rho$ = Resistivity of the material

**Q.** **A parallel plate capacitor is filled completely with a material having dielectric constant k = 5 and electrical conductivity $\sigma = 7.4 \times 10^{-12}\ \Omega^{-1}\ m^{-1}$. If the charge on the plate of the capacitor at instant t = 0 is equal to 8.85 μC, then calculate the leakage current at the instant t = 12 s.**

**Sol.** The problem deals with discharging of CR circuit, because between the plates of the capacitor, there is capacitor as well as resistance.




$$R = \frac{\rho d}{A} = \frac{d}{\sigma A} \qquad \{\text{where } \rho = \text{resistivity and } \sigma = \text{conductivity}\}$$

and $C = \frac{K\varepsilon_0 A}{d}$

The capacitive time constant of the circuit is,

$$\tau = CR = \frac{K\varepsilon_0}{\sigma}$$

Substituting the values, we have

$$\tau = \frac{5 \times 8.86 \times 10^{-12}}{7.4 \times 10^{-12}} = 5.98 \text{ s}$$

Charge at any time decrease exponentially as

$$q = q_0 e^{-t/\tau}$$

Where $q_0 = 8.85 \times 10^{-6}$ C is the charge at time t = 0. Therefore, discharging (leakage) current at time t is given by

$$I = \left(-\frac{dq}{dt}\right) = \frac{q_0}{\tau} e^{-t/\tau}$$

So, the current at t = 12 s is

$$I = \frac{(8.85 \times 10^{-6})}{5.98} e^{-12/5.98} = 0.198 \times 10^{-6} \text{ A} = 0.198 \,\mu\text{A}$$